



CV2 Oil-Immersed Vacuum On-Load Tap-Changer

Operating Manual

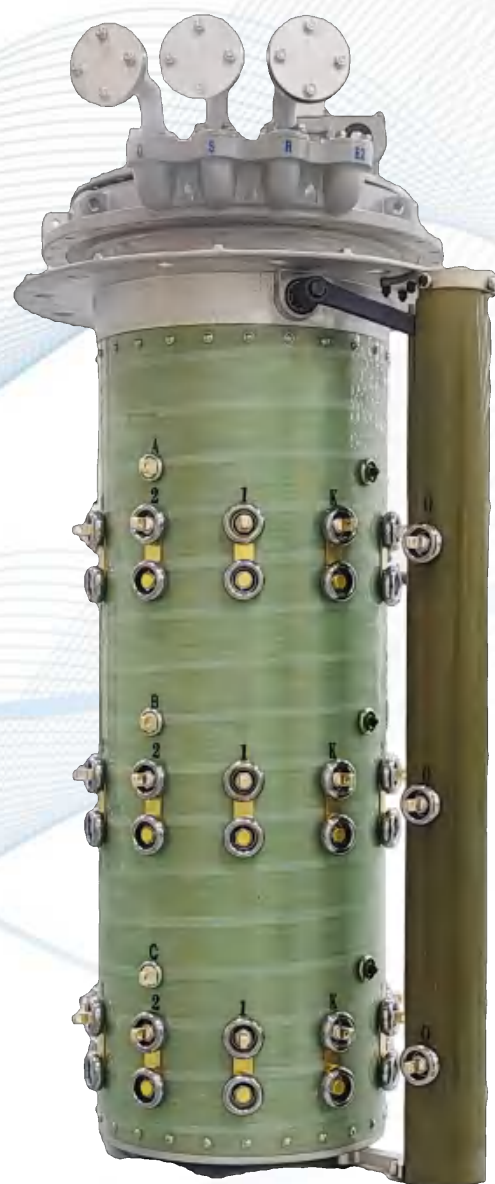
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Applicable Product Models: CV2 I/II/III

Important Note!

Read carefully before use.

Keep for future reference.



Legal notice

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Contents

1. About this manual.....	1
1.1. Target audience.....	1
1.2. Document retention.....	1
1.3. Related documents.....	1
1.4. Conventions.....	2
1.4.1. Warning messages.....	2
1.4.2. Notes.....	4
1.4.3. Procedural instructions.....	4
1.5. Abbreviations.....	5
2. Safety notes.....	7
2.1. Safety instructions.....	7
2.2. Safety Label.....	8
2.3. Intended use.....	9
2.4. Foreseeable misuse.....	11
2.5. Personnel qualifications.....	12
2.6. Personal Protective Equipment (PPE).....	13
3. Product overview.....	15
3.1. Location of On-Load Tap-Changer (OLTC) in transformer.....	15
3.2. OLTC main body.....	16
3.3. Drive mechanism components.....	17
3.4. Technical specifications.....	18
3.5. Operating conditions.....	19
3.6. Product compatibility requirements.....	19
3.7. Model description.....	20
3.7.1. Nameplate.....	20
3.7.2. Model number structure.....	20
3.7.3. Basic connection modes.....	21
4. Packaging, transportation, and storage.....	23
4.1. Supplied components.....	23
4.2. Transportation.....	24

4.2.1. Receiving inspection.....	25
4.2.2. Transshipment.....	25
4.2.3. Unpacking and inspection.....	25
4.2.4. Lifting.....	26
4.3. Storage.....	26
5. Installation of the OLTC on the transformer.....	27
5.1. Safety.....	27
5.2. Preparations.....	27
5.2.1. Transformer mounting flange.....	28
5.2.2. List of Tools.....	28
5.3. Install the On-Load Tap-Changer (OLTC) on a bell-type transformer.....	29
5.3.1. Overview.....	29
5.3.2. Remove the top cover.....	30
5.3.3. Separate the energy storage spring and oil suction pipe.....	33
5.3.4. Remove the OLTC Mounting Flange.....	34
5.3.5. Install the OLTC into the bell-type transformer.....	35
5.3.6. Install the bell-type transformer tank cover and OLTC mounting flange.....	37
5.3.7. Reinstall the energy storage spring and the oil suction pipe.....	40
5.3.8. Install the top cover.....	41
5.4. Install the On-Load Tap-Changer (OLTC) to the top-mounted transformer.....	43
5.5. Remove the selector switch insert.....	45
6. Transformer turns ratio test and DC winding resistance measurement.....	51
6.1. Turns ratio test.....	51
6.2. DC winding resistance measurement.....	54
7. Drying and oil filling.....	55
7.1. Vacuum drying.....	55
7.1.1. Vacuum drying in a drying oven.....	55
7.1.2. Vacuum drying in the transformer tank.....	55
7.2. Vapor-phase drying.....	57
7.2.1. Vapor-phase drying in a drying oven.....	58
7.2.2. Vapor-phase drying in the transformer tank.....	59
7.3. Oil filling.....	61



8. Installation of protective devices and drive unit.....	63
8.1. Install protective relay.....	63
8.2. Install pressure relief valve.....	63
8.3. Install Motor Drive Unit (MDU).....	64
8.4. Install drive mechanism components.....	65
8.4.1. Drive mechanism components.....	65
8.4.2. Adjust the top gearbox.....	66
8.4.3. Install the bevel gearbox.....	67
8.4.4. Install the intermediate support gearbox.....	69
8.4.5. Install the horizontal driving shaft and vertical driving shaft.....	70
8.4.6. Install the driving shaft guards.....	71
8.5. Installation Precautions for Three Single-phase OLTCs or I+II OLTC Configuration.....	74
8.6. Calibrate the OLTC Drive System.....	74
8.6.1. Calibration procedure.....	74
8.6.2. Calibration example.....	75
9. Factory Acceptance Tests (FAT).....	77
9.1. Test preparations.....	77
9.1.1. Degas the OLTC.....	77
9.1.2. Grounding.....	78
9.2. Tests at the transformer manufacturer's facility.....	79
9.2.1. Functional test.....	79
9.2.2. Transformer electrical tests.....	80
10. Transformer transportation and on-site commissioning.....	81
10.1. Remove the MDU for transportation.....	81
10.2. Transport with oil and no conservator.....	81
10.3. Transport without oil.....	82
10.4. On-site commissioning.....	82
11. OLTC operation monitoring.....	85
12. Inspection, maintenance, and disposal.....	87
12.1. Periodic maintenance inspection.....	88
12.2. Maintenance intervals.....	90
12.3. Disposal.....	90

13. Drawings.....	91
13.1. Overall dimensions of CV2 type OLTC with change-over selector.....	92
13.2. Overall dimensions of CV2 type OLTC without change-over selector.....	93
13.3. Dimension drawing for bell-type mounting flange of CV2 type OLTC.....	94
13.4. Bell-type installation diagram for CV2 type OLTC.....	95
13.5. Dimensional drawing of bevel gearbox.....	96
13.6. Connection layout drawing of CV2 type OLTC and SHM series MDU.....	97
13.7. Connection layout drawing of CV2 type OLTC and CMA7 MDU.....	98
13.8. OLTC bypass pipe structure drawing.....	99
13.9. Special tool for installing and removing oil suction pipe for CV2 type OLTC.....	100
13.10. Horizontal and vertical driving shaft installation.....	101
13.11. Dimension drawing of CV2 type intermediate support gearbox.....	102
13.12. CV2 type OLTC 10070 operating position table and wiring schematic.....	103
13.13. CV2 type OLTC 10071W operating position table and wiring schematic.....	104
13.14. CV2 type OLTC 10191G operating position table and wiring schematic.....	105
13.15. CV2 type OLTC 10191W operating position table and wiring schematic.....	106
13.16. CV2 type OLTC 10193G operating position table and wiring schematic.....	107
13.17. CV2 type OLTC 10193W operating position table and wiring schematic.....	108
13.18. CV2 type OLTC 12231G operating position table and wiring schematic.....	109
13.19. CV2 type OLTC 12231W operating position table and wiring schematic.....	110
13.20. CV2 type OLTC 12233G operating position table and wiring schematic.....	111
13.21. CV2 type OLTC 12233W operating position table and wiring schematic.....	112

List of figures

Figure 2-1 Safety label on the rupture disk.....	9
Figure 3-1 OLTC in the transformer.....	15
Figure 3-2 Overview of the OLTC structure.....	16
Figure 3-3 Drive mechanism components.....	17
Figure 3-4 Nameplate of the On-Load Tap-Changer.....	20
Figure 3-5 Model number description.....	20
Figure 3-6 Description of connection mode.....	21
Figure 4-1 Supply components (partial).....	23
Figure 4-2 Lift the OLTC.....	26
Figure 5-1 Transformer mounting flange.....	28
Figure 5-2 Fix the top gearbox limit plate.....	30
Figure 5-3 Remove the top cover bolts.....	31
Figure 5-4 Remove the top cover.....	32
Figure 5-5 Separate the energy storage spring from the OLTC mounting flange.....	33
Figure 5-6 Separate the oil suction pipe from the OLTC mounting flange.....	33
Figure 5-7 Remove the nuts on the OLTC mounting flange.....	34
Figure 5-8 Diagram of lifting off the mounting flange.....	34
Figure 5-9 OLTC placed on temporary support.....	35
Figure 5-10 Connect the tap selector terminal to the transformer regulation winding.....	36
Figure 5-11 Sealing ring of the support flange.....	37
Figure 5-12 Bolts and sealing ring of the transformer mounting flange.....	38
Figure 5-13 Align the mounting flange.....	38
Figure 5-14 Install the OLTC mounting flange.....	39
Figure 5-15 Reinstall the oil suction pipe.....	40
Figure 5-16 Reinstall the energy storage spring.....	40
Figure 5-17 Install the sealing ring onto the top cover.....	41
Figure 5-18 Align the bevel gear box pin holes.....	41
Figure 5-19 Install the top cover.....	42
Figure 5-20 Bolts and sealing ring of the transformer mounting flange.....	43
Figure 5-21 Lift the OLTC into the transformer tank.....	43

Figure 5-22 Connect the tap selector terminal to the transformer regulation winding.....	44
Figure 5-23 Remove the specific tap positions for selector switch insert.....	45
Figure 5-24 Remove the oil suction pipe from the selector switch insert.....	46
Figure 5-25 Insert the locating pin into the pin hole.....	47
Figure 5-26 Remove the nuts that secure the selector switch insert.....	47
Figure 5-27 Fasteners on the oil suction pipe.....	48
Figure 5-28 Pull the oil suction pipe out from the selector switch insert.....	48
Figure 5-29 Lift the selector switch insert out vertically.....	49
Figure 6-1 Basic structure of the OLTC head.....	52
Figure 6-2 Location of the OLTC inspection window.....	53
Figure 7-1 Remove the pipe connectors from the OLTC top.....	56
Figure 7-2 Connect E2 and R with a bypass pipe.....	57
Figure 7-3 Loosen the oil drain bolt.....	58
Figure 7-4 Loosen the oil drain bolt.....	59
Figure 7-5 Remove the pipe connectors from the OLTC top.....	60
Figure 8-1 Red protection strip on pressure relief valve.....	63
Figure 8-2 Drive mechanism components.....	65
Figure 8-3 Top gearbox adjustment and fixing.....	66
Figure 8-4 Align the shaft of the bevel gearbox and the shaft of the MDU.....	67
Figure 8-5 Install the bevel gearbox.....	68
Figure 8-6 Install the vertical shaft intermediate support gearbox.....	69
Figure 8-7 Install the horizontal shaft intermediate support gearbox.....	69
Figure 8-8 Install the bevel gearbox and the driving shaft.....	70
Figure 8-9 Install the horizontal guard.....	71
Figure 8-10 Installation steps for the vertical guard.....	72
Figure 8-11 Install the bushing.....	73
Figure 8-12 Install the vertical guard.....	73
Figure 9-1 Bleeder valve exhaust operation.....	77
Figure 9-2 Exhaust at pipe connector S.....	78
Figure 9-3 Grounding wire connection.....	78

List of tables

Table 1-1 Classification of signal words.....	2
Table 1-2 Safety symbols.....	3
Table 1-3 Abbreviations and full forms.....	5
Table 2-1 Potential danger and preventive measures.....	7
Table 2-2 Intended use.....	10
Table 2-3 Personnel qualifications.....	12
Table 2-4 Personal Protective Equipment.....	13
Table 12-1 Inspection guidelines.....	88
Table 12-2 Oil sample limits.....	89
Table 12-3 Maintenance interval for the tap-changer.....	90

1. About this manual

This manual provides an overview and essential safety instructions for the CV2 Oil-Immersed Vacuum On-Load Tap-Changer (OLTC). It also offers comprehensive instructions on transporting, installing, commissioning, and maintaining the OLTC to ensure proper integration with the corresponding transformer.

1.1. Target audience

This manual is intended for qualified personnel involved in the installation, commissioning, operation, and maintenance of the OLTC.

Roles	Responsibilities
Technical Field Assistant (TFA) or Commissioning Engineer	• Installing and commissioning OLTC • Integrating OLTC into transformers • Verifying functional and safety compliance
Operator	• Monitoring operating parameters • Performing routine tap-change operation • Reporting abnormal conditions
Maintenance Technician	• Performing inspections, oil sampling, and maintenance • Conducting mechanical/electrical safety checks • Conducting minor repair operations



Important:

This equipment must only be operated by qualified technical personnel. Untrained personnel shall not operate, maintain, or modify OLTC. For required personnel qualifications, refer to [2.5 Personnel qualifications](#).

1.2. Document retention

Keep this manual and all related documents in a safe place for future reference.

1.3. Related documents

This manual should be used in conjunction with the following documents. Reading this manual in isolation may not provide a complete understanding of the product.

- Wiring Diagram
- Order Documentation
- Routine Test Report
- Packing List
- Motor Drive Unit (MDU) Manual
- Protective Relay Manual
- Supplementary Documents (if any)

1.4. Conventions

1.4.1. Warning messages

The warning messages in this manual follow the format below:

 DANGER	
	Identify type of hazard Identify consequences of not avoiding the hazard ✓ Identify how to avoid the hazard
 WARNING	
	Identify type of hazard Identify consequences of not avoiding the hazard ✓ Identify how to avoid the hazard
 CAUTION	
	Identify type of hazard Identify consequences of not avoiding the hazard ✓ Identify how to avoid the hazard
NOTICE	
Identify type of risk Identify consequences of not avoiding the risk ✓ Identify how to avoid the risk	

Table 1-1 Classification of signal words

Signal word	Hazard level	Potential consequences
 Danger	Imminent hazard	Death or serious injury
 Warning	Potential hazard	Death or serious injury
 Caution	Potential hazard	Minor or moderate injury
 Notice	Potential risks	Damage to the supported product (or to other property)

Table 1-2 Safety symbols

Symbol	Meaning
	Danger: Indicates a hazard that will result in serious injury or death if not avoided.
	Warning: Indicates a hazard that could result in serious injury or death if not avoided.
	Overhead load
	High voltage
	Explosive material
	Flammable material

1.4.2. Notes

The following symbols are used to highlight information worth noting:



Important: Indicates key information and requires attention.



Note: Indicates an ordinary note.

1.4.3. Procedural instructions

Procedural instructions are presented as single-step or multi-step operations.

Single-step operations

Pre-operation preparation (Optional)

Objective:

This is a step.

1. This is a substep.
2. This is a substep.

Specific instructions/Expected results (Optional)

Outcome (Optional)

Multi-step operations

Pre-operation preparation (Optional)

Objective:

1. This is a step.
2. This is a step.
 - a. This is a substep.
 - b. This is a substep.
 - c. This is a substep.

Specific instructions/Expected results (Optional)

Outcome (Optional)

1.5. Abbreviations

The following table lists the abbreviations used in this manual and their full forms.

Table 1-3 Abbreviations and full forms

Abbreviations	Full form
FAT	Factory Acceptance Test
MDU	Motor Drive Unit
OLTC	On-Load Tap-Changer
PPE	Personal Protective Equipment
TFA	Technical Field Assistant

2. Safety notes

This manual is an integral part of the product.

Read this manual thoroughly to familiarize yourself with the product.

Pay close attention to all warning and cautionary information in this manual to prevent hazards.

Read and follow all safety instructions in this chapter.

While this product is designed and manufactured using state-of-the-art technology, improper use may result in personal injury, equipment damage, or property loss.

2.1. Safety instructions

To prevent possible injury, read and follow the instructions in the table below.

Table 2-1 Potential danger and preventive measures

Potential hazards	Description	Instructions
Electric shock	Contact with live electrical parts can cause serious injury or death.	- Before inspection or maintenance, take the following safety actions: - Cut off the power supply for the equipment. - Prevent the equipment from re-energization. - Verify that no voltage is present. - Discharge, ground, and short circuit as required. - Isolate adjacent live parts to prevent accidental contact. - For the specific task, wear the proper Personal Protective Equipment (PPE), for example, safety helmets and work gloves.
Improper grounding	Improper grounding can cause dangerous potential difference and equipment fault.	Before installation, operation, or testing, ensure correct grounding.
Arcing and flashover	Arcing during switching or insulation failure can cause burns or fires.	- Maintain insulation integrity. - Wear arc-rated PPE. - Maintain safe distance during operation.

Table 2-1 Potential danger and preventive measures (continued)

Potential hazards	Description	Instructions
Explosion or fire	Highly flammable or explosive gases, vapors, and dust can cause serious explosions and fires.	• Operate the product only in a non-explosive environment. • Keep oil away from open flames, sparks, or any potential fire sources. • In case of a fire, use dry powder, foam, or carbon dioxide extinguishers. Never use water.
Mechanical hazards	Moving or rotating parts can cause entanglement risks.	• Wear well-fitting clothing. • Do not wear rings, necklaces, or other jewelry. • Those having long hair must wear a hairnet.
Chemical hazards	Transformer oil degradation may release harmful substances.	• Use approved transformer oil. • Equip with appropriate PPE such as work gloves to avoid direct contact with transformer oil. • Ventilate area during operation.
Work area hazards	Poor lighting and a cluttered area can increase the risk of accidents.	• Keep the work area clean and well-lit. • Obey national accident prevention regulations.
Noise hazards	Tap-change operation can produce high noise levels.	Wear hearing protection as required.
Waste disposal	Incorrect disposal of waste materials, such as insulating oil, packaging, or removed parts, can harm the environment and lead to health risks.	• Collect used transformer oil in designated containers and store them properly. • Dispose of waste according to local regulations. • Different waste categories require specific disposal methods. Improper waste disposal can harm the environment and human health.

2.2. Safety Label

The rupture disk of the On-Load Tap-Changer (OLTC) selector switch is affixed with the following safety label: "NO STEP". It indicates that it is strictly forbidden to step on the rupture disk, as it may lead to structural damage or personal injury.

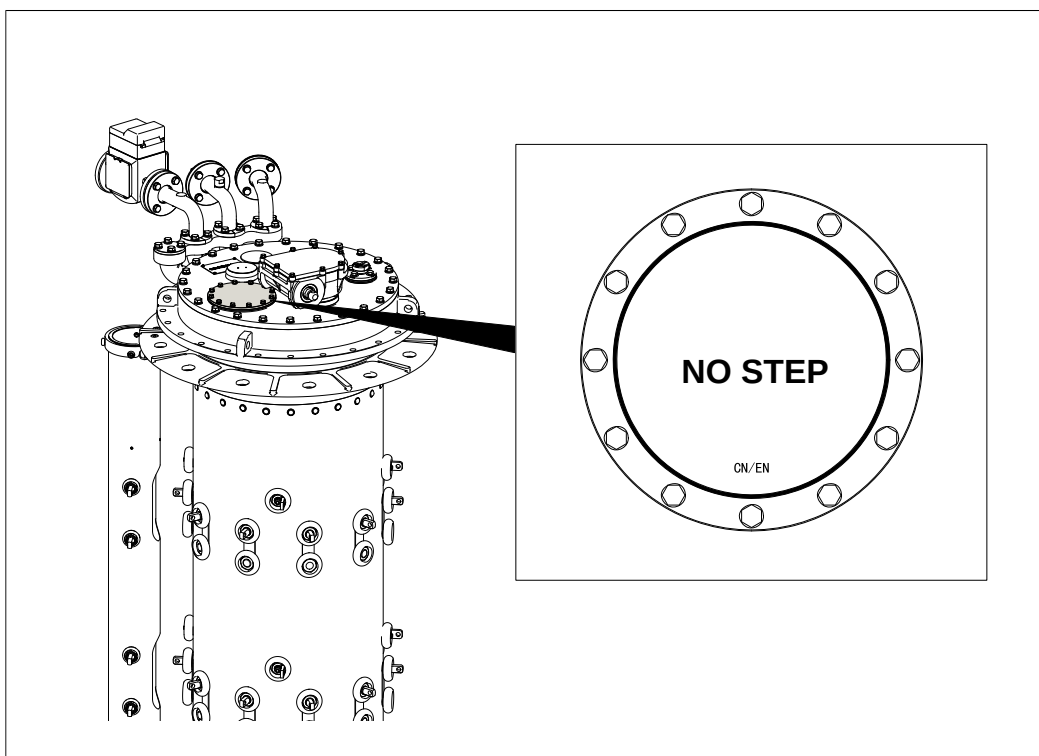


Figure 2-1 Safety label on the rupture disk

⚠ WARNING	
	Risk of structural damage or personal injury! Hazards may not be identified if safety labels are missing.
	✓ Keep all safety labels intact and legible.
	✓ If a label is missing or damaged, replace it immediately.

2.3. Intended use

This product (CV2 Oil-Immersed Vacuum On-Load Tap-Changer) is designed for regulating the transformer turns ratio while the transformer is energized.

It is intended for use in electrical power systems and related installations.

When operated in accordance with the instructions, conditions, and safety warnings provided in this manual and on the product itself, it does not pose a hazard to personnel, equipment, or the environment.

This assurance applies throughout the product's entire lifecycle—from delivery and installation to operation, maintenance, and final disposal.

The following table shows the intended use of this product.

Table 2-2 Intended use

Aspect	Requirements & specifications
Equipment compatibility	Install and operate this product only with the transformer and the Motor Drive Unit (MDU) as specified in the order documentation. If this product and accessories are supplied as a complete set, verify that the serial numbers of all components match and are compatible.
Product identification	Use the nameplate data (compliance standards, model, serial number, year of manufacture) to verify specifications before installation. For the location of this nameplate, see 3.7.1 Nameplate .
Operating conditions	Operate this product strictly under the environmental and electrical conditions defined in 3.4 Technical specifications and 3.5 Operating conditions . These include ambient temperature, humidity, and voltage range.
Personnel qualification	All installation, commissioning, inspection, maintenance, and disposal work must be performed by personnel qualified as described in 2.5 Personnel qualifications. Operators must ensure: • Regular safety training regarding procedures and guidelines. • Safety instructions and emergency protocols are displayed and accessible in work area.
Operating procedures	Operators must: • Strictly follow all warnings and safety instructions as described in this manual. • Strictly follow procedures for transportation, installation, commissioning, testing, and disposal as described in this manual. • Operate this product exclusively only when in good working condition. Regularly inspect safety devices to ensure reliable operation. • Provide all necessary tools and PPE for each task as described in this manual.
Tools and equipment	Use the provided tools and equipment solely for their intended purpose and in accordance with the instructions in this manual. Do not modify or misuse the tools.
Auxiliary equipment	The use of auxiliary equipment such as an oil purification system is optional and depends on the specific application and customer requirements. Only use auxiliary equipment approved by the manufacturer.
Spare parts	Only use spare parts and consumables that have been approved by the manufacturer to ensure continued safety and functionality.
Modifications	Any modification or alteration of the product that has not been authorized by the manufacturer is strictly prohibited. Unauthorized modifications may lead to personal injury, equipment failure, or loss of warranty.

2.4. Foreseeable misuse

Any use of the product outside the scope described in [2.3 Intended use](#) is misuse. Misuse can cause serious hazards to the personnel, equipment and environment.

Types of foreseeable misuse include, but not limited to:

- Operation with incompatible equipment: Using the product with transformers, motor drive units (MDUs), or accessories not specified in the order documents.
- Operation in unspecified environmental conditions: Operating the product in conditions that exceed the permitted temperature, humidity, or other limits stated in the [3.5 Operating conditions](#).
- Unauthorized modifications: Making mechanical or electrical modifications to the product or its components without written approval from the manufacturer.
- Bypassing safety devices: Disabling or interfering with protective relays or warning systems integrated in the product or related equipment.
- Incorrect electrical connections: Incorrect wiring or applying voltages that do not match the connection diagram or rated electrical values.
- Ignoring maintenance and inspection intervals: Not performing the specified inspection and maintenance tasks. This can cause unsafe operation or early failure.
- Unqualified personnel: Allowing untrained or unauthorized personnel to install, commission, maintain or operate the product.
- Stepping on or placing heavy objects on critical components: Failure to observe the product safety labels (such as the "No Stepping" safety label on the rupture disk of the selector switch) may result in mechanical damage or personal injury.
- Unauthorized or incorrect changes to the product can cause personnel injury, material damage and operation fault.

2.5. Personnel qualifications

All personnel involved in the installation, commissioning, operation, and maintenance of the product must meet the qualification levels in the table below.

Table 2-3 Personnel qualifications

Qualification level	Applicable positions	Core competencies and requirements
Electrical Technical Personnel	• Technical Field Assistant (TFA) • Commissioning Engineer • Maintenance Technician	• Holding valid technical certifications as required by national safety regulations (e.g., Certified Electrician/Registered Engineer) • Possessing specialized knowledge in identifying and avoiding electrical hazards (e.g., electric arc and high-voltage risks) • Demonstrating proficiency in electrical system operation and maintenance • Understanding local electrical codes and accident prevention regulations • Receiving complete training for operation in specific environments • Completing internal training on transformer installation. (TFA and Commissioning Engineers)
Trained Electrical Personnel	Operator (working under supervision)	• Receiving task-specific training from electrical technical personnel, covering: ◦ Equipment-specific hazards ◦ Safety procedures and protective measures ◦ High-voltage equipment operation • Performing work under the direct supervision of Electrical Technical Personnel
Maintenance Technician	Maintenance Technician	Holding certification in Mechanical and Electrical Safety Regulations
All positions	All personnel	Possessing fundamental knowledge of transformers and On-Load Tap-Changer systems

2.6. Personal Protective Equipment (PPE)

The following table lists the mandatory personal protective equipment during operation and maintenance to mitigate potential risks.

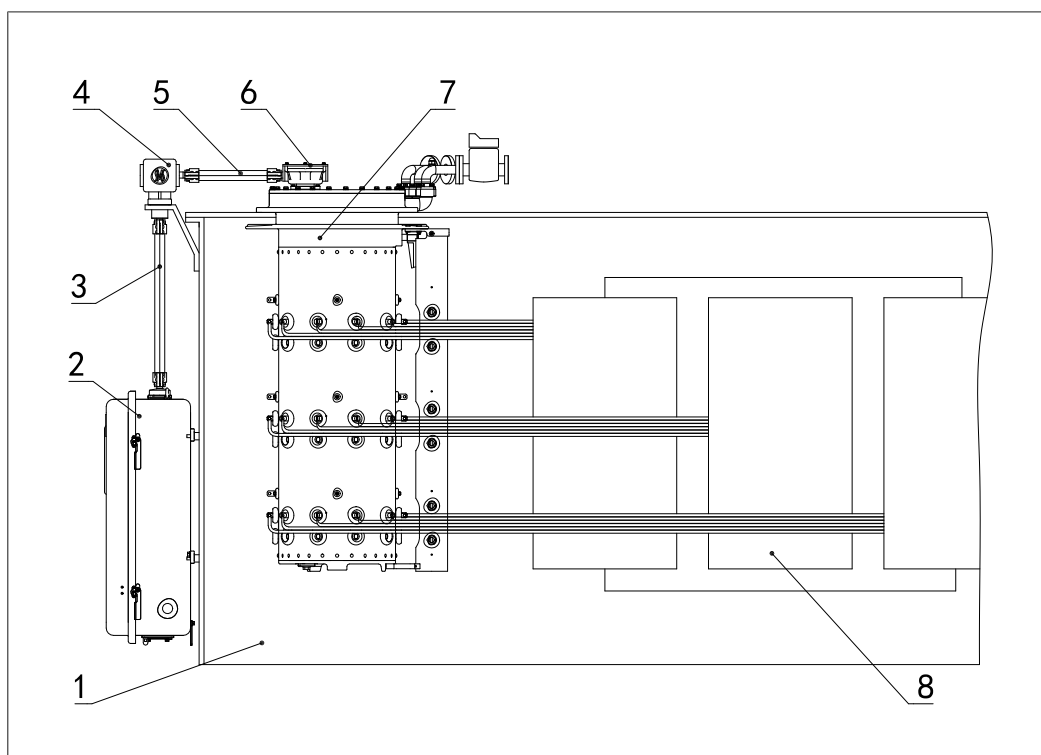
Table 2-4 Personal Protective Equipment

	Wear close-fitting workwear made of low-tear-strength material, with tight cuffs and no protruding parts, to minimize the risk of being caught by moving mechanical components.
	Wear safety shoes with puncture-resistant and slip-resistant soles.
	Use protective goggles to shield the eyes and the face from impact, dust, splashes, and high-temperature hazards.
	Wear a safety helmet to protect the head against impact.
	Use hearing protection in noisy environments to prevent hearing impairment. Certain operations of On-Load Tap-Changers, especially during switching sequences, functional tests or fault conditions, may generate impulse noise levels exceeding 85 dB(A). Repeated or prolonged exposure to such noise levels can cause permanent hearing impairment.
	Wear insulated gloves to protect your hands from mechanical, thermal and electrical hazards.

3. Product overview

3.1. Location of On-Load Tap-Changer (OLTC) in transformer

The CV2 type oil-immersed vacuum On-Load Tap-Changer is securely fastened to the transformer tank cover by using the mounting flange. It is connected to the motor drive unit (MDU) through the top gearbox, bevel gearbox as well as the horizontal and vertical driving shafts, enabling electric and remote tap-change operations.

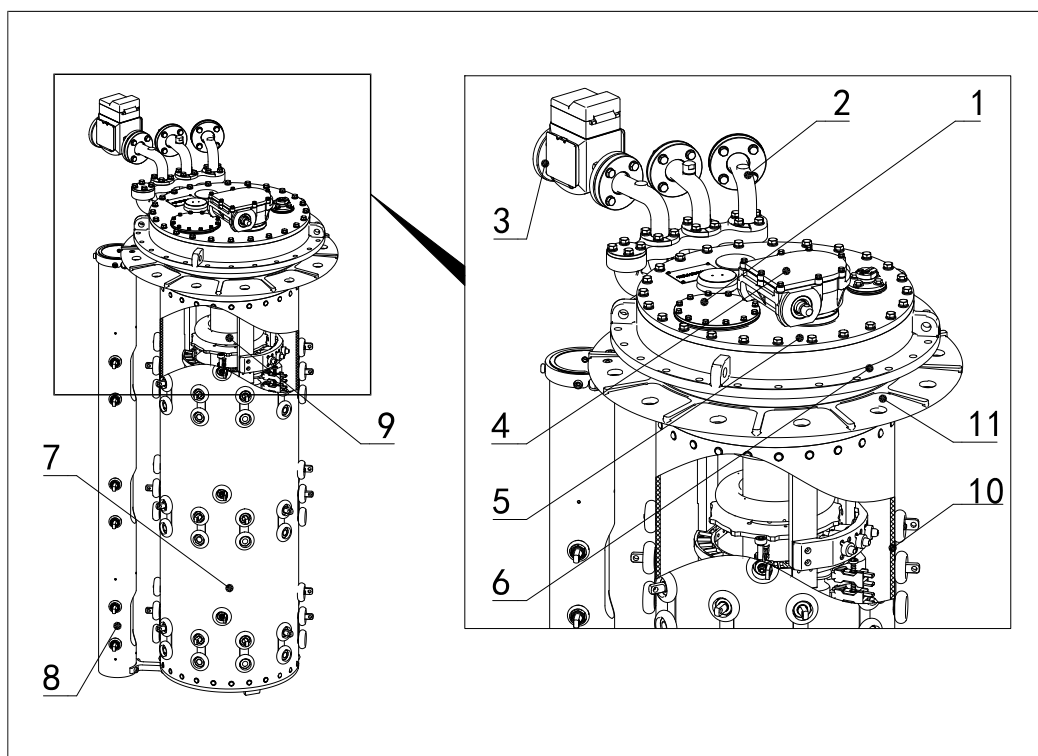


1. Transformer tank	2. Motor drive unit (MDU)
3. Vertical driving shaft	4. Bevel gearbox
5. Horizontal driving shaft	6. Top gearbox
7. OLTC main body	8. Transformer winding

Figure 3-1 OLTC in the transformer

3.2. OLTC main body

The CV2 type oil-immersed vacuum On-Load Tap-Changer is a composite On-Load Tap-Changer. A selector switch does not have an independent tap selector; one set of contacts completes both tap selection and current switching.



1. Rupture disk	2. Oil pipe
3. Protective relay	4. Top gearbox
5. Top cover	6. Mounting flange
7. Selector switch	8. Change-over selector
9. Selector switch insert	10. Oil compartment
11. Support flange	

Figure 3-2 Overview of the OLTC structure

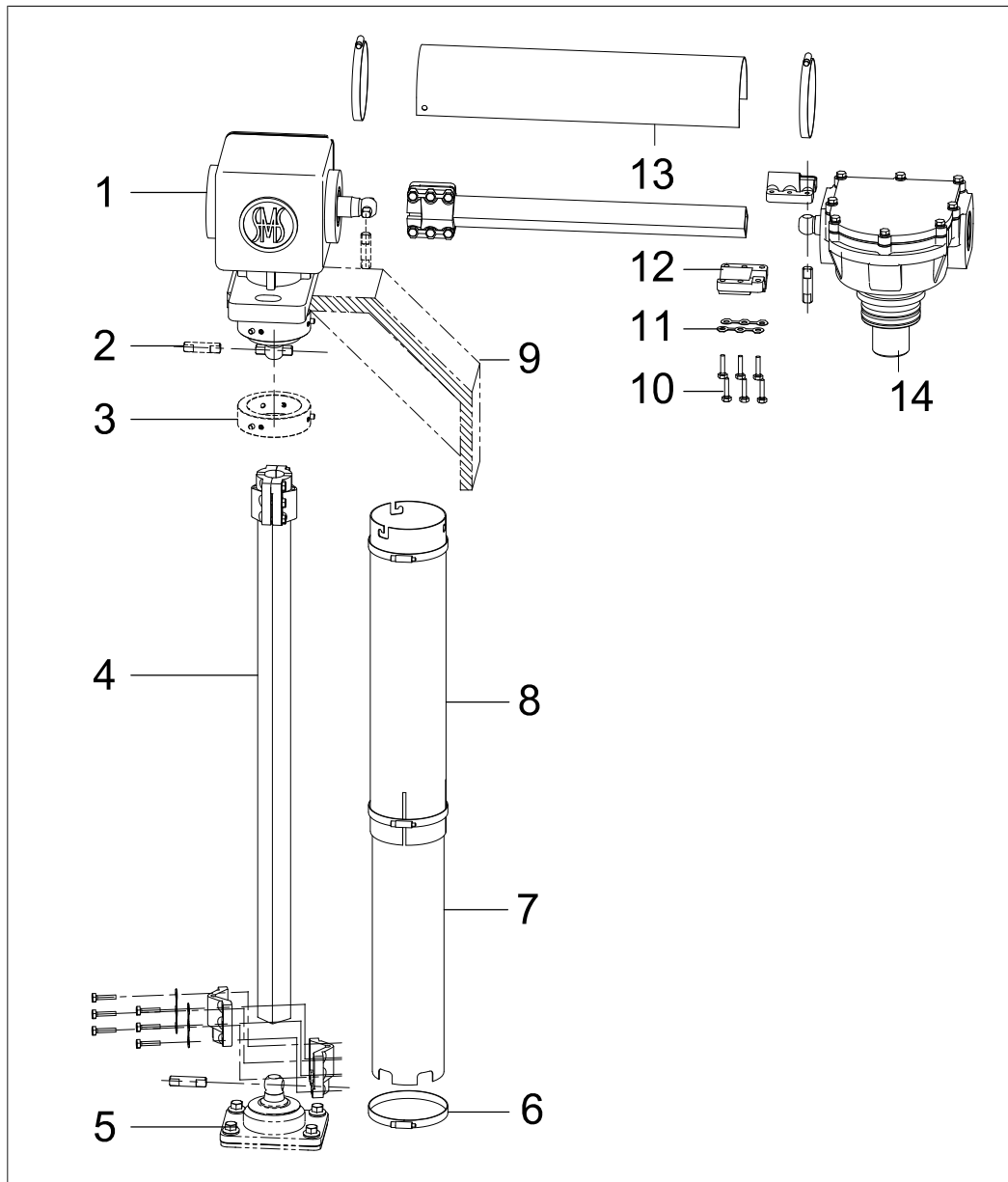


Important:

It is recommended to adopt a protective relay with gas detection signals for the vacuum OLTC.

3.3. Drive mechanism components

Drive mechanism components consist of the bevel gearbox, driving shafts, couplings and other accessories.



1. Bevel gearbox	2. Cylindrical pin	3. Bushing
4. Driving shaft	5. Motor Drive Unit (MDU)	6. Clamp
7. Vertical guard 1	8. Vertical guard 2	9. Gearbox bracket (transformer)
10. M6 screw	11. Lock washer	12. Coupling
13. Horizontal guard	14. Top gearbox	

Figure 3-3 Drive mechanism components

3.4. Technical specifications

The tap-changer complies with the standards of IEC 60214-1:2014 and GB/T 10230.1-2019.

Item	Model		CV2 III 350 CV2 I 350			CV2 III 600 CV2 I 600		
1	Maximum rated through-current (A)		350			600		
2	Rated frequency (Hz)		50/60					
3	Number of phases		I	III		I	III	
4	Connection mode		-	Y	D	-	Y	D
5	Maximum rated step voltage (V)		• 10 contacts: 2000 • 12 contacts: 1500					
6	Rated step capacity (kVA)		700			800		
7	Short-circuit withstand capability (kA)	Thermal stability (3 s)	5			8		
		Dynamic stability (peak)	12.5			20		
8	Maximum number of tap positions		• Without change-over selector: 12 • With change-over selector: 23					
9	Ground insulation level (kV)	Highest equipment voltage	40.5	72.5		126	145	
		Rated power-frequency withstand voltage (50Hz, 1min)	95	140		230	275	
		Rated lightning impulse withstand voltage (peak) full wave (1.2/50μs)	250	350		550	650	
		Rated lightning impulse withstand voltage (peak) chopped wave (1.2/50μs)	275	285		605	715	
10	Mechanical lifespan		1,500,000 operations					
11	Maintenance-free operation times		300,000 operations					
12	Switch oil compartment	Operating pressure	0.03 MPa					
		Sealing performance	No leakage for 24 hours under 0.1 MPa pressure					
		Overpressure protection	Explosion pressure of the rupture disk: 0.3 ± 20% MPa					
		Protective relay	Set oil flow rate: 1.0 m/s±10%					



Note: Step capacity equals the product of step voltage and load current. Rated step capacity is the maximum allowable step capacity under continuous operating conditions.

3.5. Operating conditions

The On-Load Tap-Changer (OLTC) must operate under the following conditions:

Operating conditions	Requirements
Allowable operating temperature range	-25°C to +105°C
Ambient temperature range	-25°C to +40°C
Relative humidity range	≤ 85%  Explosion risk! Condensation water may affect electrical performance and damage equipment. In extreme cases, it may even lead to an explosion. ✓ It is not allowed to install the On-Load Tap-Changer in a place with a relative humidity of over 85%.
Ground verticality deviation of the OLTC after installation on the transformer	≤ 2%
Requirements for installation location of the OLTC	No excessive dust, explosive gases or corrosive gases



Important:

If customized solutions are required for special environment conditions, please contact Huaming.

3.6. Product compatibility requirements

To ensure safe and reliable operation, the On-Load Tap-Changer (OLTC) must be correctly matched with the key components of the system. Compatibility with the transformer, protective relay, and installation ports is crucial for achieving optimal performance and avoiding damage or safety hazards.

- MDU: HM CMA7 or SHM series
- Protective relay: QJ-25 or equivalent products

3.7. Model description

3.7.1. Nameplate

The nameplate of the On-Load Tap-Changer is located on the top cover. The nameplate provides key identification information, including the serial number, date of manufacture, model, technical specifications, and applicable compliance standards.

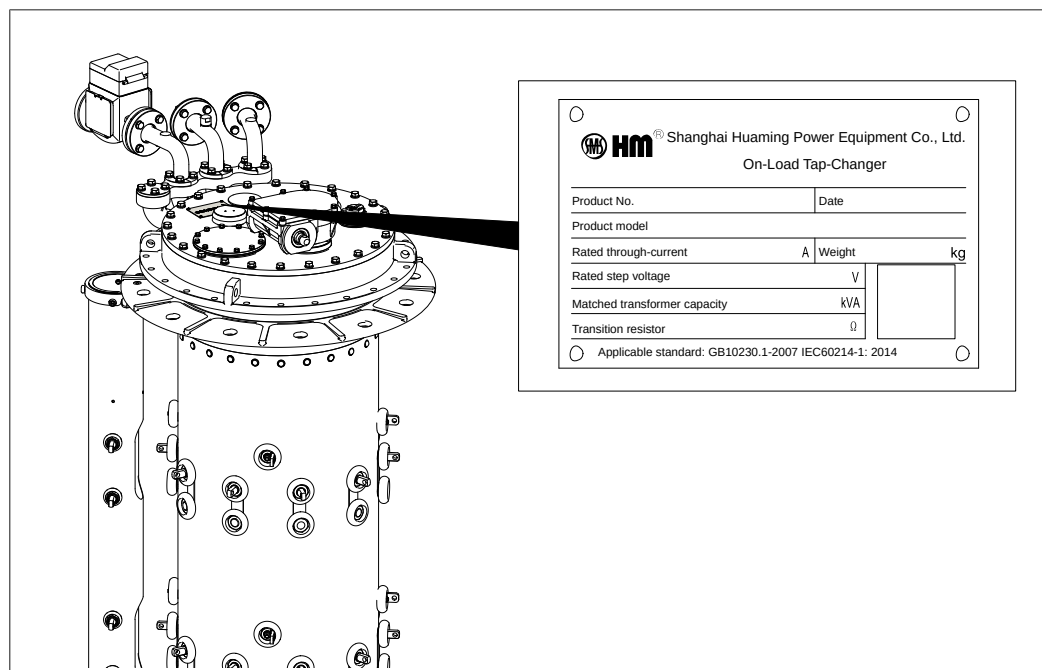


Figure 3-4 Nameplate of the On-Load Tap-Changer

3.7.2. Model number structure

The model number of an On-Load Tap-Changer reflects its key specification parameters, including the number of phases, maximum rated through-current, maximum equipment voltage, insulation class of the selector switch, and connection mode.

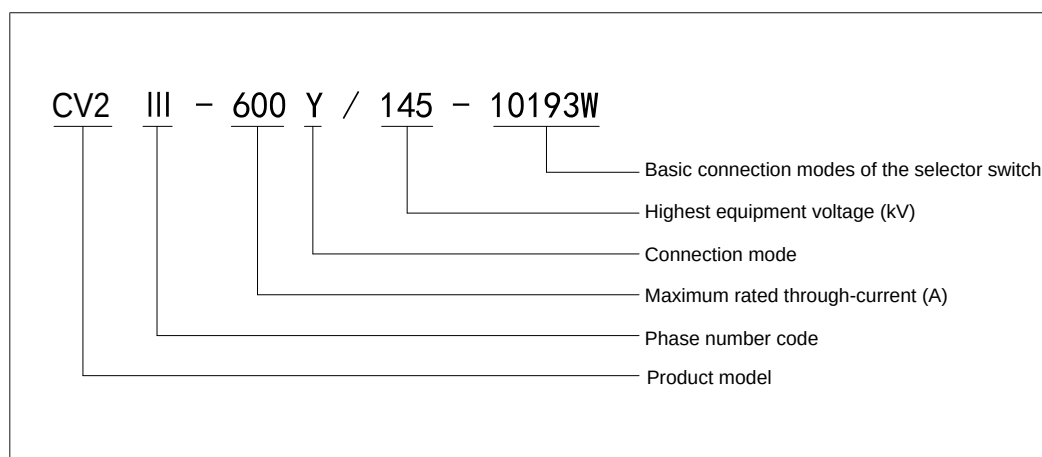


Figure 3-5 Model number description

3.7.3. Basic connection modes

Multiple selector switch configurations are available to accommodate different voltage regulation ranges and winding connection modes. The selector switch type is determined by the number of contacts, operating tap positions, intermediate tap positions, and whether a change-over selector is equipped.

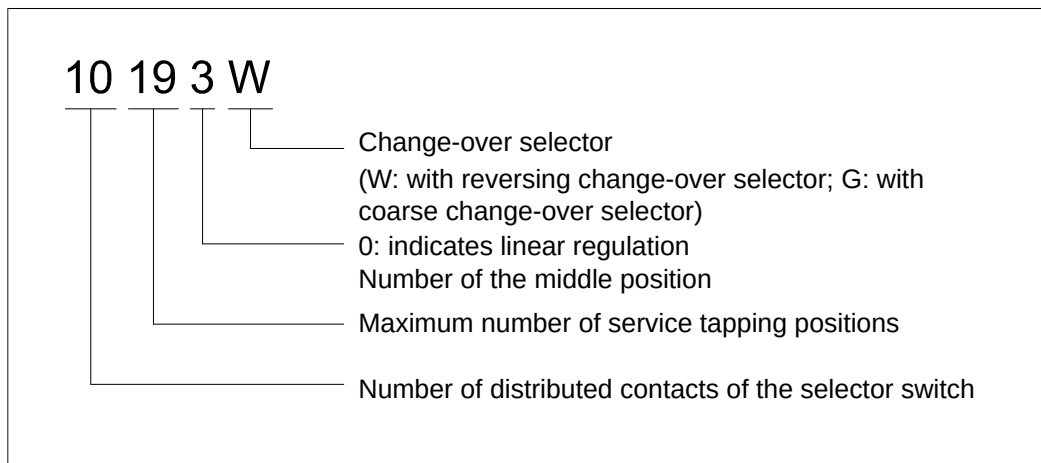


Figure 3-6 Description of connection mode

The following examples illustrate how the numbering indicates the basic connection mode:

- 10090: Indicates the selector switch has 10 contacts, and 9 operating tap positions, without a change-over selector.
- 10191W: Indicates the selector switch has 10 contacts, 19 operating tap positions, 1 intermediate tap position, and is equipped with a reversing change-over selector (W).
- 10191G: Indicates the selector switch has 10 contacts, 19 operating tap positions, 1 intermediate tap position, and is equipped with a coarse change-over selector (G).

4. Packaging, transportation, and storage

4.1. Supplied components

The On-Load Tap-Changer is delivered as a complete set, including the following components:

1. Selector switch
2. Motor drive unit (MDU)
3. Protective relay
4. Bevel gearbox
5. Driving shaft, coupling and other accessories
6. Special tools and accessories for OLTC assembly
7. Customer-required special accessories (if any)

Figure 4-1 It shows the appearance of items 1 to 4.

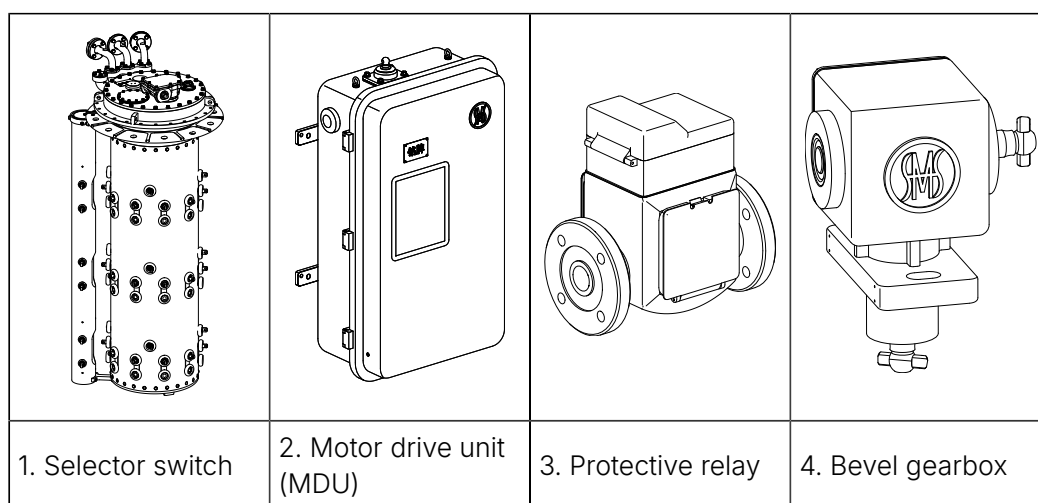


Figure 4-1 Supply components (partial)



Important:

The actual supplied components shall be based on the packing list provided with the goods.

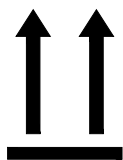
4.2. Transportation

⚠ WARNING	
	Risk of Toppling! A falling or toppling package may cause severe personal injury and equipment damage. ✓ Only trained and designated personnel may select slings and secure loads. ✓ Do not stand under a suspended load. ✓ Use transport and lifting equipment with a load capacity greater than 1,000 kg.

⚠ CAUTION	
	Risk of improper transport! Improper handling may lead to product damage, functional failure, or secondary safety incidents. ✓ Transport and lifting must be carried out by qualified personnel. ✓ Only use the lifting and transport equipment with a rated load capacity not below 1,000 kg. ✓ Only use the designated lifting points marked on the package. ✓ Secure the packaging case firmly during the transport and lifting process. Prevent vibration, impact, tilting, falling, or collision to avoid product damage. ✓ If the package has dropped or sustained a severe impact, check whether OLTC is damaged. If necessary, contact Huaming for inspection and repair.

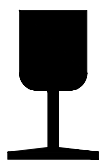
The factory packaging is suitable for multiple transport methods. Always follow these instructions during transport:

- Always position the package according to the mark of the center of gravity.
- During storage, stack packages only when the load on the top surface does not exceed 500 kg/m².
- Always follow the markings on the package during transport (as shown in the figure below).



This side up

Gross weight (G.W.)



Handle with care

Net weight (N.W.)



**Protect from rain
and moisture**



Stacking layer limit

Kilogram (KG)

4.2.1. Receiving inspection

Upon receiving the packages, inspect each package and verify that the packaging is intact and undamaged. If any damage is found (for example dents, tears, moisture), perform these steps:

1. Refuse delivery and immediately notify the carrier.
2. Take photos of the damaged packaging and contents if possible.
3. If the issue cannot be resolved with the carrier, contact Huaming After-Sales Service in writing.

4.2.2. Transshipment

After the receiving inspection, you may need to transship the packages to another location.

1. Unload and handle the packaged unit in accordance with the instructions on the packaging.
2. Use the designated lifting points for hoisting or follow the forklift markings when transporting with a forklift.



Important:

Select an appropriate forklift and/or crane based on the dimensions and gross weight of the crate or package.

4.2.3. Unpacking and inspection

To unpack and inspect the packages, perform these steps:

1. Before unpacking, move the package to the installation site.



Important: Take care not to damage the original packaging during unpacking and inspection.

2. Inspect each package against the packing list and Confirmation of Acceptance form. Verify the following:
 - The quantity of items matches the packing list.
 - The product specifications match the order documentation.



Important:

- Missing components may affect subsequent operation. If any item or accessory does not match the packing list, contact Huaming immediately.
- If the package has been dropped or subjected to impact after delivery, assume potential damage. Thoroughly inspect the OLTC. If necessary, contact Huaming to arrange inspection and/or repair by qualified personnel.

3. Before installation, store all components in a dry location and keep OLTC sealed in its original packaging.

4.2.4. Lifting

⚠ WARNING



The lifting process involves risks of personal injury and equipment damage!

Improper lifting may lead to load falling or instability, causing serious personal injury or equipment damage.

- ✓ Use the lifting equipment with a rated load capacity not below 1,000 kg.
- ✓ According to the regulations, always use at least two horizontal lifting points or all the designated lifting points.
- ✓ Avoid sudden movement or shaking when lifting or lowering the load.
- ✓ Never stand or walk under a suspended load.

Use all the designated lifting points on the mounting flange to vertically lift the OLTC.

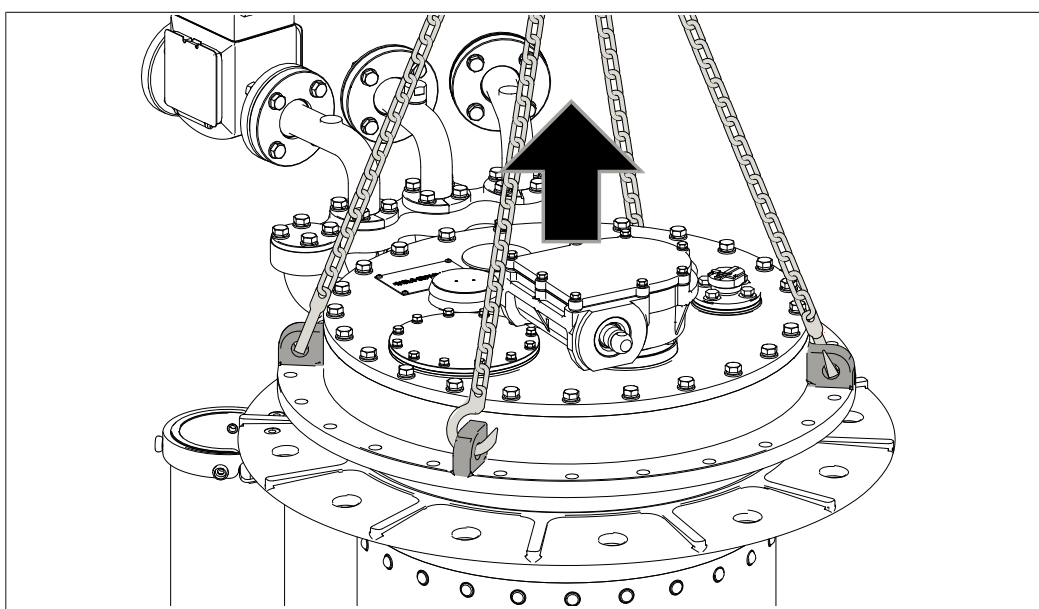


Figure 4-2 Lift the OLTC

4.3. Storage

Specially-sealed units can be stored outdoors if the following conditions are met:

- The ambient temperature is between -25°C and +40°C.
- The storage area is away from flammable, explosive, or corrosive gases.
- Protect the OLTC against moisture, dust, rodents, and insects.
- Provide the corresponding protection, to protect the OLTC against moisture, dust accumulation, rodents and insects.
- Regularly check the stored equipment for any abnormalities.
- Regularly replace desiccants during long-term storage and ensure the packaging remains sealed.

5. Installation of the OLTC on the transformer

5.1. Safety

⚠ WARNING



Risk of injury from unstable placement!

- ✓ Placing components (e.g., top cover, OLTC mounting flange) that are removed from the OLTC on uneven or unstable surfaces may cause them to tilt or fall, leading to injury.
- ✓ Always place components on a level, stable, and clean surface.

NOTICE

Risk of equipment damage from dropped fasteners!

Dropped fasteners (e.g., bolts, nuts) can contaminate the insulating oil, damage internal components, and lead to malfunction or abnormal operation.

- ✓ Store all fasteners securely.
- ✓ Take precautions to prevent fasteners from falling into the oil compartment.

NOTICE

Risk of sealing failure!

Improper sealing may occur if the component's sealing ring or sealing surfaces are damaged, potentially leading to equipment failure or leakage.

- ✓ Preserve the sealing rings during handling and installation.
- ✓ Avoid damaging the sealing surfaces of the top cover and the flanges.

5.2. Preparations



Important:

- Each OLTC is matched with a specific transformer.
- Carefully inspect the OLTC and its accessories before installation.
- Verify that the OLTC specifications meet the transformer requirements.

Before installing the OLTC onto the transformer, prepare the following items:

- Selector switch and accessory box
- Tap selector and accessory box (if applicable)
- Transformer mounting flange
- Temporary support structure for temporarily supporting the OLTC during installation
- Tools to be prepared by the customer (see [5.2.2 List of Tools](#))

5.2.1. Transformer mounting flange

The OLTC is installed onto the transformer tank cover via the transformer mounting flange. This flange is not included in the OLTC scope of supply and must be manufactured by the customer according to the drawings provided by Huaming.

The fabrication of this flange must meet the following requirements (see [Figure 5-1](#)):

- The transformer mounting flange must match the sealing surface of the OLTC mounting flange.
- The stud bolts (M12, maximum length 45 mm) must be precisely positioned.

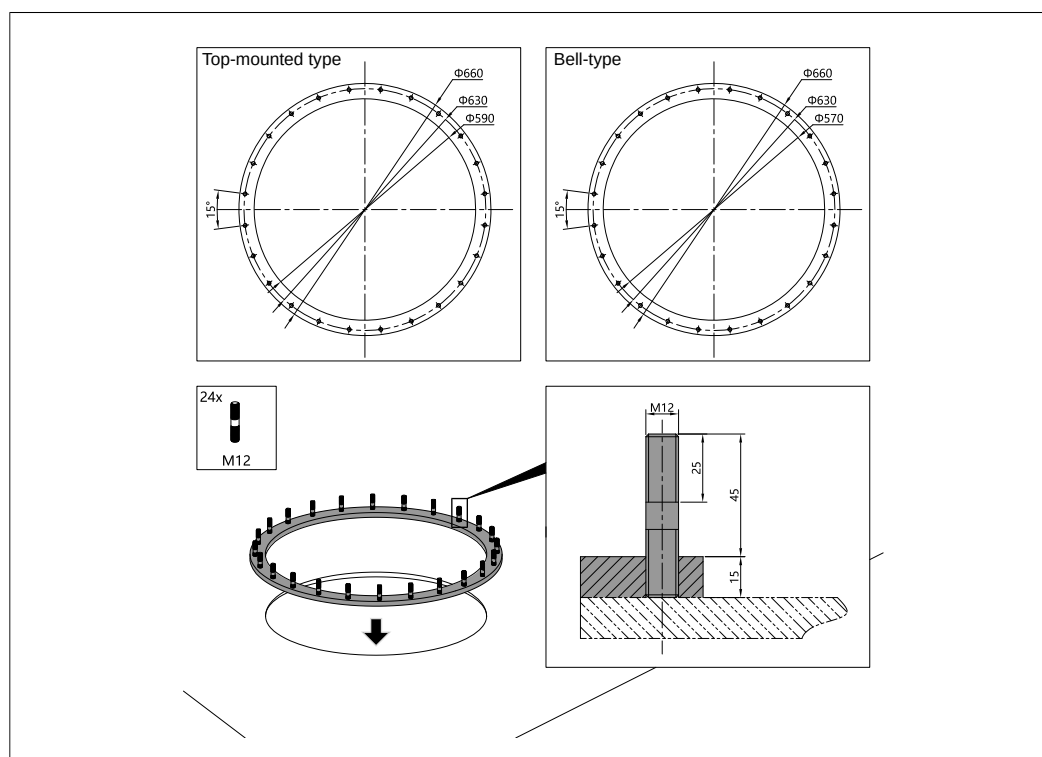


Figure 5-1 Transformer mounting flange

5.2.2. List of Tools

Tools required for the entire installation process:

- Lifting equipment with a rated load capacity not below 1,000 kg
- Special lifting tools, such as lifting rings
- 5 mm hex key
- 10 mm open-end/socket wrench
- 13 mm open-end/socket wrench
- 13 mm open-end wrench
- 17 mm open-end wrench/socket wrench
- 19 mm open-end wrench/socket wrench
- 24 mm open-end wrench
- 38 mm open-end wrench
- M4 bolt
- Adjustable open-end wrench
- Flat-head screwdriver

5.3. Install the On-Load Tap-Changer (OLTC) on a bell-type transformer

5.3.1. Overview

Installation of the OLTC on the transformer includes the following steps:

No.	Step	Reference
1	Remove the top cover	5.3.2 Remove the top cover
2	Separate the energy storage spring and oil suction pipe	5.3.3 Separate the energy storage spring and oil suction pipe
3	Remove OLTC mounting flange	5.3.4 Remove the OLTC Mounting Flange
4	Install the OLTC into the bell-type transformer	5.3.5 Install the OLTC into the bell-type transformer
5	Install the bell-type transformer tank cover and OLTC mounting flange.	5.3.6 Install the bell-type transformer tank cover and OLTC mounting flange.
6	Reinstall the energy storage spring and the oil suction pipe	5.3.7 Reinstall the energy storage spring and the oil suction pipe
7	Install the top cover	5.3.8 Install the top cover

5.3.2. Remove the top cover

Remove the top cover following these steps:

1. Check that the limit plate of the OLTC top gearbox is fixed as required in the figure below.

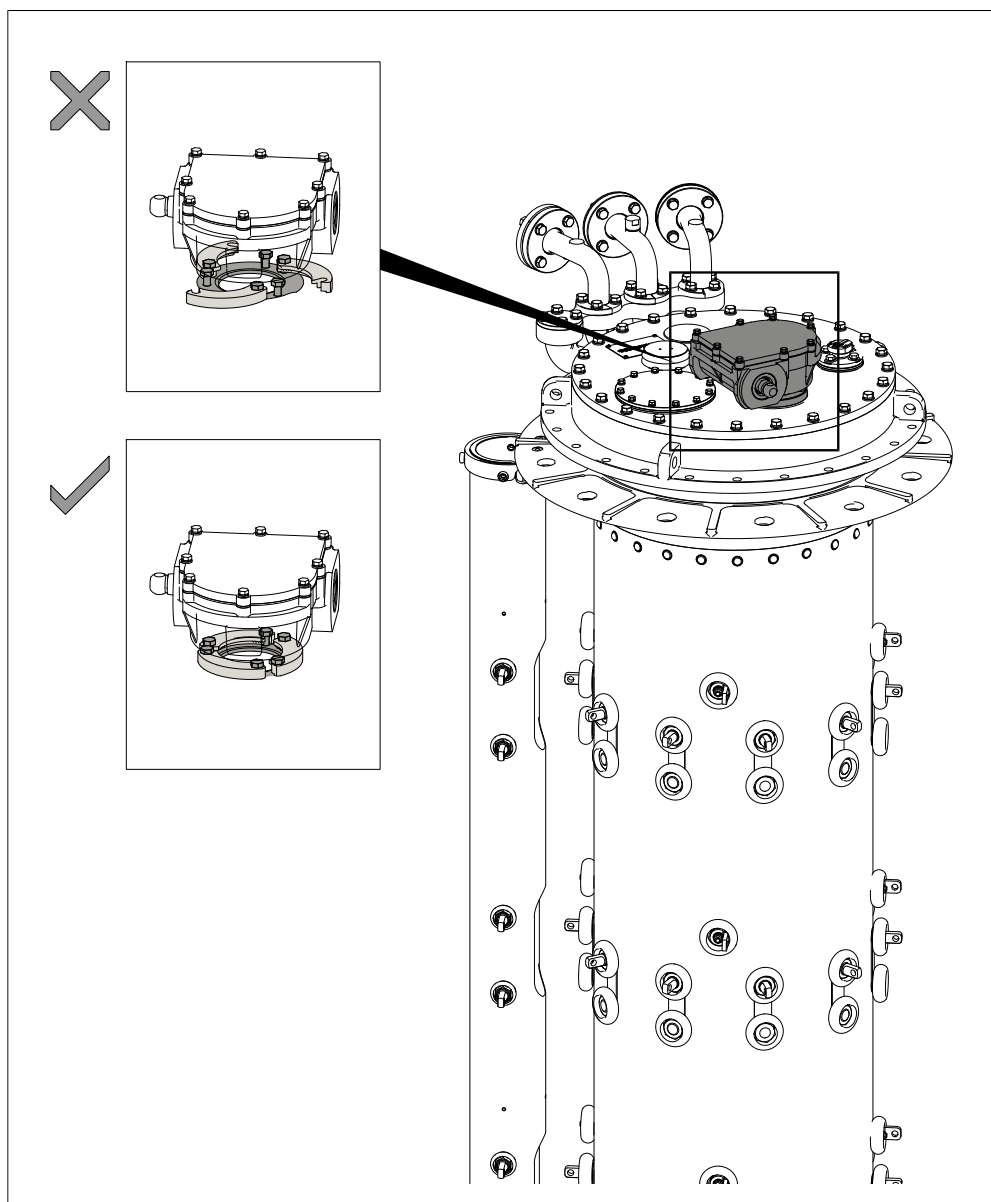


Figure 5-2 Fix the top gearbox limit plate

2. Use a 17 mm open-end/socket wrench to remove 24 sets of M10 bolts, each set with spring washer x1 and flat washer x1, and remove the top cover bolts from the OLTC mounting flange.



Notice: Properly store all fasteners. Prevent fasteners from falling into the oil compartment.

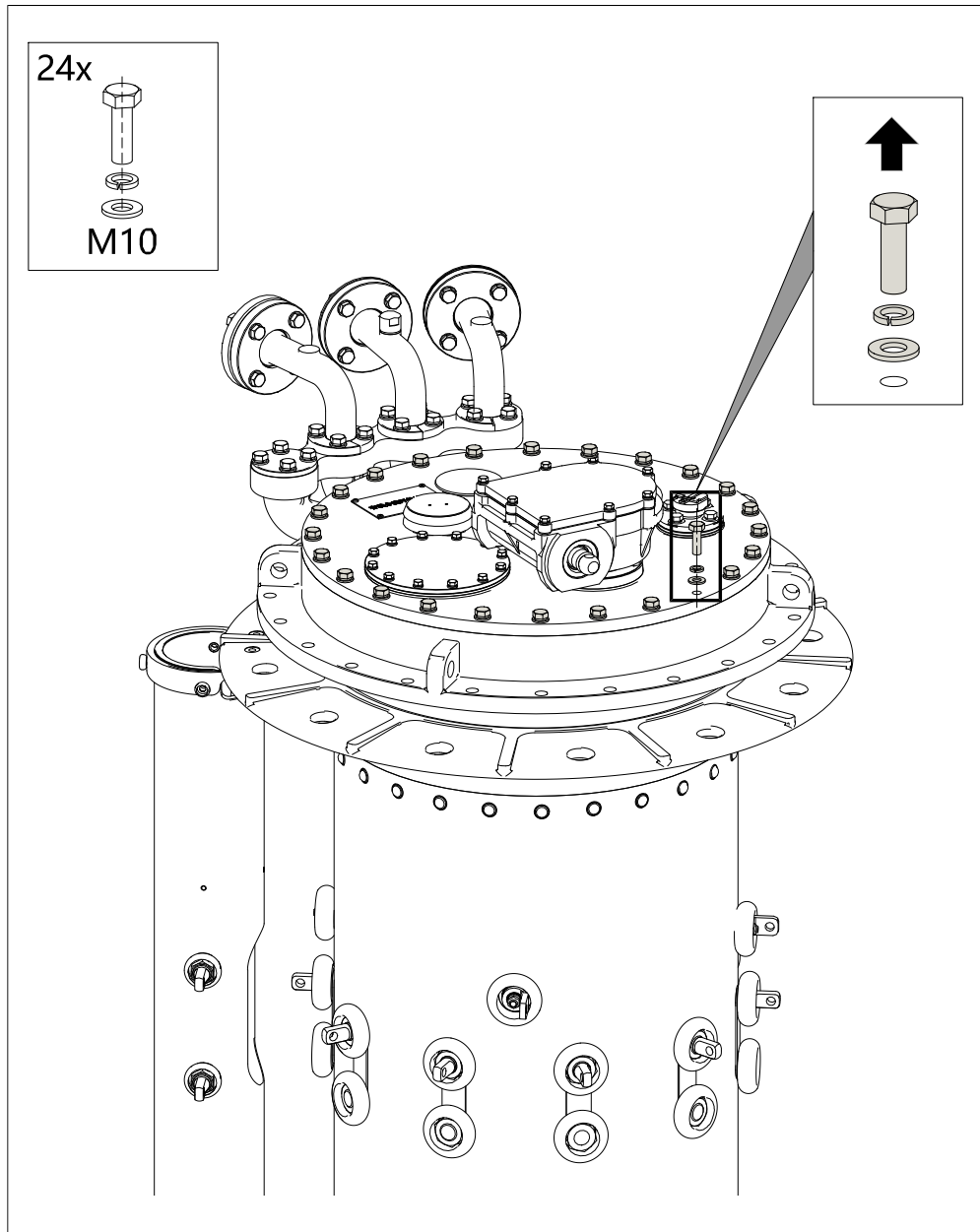


Figure 5-3 Remove the top cover bolts

3. Remove the OLTC top cover.



Notice: Place the top cover on a level, stable, and clean surface. Keep the top cover sealing ring intact and clean.

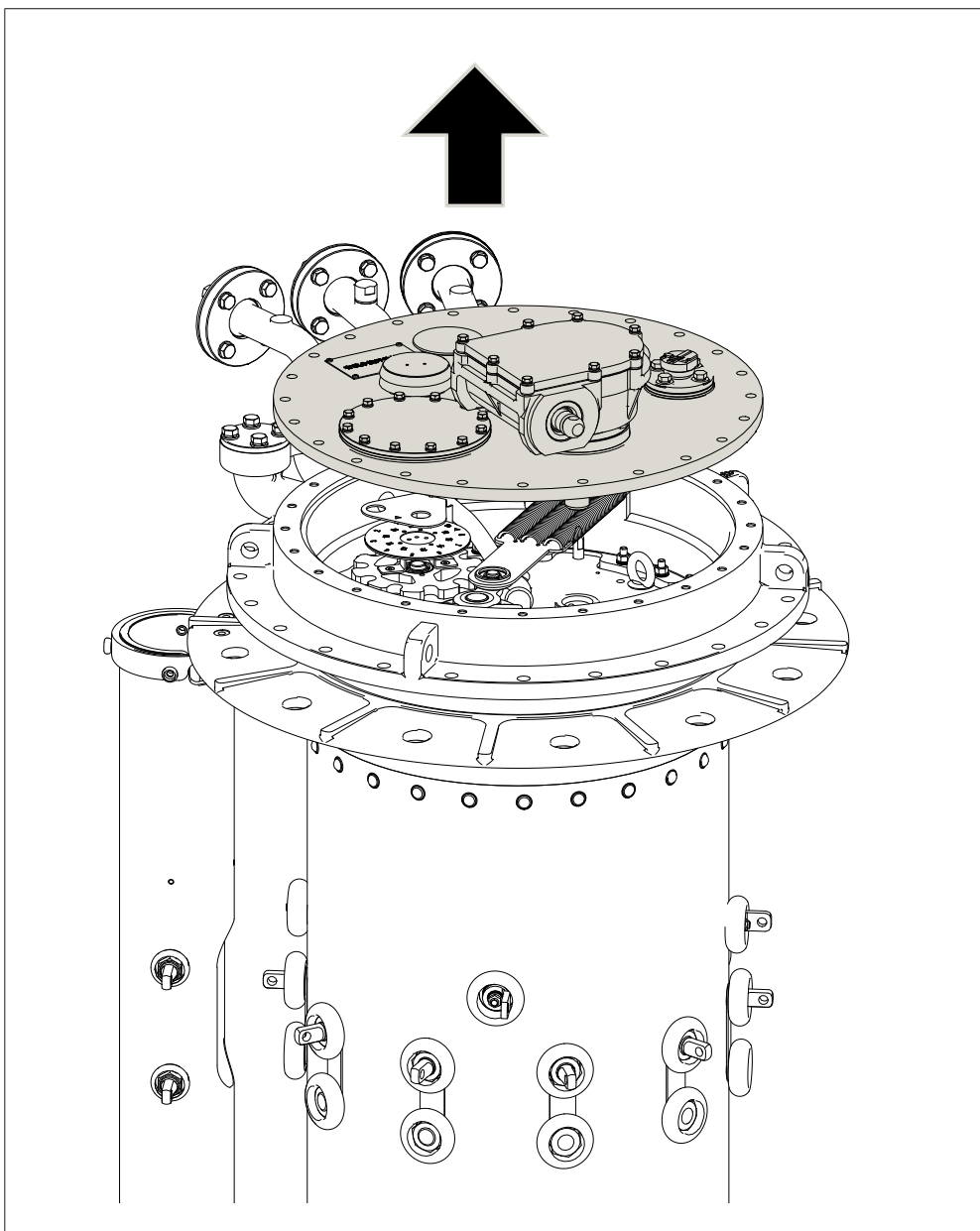


Figure 5-4 Remove the top cover

5.3.3. Separate the energy storage spring and oil suction pipe

Perform these steps to separate the energy storage spring and the oil suction pipe:

1. Using M4 bolts, remove 3 sets of tension spring pins and pull them out vertically from the OLTC mounting flange.



Notice: Properly store all fasteners. Prevent fasteners from falling into the oil compartment.

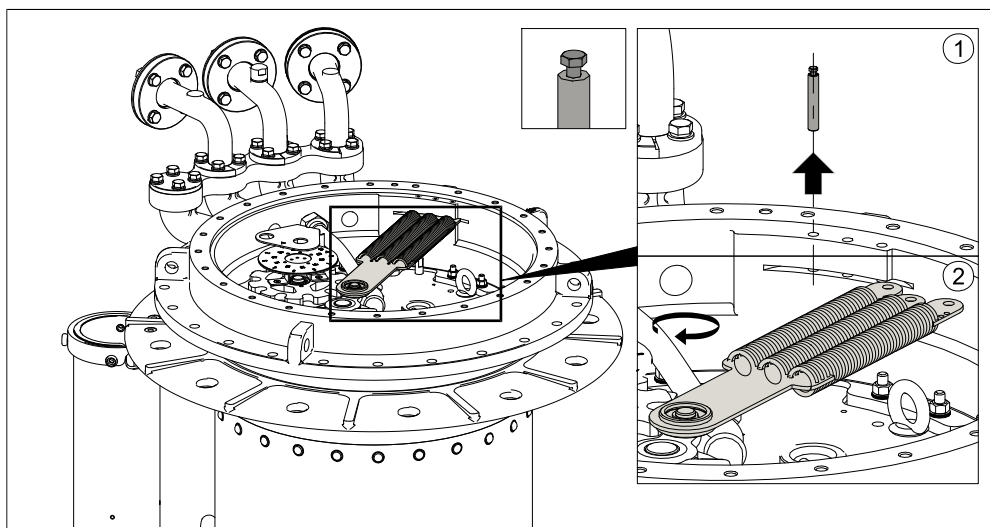


Figure 5-5 Separate the energy storage spring from the OLTC mounting flange

2. After loosening the elbow nut left by using a 38 mm open-ended wrench, rotate the oil suction pipe until it does not interfere with the OLTC mounting flange.

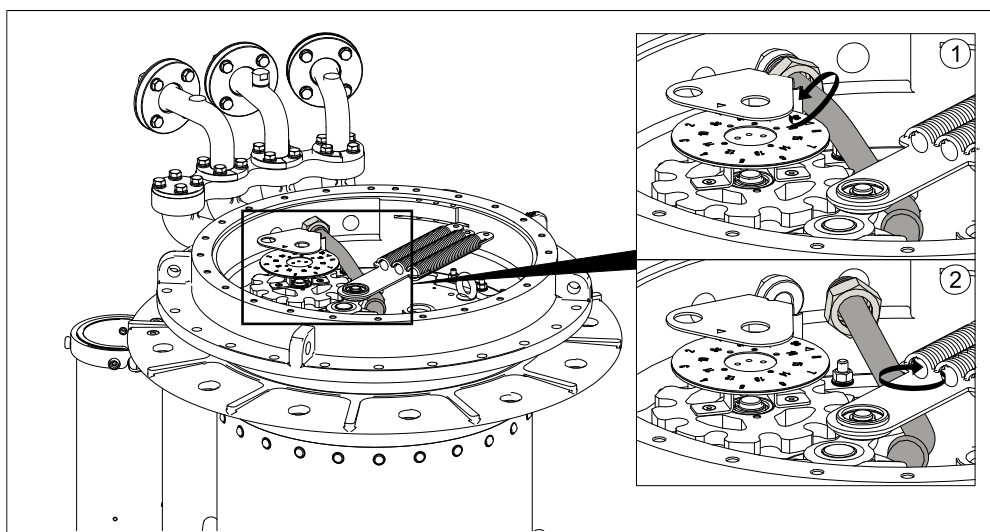


Figure 5-6 Separate the oil suction pipe from the OLTC mounting flange

5.3.4. Remove the OLTC Mounting Flange

Remove the OLTC mounting flange from the selector switch following these steps:

1. Use a 13 mm open-end/socket wrench to remove 16 sets of M8 nuts, each set with belleville washer x4.



Notice: Properly store all fasteners. Prevent fasteners from falling into the oil compartment.

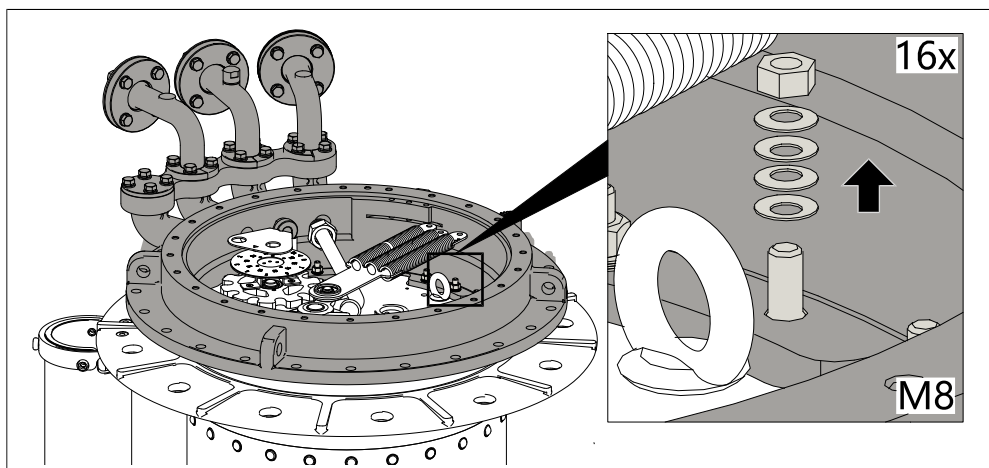


Figure 5-7 Remove the nuts on the OLTC mounting flange

2. Carefully lift the mounting flange off the support flange.

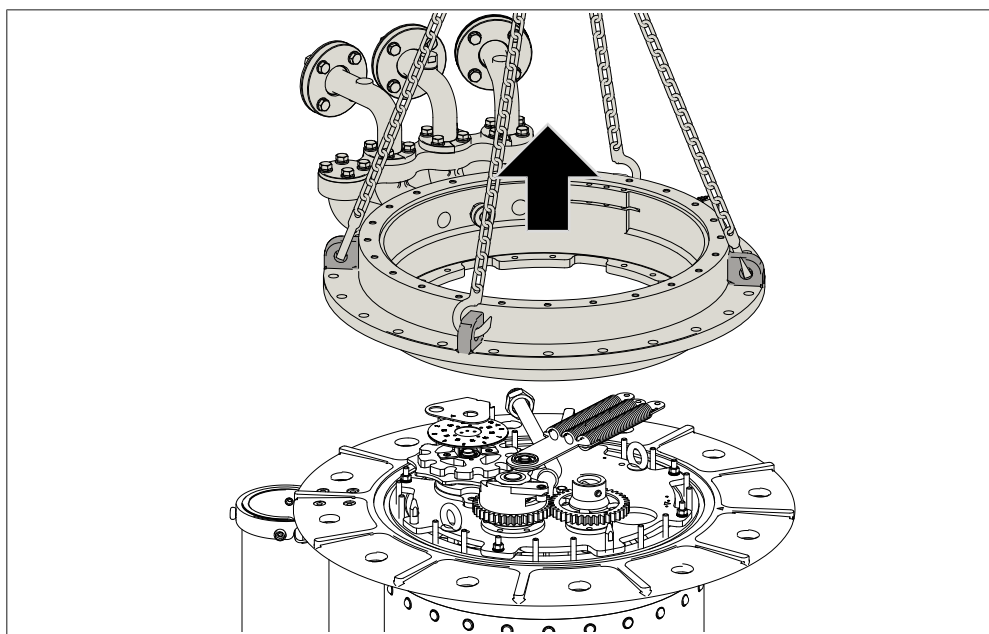


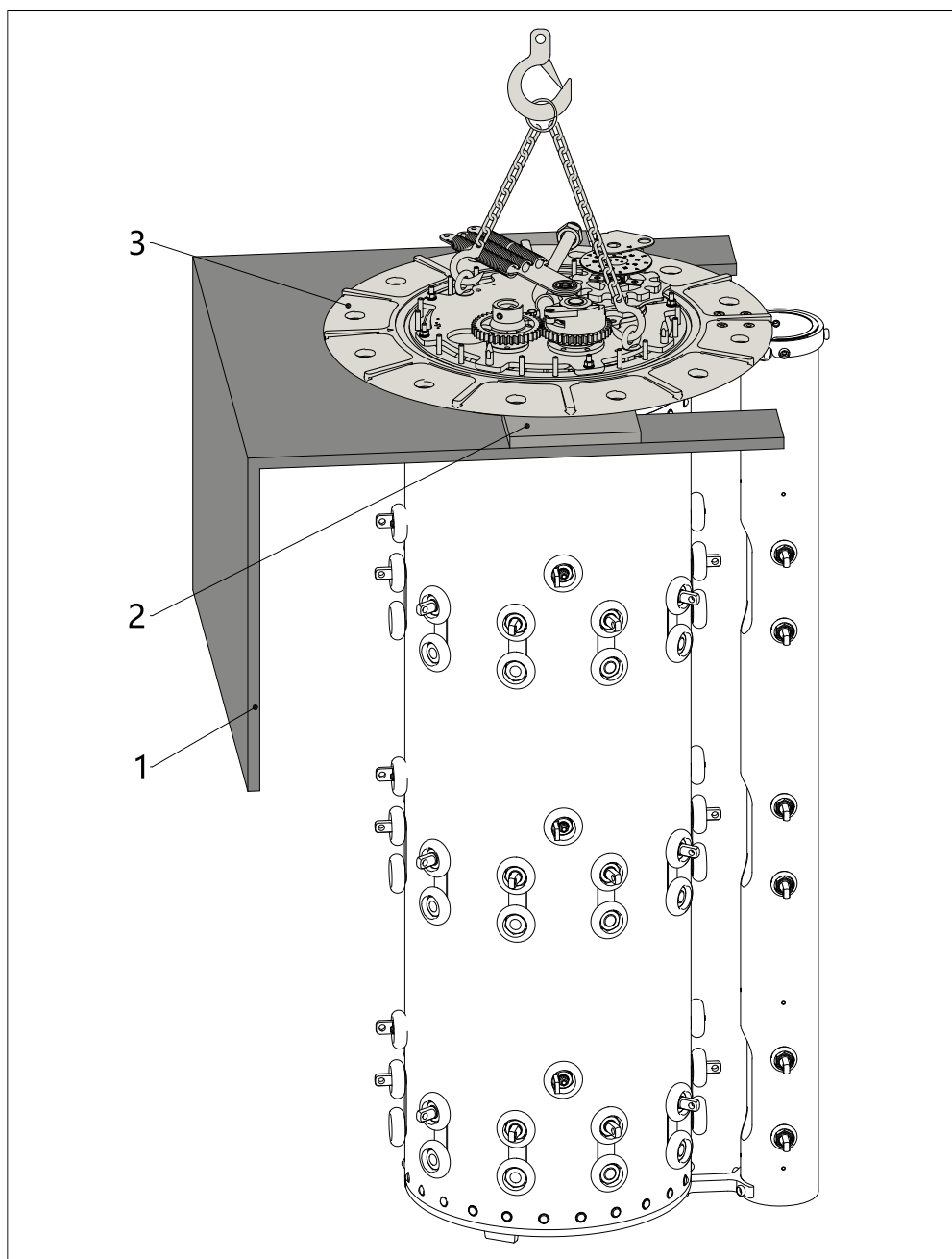
Figure 5-8 Diagram of lifting off the mounting flange

3. Place the OLTC mounting flange on a level, stable, and clean surface.

5.3.5. Install the OLTC into the bell-type transformer

Install the On-Load Tap-Changer to the bell-type transformer following the steps below:

1. Lift the OLTC and carefully place it on the support structure. Use adjusting shims to adjust its height to the final mounting height (raised by approximately 5-20 mm).



1. Support structure

2. Adjusting shim

3. Support flange

Figure 5-9 OLTC placed on temporary support

2. After checking that the OLTC reaches the final mounting height, pass bolts through the mounting holes of the support flange to fix it to the temporary support structure, so as to prevent it from moving.

3. Use a 17 mm open-end/socket wrench to install the M10 bolt fasteners, each set with spring washer x1, flat washer x2, dished washer x1 and screening cap x2, torque: 28 to 34 Nm. Install the voltage regulating winding lead of the transformer as illustrated.



Notice: Screening caps must be installed for OLTCs of 35 kV and above.

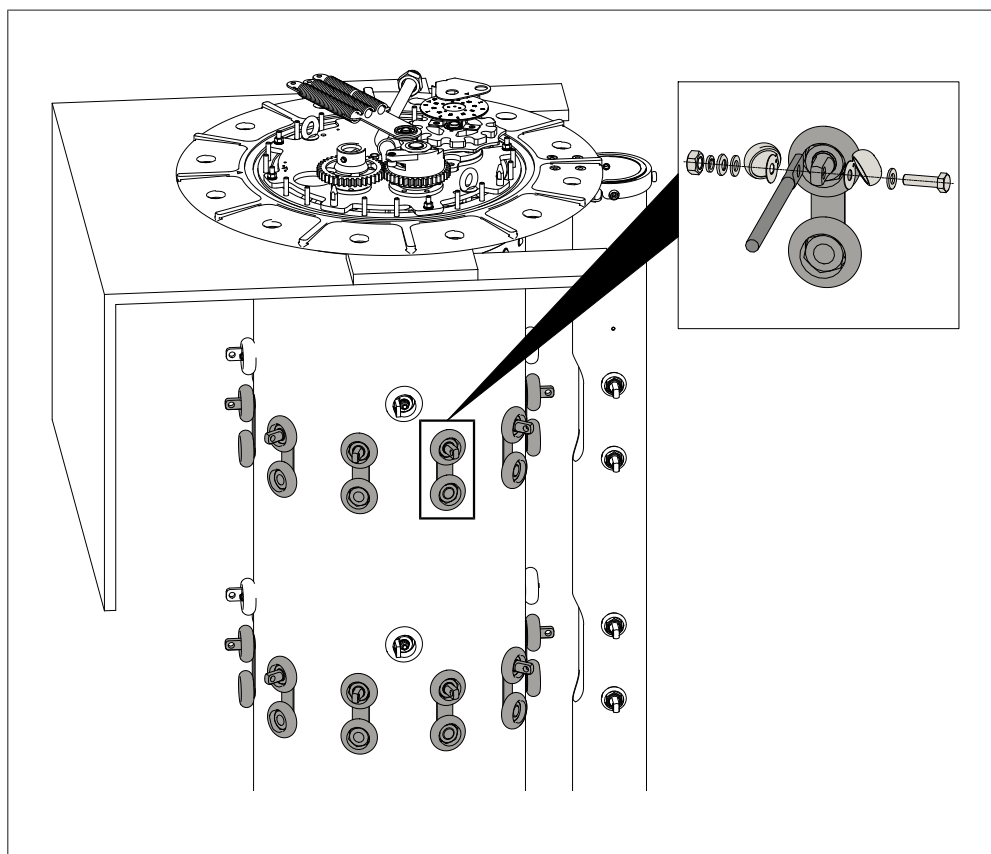


Figure 5-10 Connect the tap selector terminal to the transformer regulation winding

4. Check the lead tension to ensure that the OLTC is not under stress.



Important:

- The tap selector must not be subjected to the pulling force or stress from the leads. Improper lead routing or excessive pulling force may cause the components to deform and result in faults.
- Route the transformer winding leads out from both sides to avoid applying unbalanced force to the tap selector.

5. Remove the adjusting shims.

5.3.6. Install the bell-type transformer tank cover and OLTC mounting flange.

Perform the following steps to install the bell-type transformer tank cover onto the OLTC support flange:

1. Clean the sealing surface of the OLTC support flange. Install the sealing ring onto the support flange.

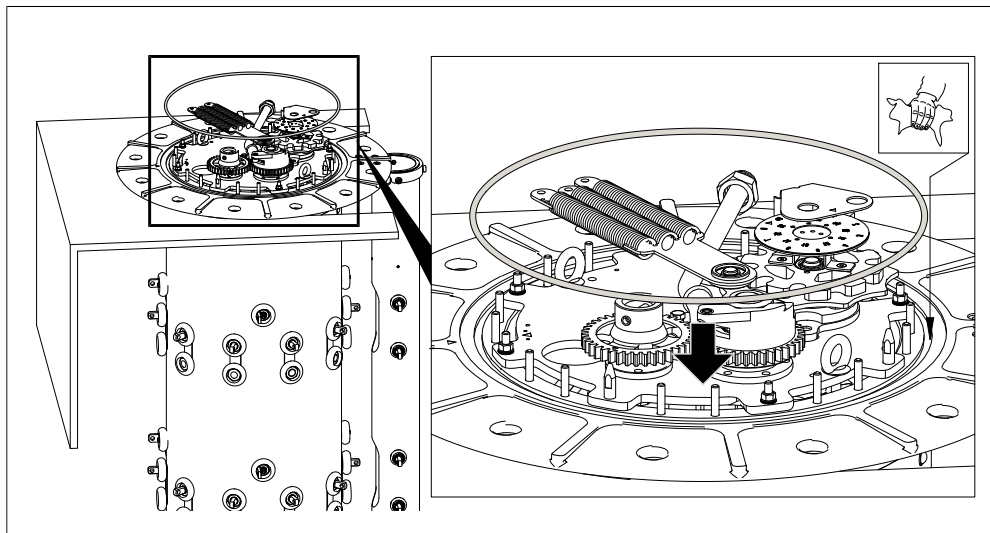


Figure 5-11 Sealing ring of the support flange

2. Lift the bell-type transformer tank cover above main body of the transformer, and securely assemble it onto the transformer.
3. Clean the sealing surfaces of the OLTC mounting flange and the transformer mounting flange.
4. Place the sealing ring on the transformer mounting flange.



Notice: To avoid oil leakage and/or equipment fault, make sure that the sealing ring is intact without damage.

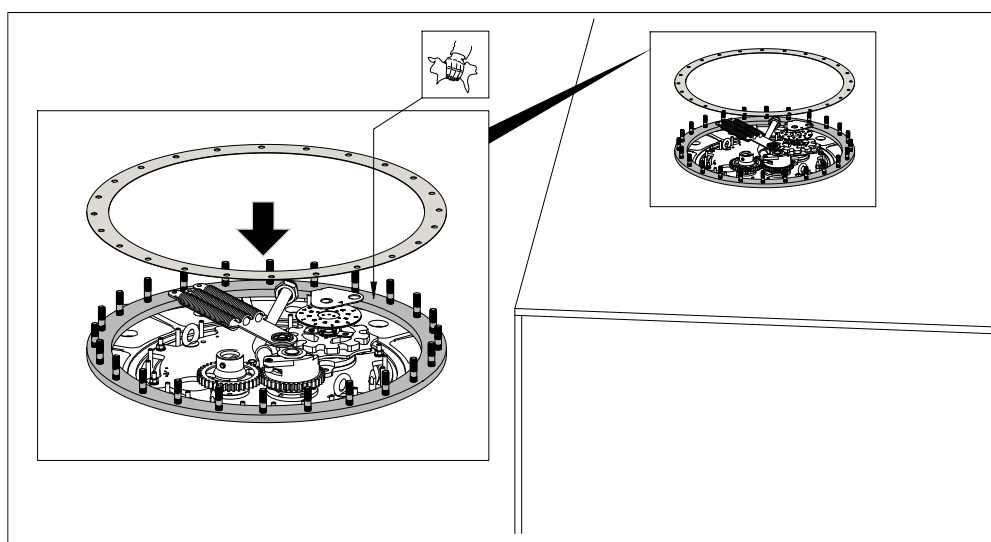


Figure 5-12 Bolts and sealing ring of the transformer mounting flange

5. Lift the OLTC mounting flange, align its holes with the bolts of the transformer mounting flange. Then lower the OLTC mounting flange. Make sure that it is firmly fixed onto the transformer mounting flange.



Notice: Prevent fasteners from falling into the oil compartment.

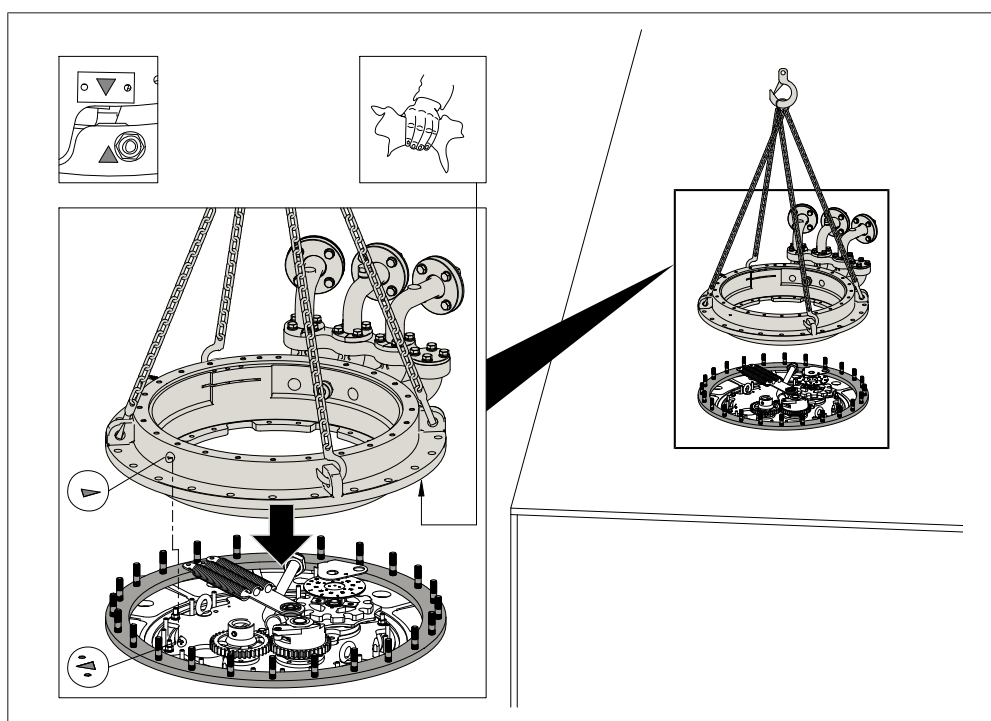


Figure 5-13 Align the mounting flange

6. Use a 13 mm open-end/socket wrench to install 16 sets of M8 nuts, each set with belleville washer x4. Torque: 23 to 26 Nm. Lift the OLTC support flange and align it with the OLTC

mounting flange. When the support flange makes full contact with the mounting reference surface of the mounting flange, install the nuts in the area of the mounting flange ring.

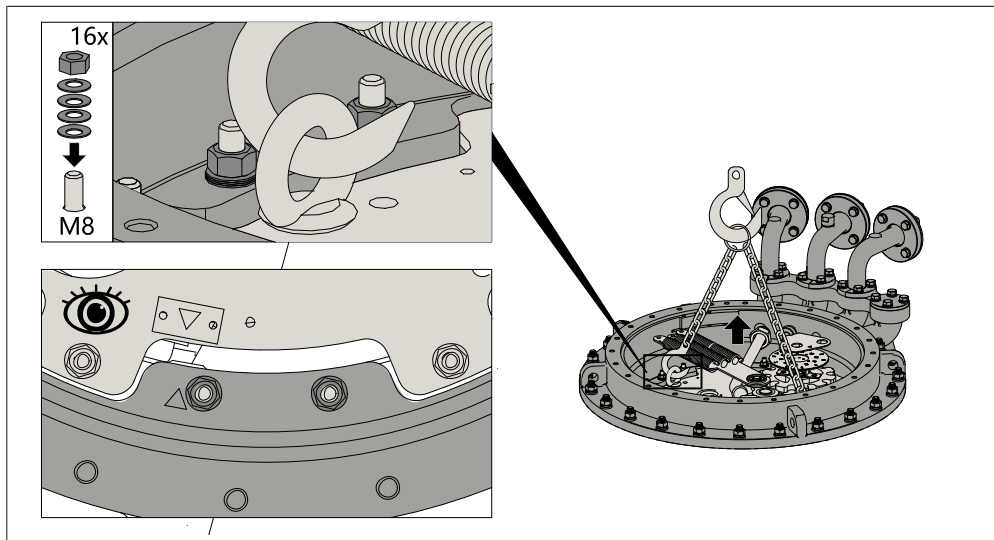


Figure 5-14 Install the OLTC mounting flange

5.3.7. Reinstall the energy storage spring and the oil suction pipe

Follow these steps to reinstall the energy storage spring and the oil suction pipe:

1. Rotate the oil suction pipe into the installation position, and use a 38 mm open-end wrench to tighten the pipe nut by turning it clockwise.



Notice: To avoid oil leakage and/or equipment failure, make sure that the seal rings on the pipe joints are intact without damage.

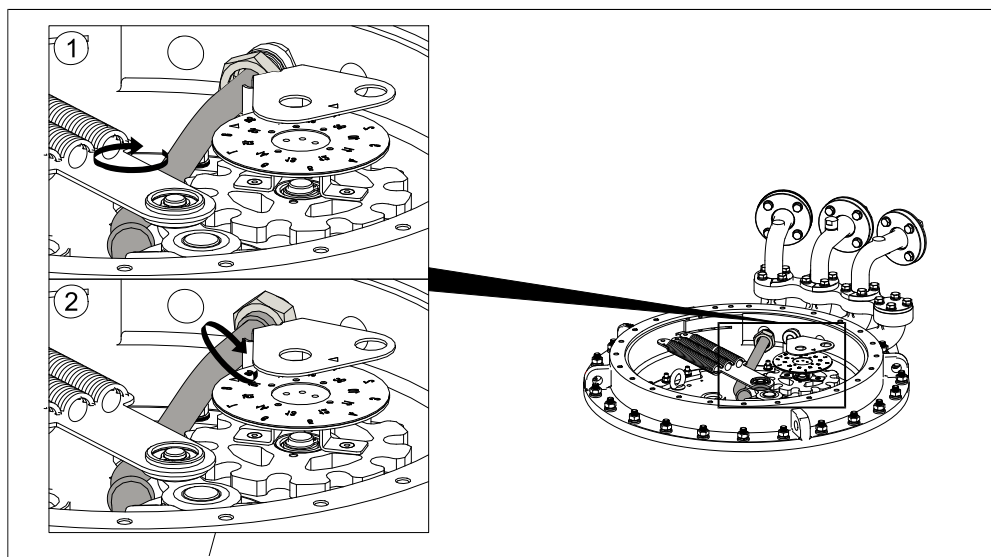


Figure 5-15 Reinstall the oil suction pipe

2. Rotate the energy storage spring into the installation position, and insert the tension spring pin vertically through the OLTC mounting flange.

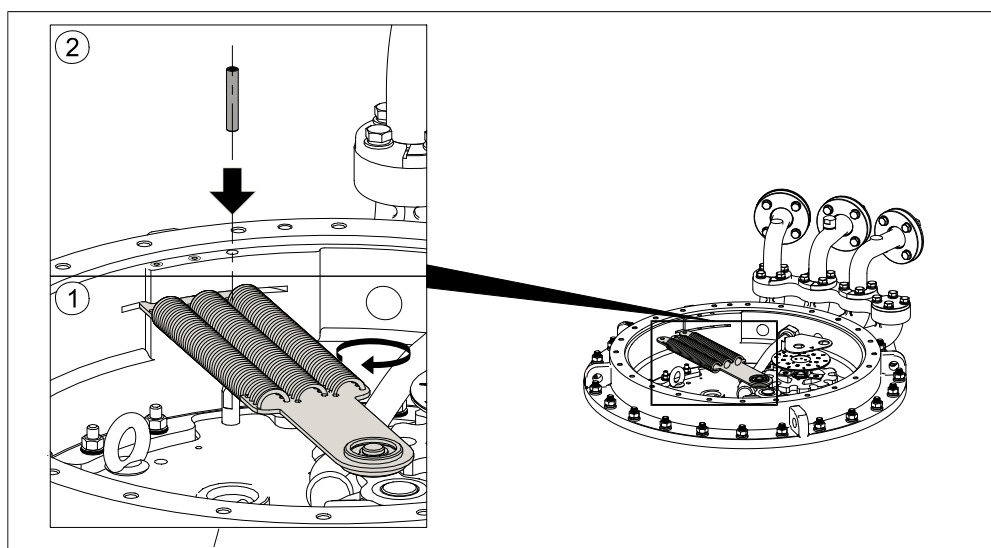


Figure 5-16 Reinstall the energy storage spring

5.3.8. Install the top cover

Perform the following steps to install the top cover onto the OLTC mounting flange:

1. Before installing the top cover, clean its sealing surface. Install the sealing ring onto the top cover. Make sure that the sealing ring is intact and installed in the correct position.

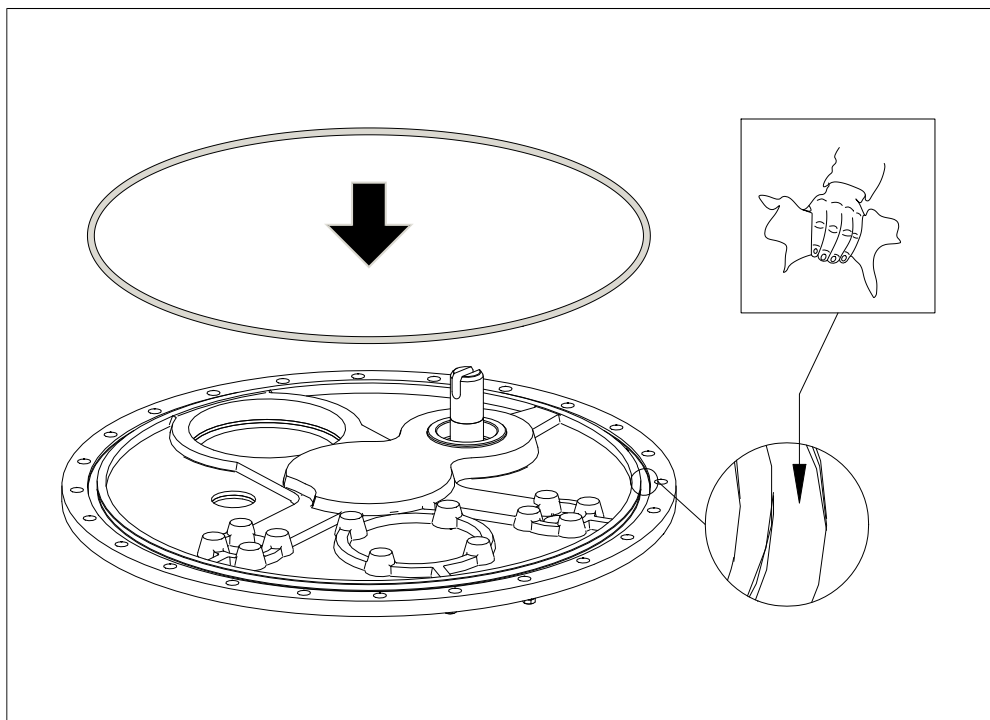


Figure 5-17 Install the sealing ring onto the top cover

2. Check the bevel gear box pins on the top gearbox and the selector switch insert.

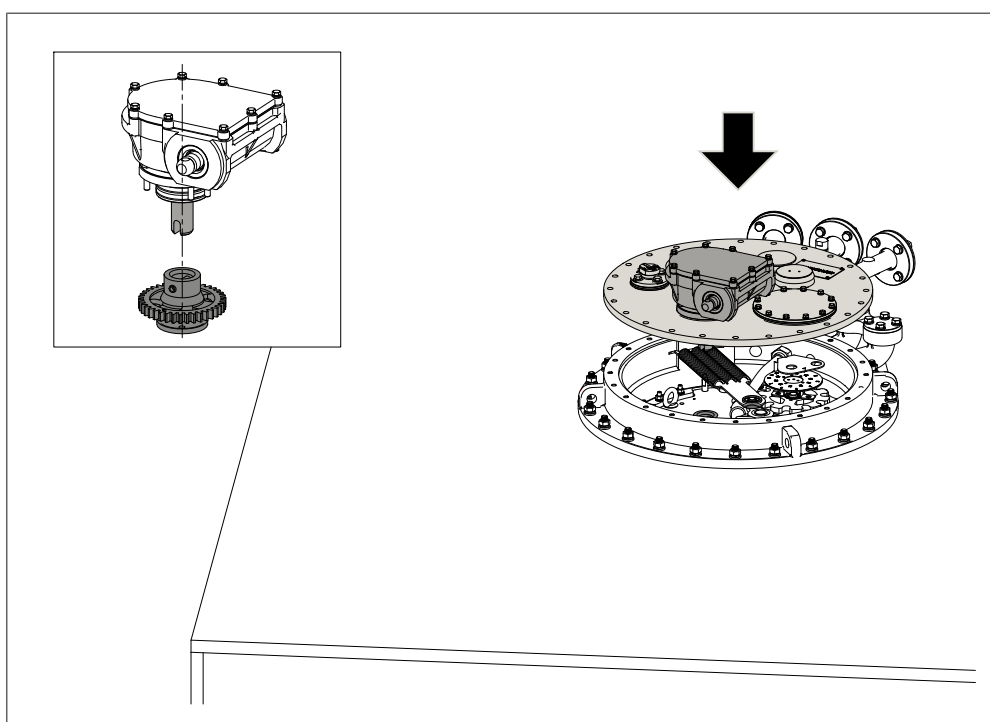


Figure 5-18 Align the bevel gear box pin holes

3. Place the top cover onto the OLTC mounting flange.
4. Use a 17 mm open-end/socket wrench to install 24 sets of M10 bolts, each set with spring washer x1 and flat washer x1. Torque: 28 to 34 Nm, and secure the top cover to the OLTC mounting flange.



Notice: Prevent fasteners from falling into the oil compartment.

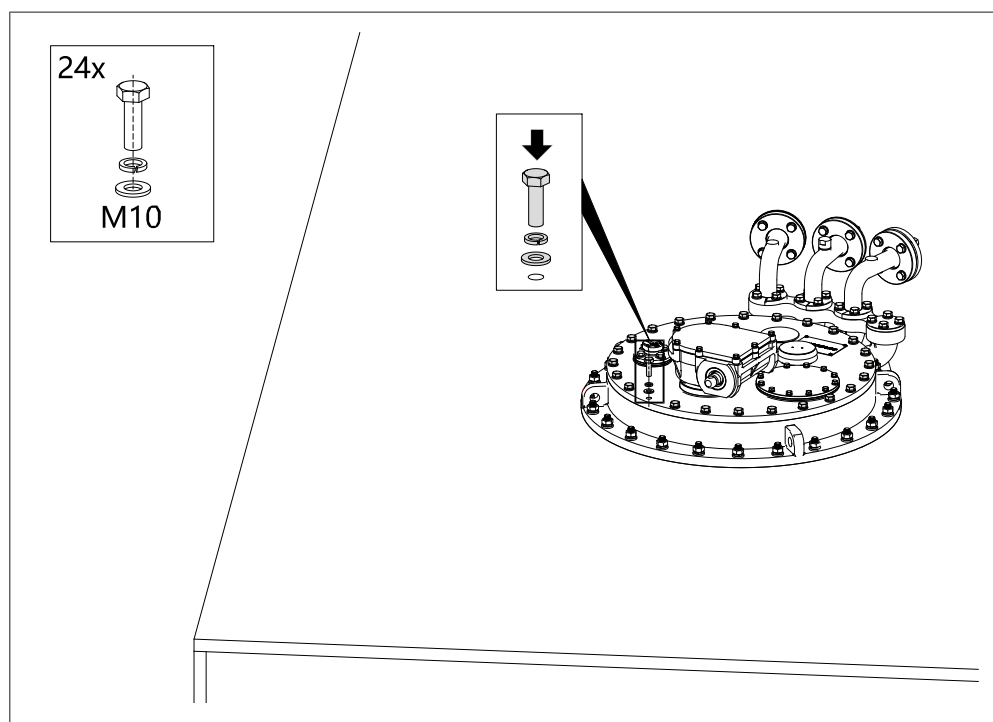


Figure 5-19 Install the top cover

5.4. Install the On-Load Tap-Changer (OLTC) to the top-mounted transformer

Install the selector switch onto a top-mounted transformer following these steps:

1. Clean the sealing surfaces of the OLTC mounting flange and the transformer mounting flange. Place the sealing ring on the transformer mounting flange.



Notice: To avoid oil leakage and/or equipment failure, make sure that the seal ring is intact without damage.

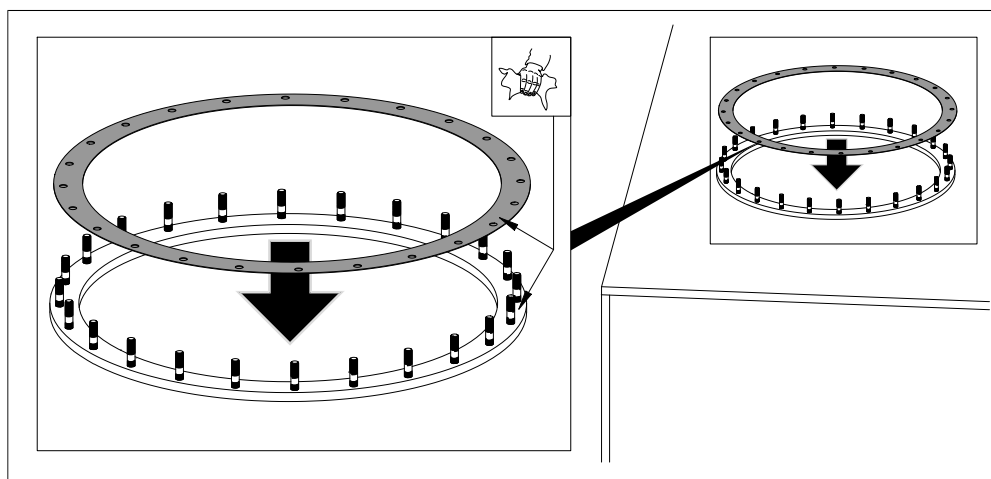


Figure 5-20 Bolts and sealing ring of the transformer mounting flange

2. Carefully and slowly lower the OLTC through the transformer mounting flange and lift it into the transformer tank, then secure it with bolted fasteners to prevent movement.

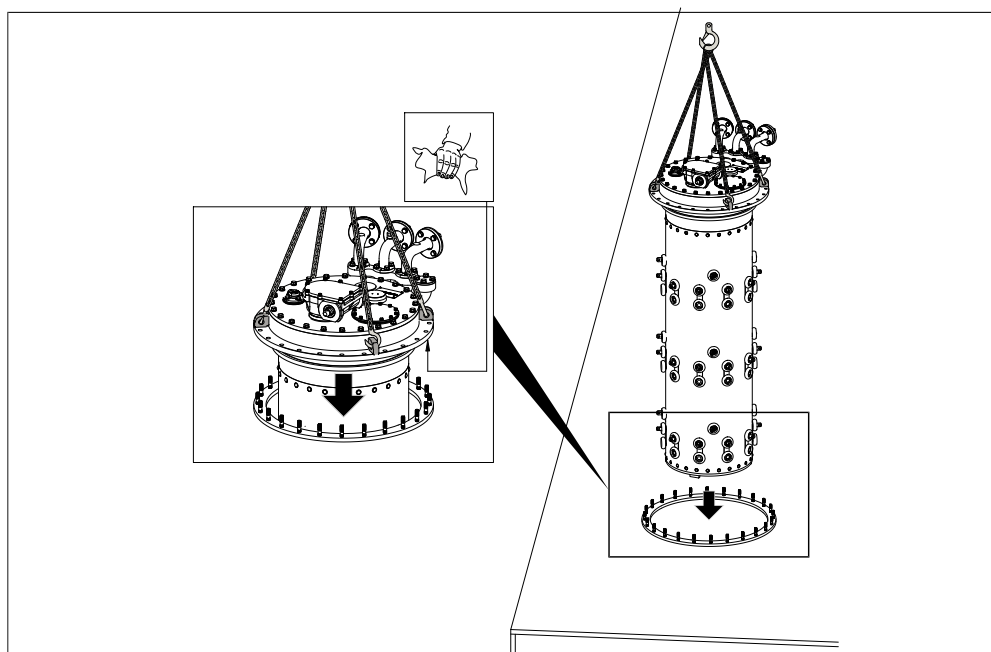


Figure 5-21 Lift the OLTC into the transformer tank

3. Use a 17 mm open-end/socket wrench to install M10 bolt fasteners, each set with spring washer x1, flat washer x2, dished washer x1 and screening cap x2. Torque: 28 to 34 Nm. Install the transformer regulating winding leads according to the diagram.



Notice: Screening caps must be installed for OLTCs of 35 kV and above.

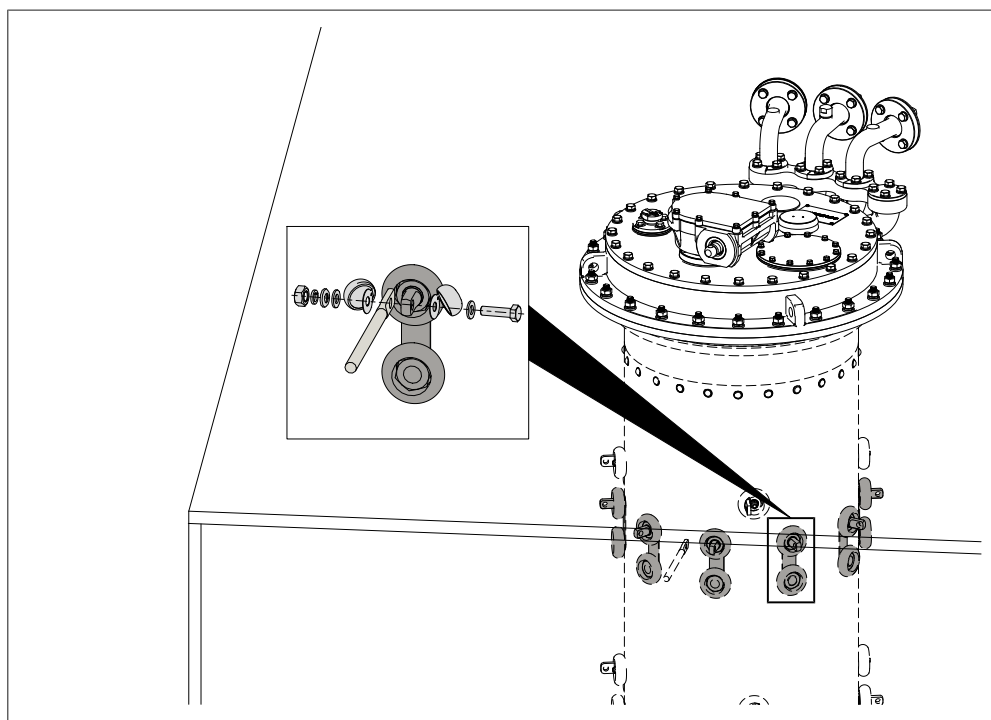


Figure 5-22 Connect the tap selector terminal to the transformer regulation winding

4. Check the lead tension to ensure that the OLTC is not under stress.



Important:

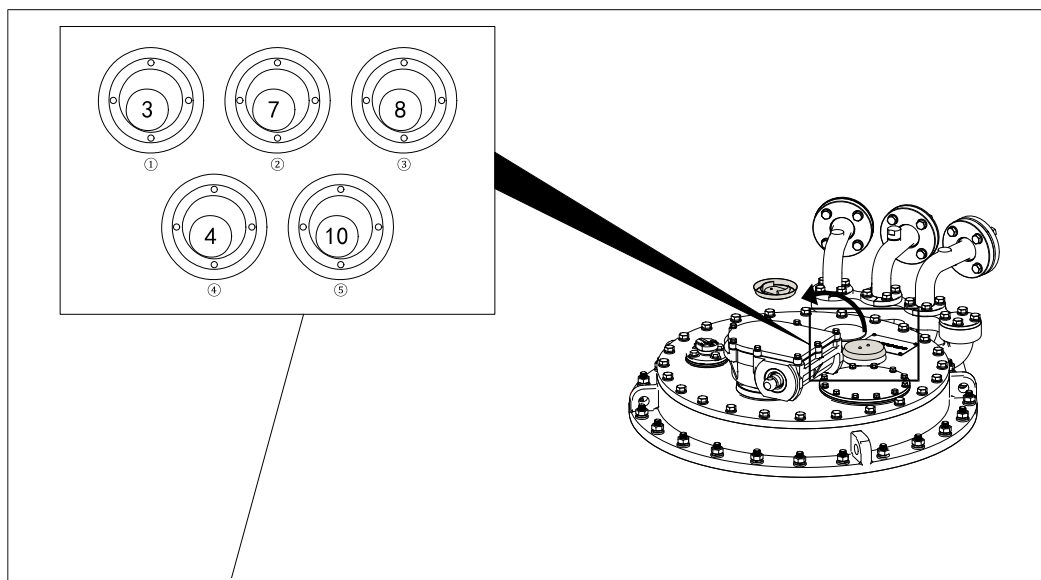
The tap selector must not be subjected to the pulling force or stress from the leads. Improper lead routing or excessive pulling force may cause the components to deform and result in faults.

Route the transformer winding leads out from both sides to avoid applying unbalanced force to the tap selector.

5.5. Remove the selector switch insert

During normal installation of the Type CV2 oil-immersed vacuum On-Load Tap-Changer, it is not necessary to remove the selector switch insert. This procedure is only necessary when the switch malfunctions or specific testing is performed.

! Important: The selector switch insert can only be removed when the tap-changer is set to a specific tap position.



Switch model	Tap position	Remarks
10040, 10050...10100, 10191W/G, 10193W/G	Tap position 3	①
10071W	Tap position 7	②
10091W	Tap position 8	③
12110, 12120, 12231W/G, 12233W/G	Tap position 4	④
12111W	Tap position 10	⑤

Figure 5-23 Remove the specific tap positions for selector switch insert

Remove the selector switch insert by following these steps:

1. Lower the oil level: close the valve between the oil conservator and the selector switch, open the bleeder valve E1 on the top cover, then open the oil drain bolt of the selector switch to lower the oil level inside the switch.
2. Remove the top cover. For detailed instructions, see [5.3.4 Remove the OLTC Mounting Flange](#).

3. Disconnect the energy storage spring and the oil suction pipe. For detailed instructions, see [5.3.3 Separate the energy storage spring and oil suction pipe](#).
4. Remove the oil suction pipe.

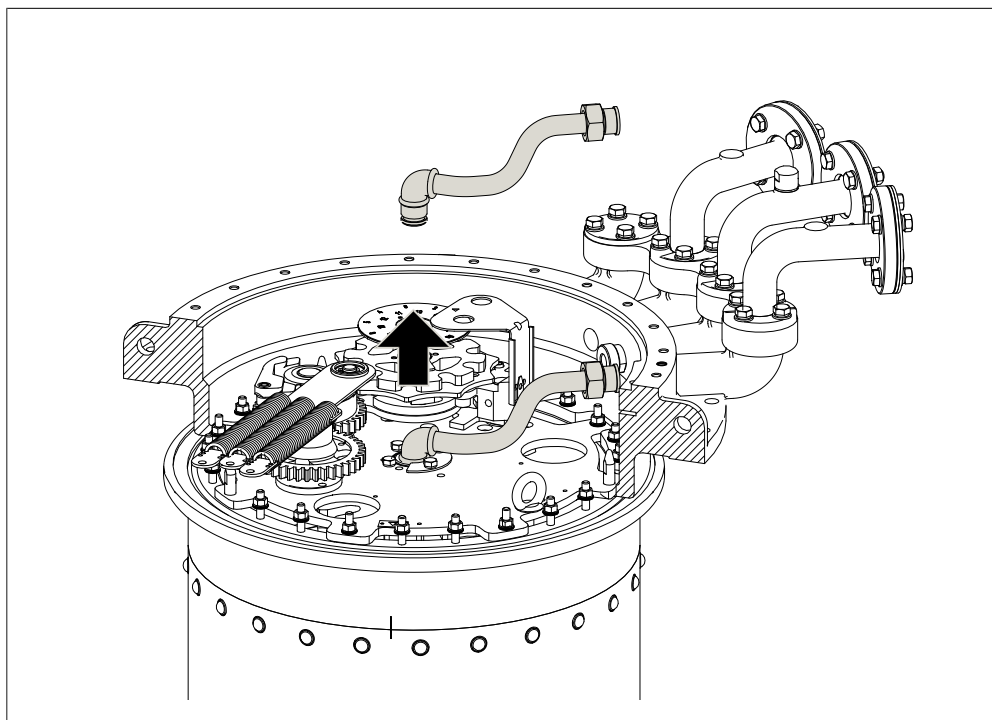


Figure 5-24 Remove the oil suction pipe from the selector switch insert

5. Move the moving contact of the selector switch insert to between the two fixed contacts in the oil compartment.
 - a. Use a special tool to rotate the drive gear counterclockwise until the two holes align.
 - b. Use a flat-head screwdriver to remove the locating pin.
 - c. Insert the locating pin through the two aligned holes.

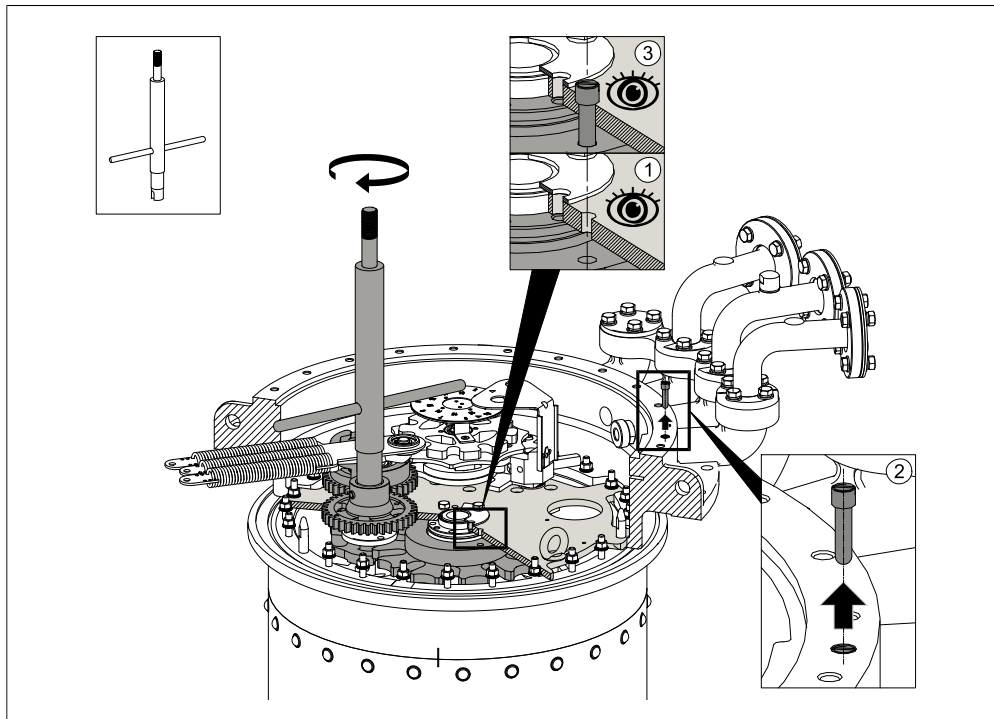


Figure 5-25 Insert the locating pin into the pin hole

6. Use a 13 mm open-end/socket wrench to remove 5 sets of M8 nuts, each set with belleville washers x4, to detach the nuts securing the selector switch insert.

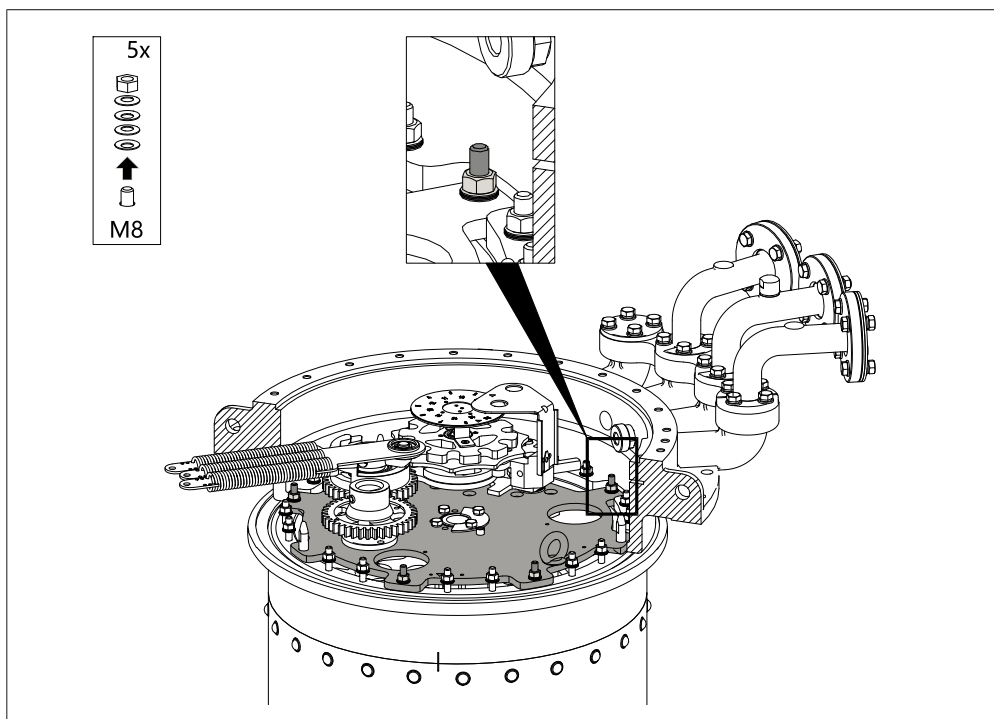


Figure 5-26 Remove the nuts that secure the selector switch insert

7. Pull out the central oil suction pipe.

- a. Use a 13 mm open-end/socket wrench to remove 2 sets of M8 bolts, each set with a spring washer x1.
- b. Remove the red semicircular baffle.
- c. Reinstall the bolts.

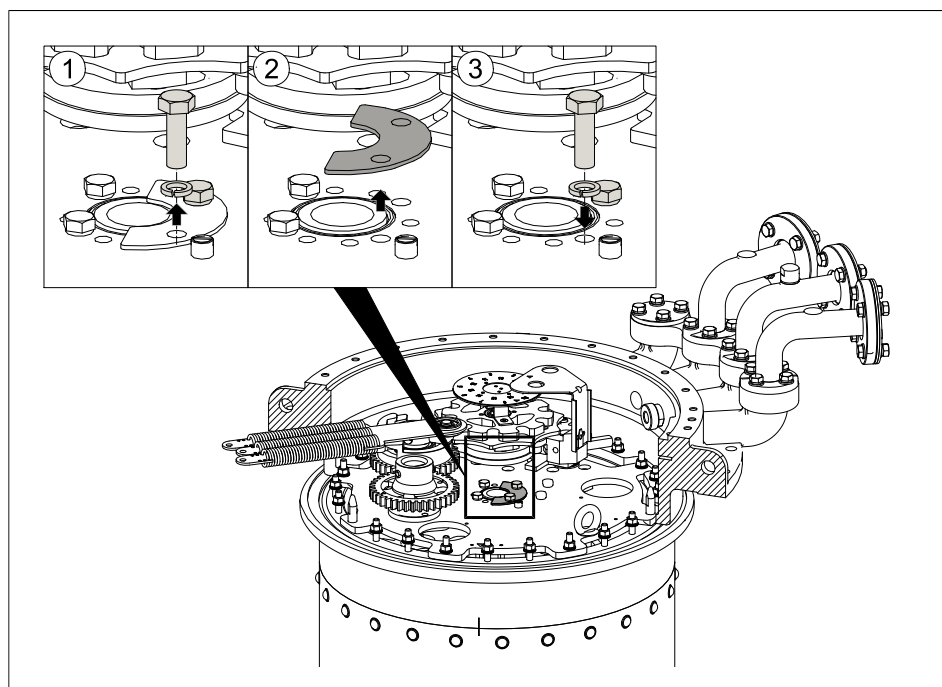


Figure 5-27 Fasteners on the oil suction pipe

- d. Use a special tool to pull out the oil suction pipe.

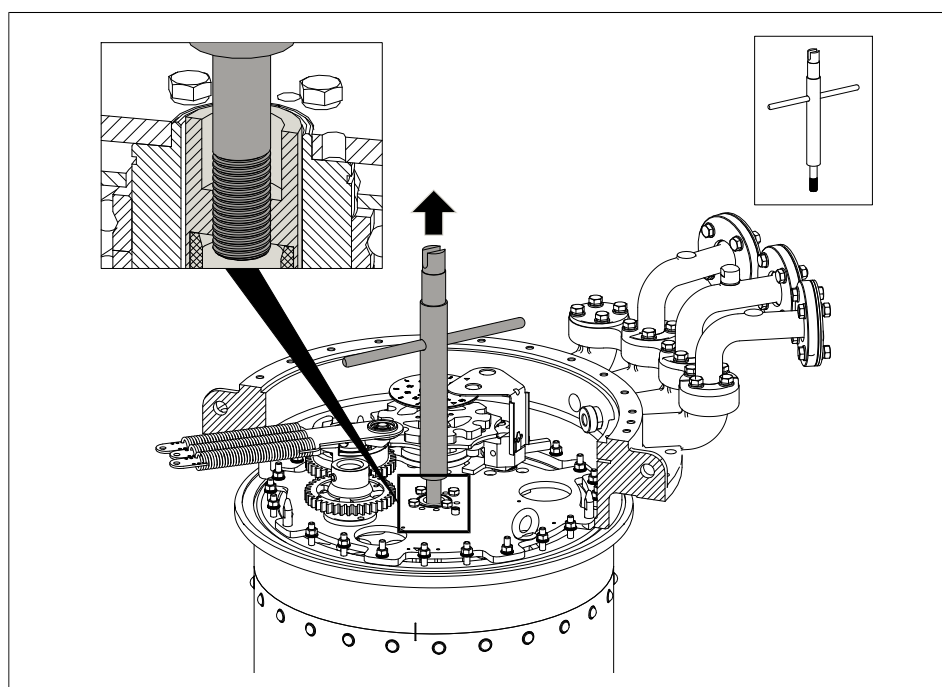


Figure 5-28 Pull the oil suction pipe out from the selector switch insert

8. Lift the selector switch insert out slowly and vertically, and place it on a flat, clean surface.

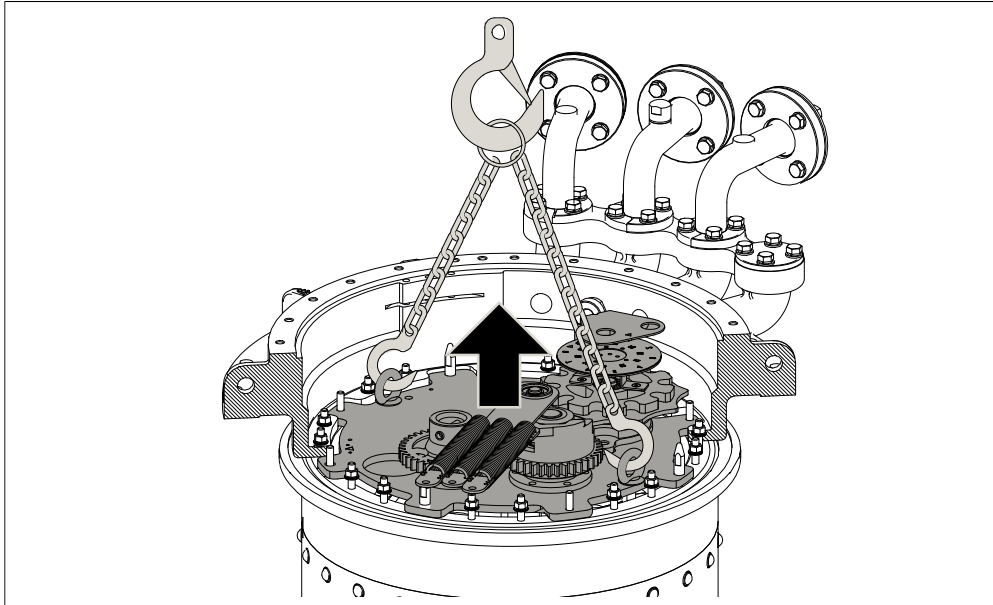


Figure 5-29 Lift the selector switch insert out vertically

9. To reinstall the selector switch insert, perform the above steps in reverse order. Failure to do so will damage the On-Load Tap-Changer.

5 - Installation of the OLTC on the transformer

6. Transformer turns ratio test and DC winding resistance measurement

! **Important:** It is recommended to perform the turns ratio test and DC winding resistance measurement before the drying process.

6.1. Turns ratio test

NOTICE

Risk of OLTC damage!

Improper operation may cause OLTC damage and fault.

- ✓ Before drying, do not operate the OLTC in an oil-free condition for more than 10 cycles. Operating the OLTC in an oil-free condition will cause damage.
- ✓ If the turns ratio test is performed after drying, ensure that the tap selector is completely immersed in oil and that the selector switch oil compartment is filled with oil before operating the OLTC.
- ✓ Each tap-change operation must be performed completely.

! **Important:** Before testing, ensure that the OLTC mounting flange, the selector switch and top cover are all properly installed in place.

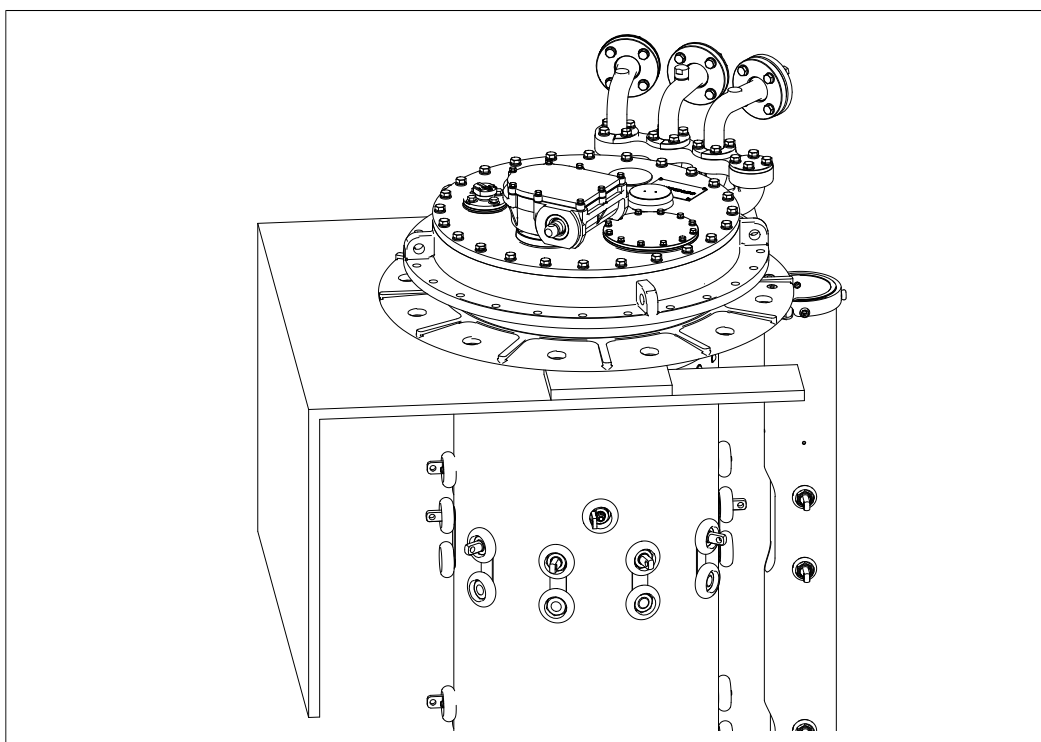


Figure 6-1 Basic structure of the OLTC head

To perform the turns ratio test, perform these steps:

1. Use a short pipe with an internal diameter of 25 mm (with a 12 mm diameter bolt inserted inside) together with a handwheel or crank to rotate the driving shaft of the top gearbox.
2. For parallel-installed groups of On-Load Tap-Changers (two or three units), connect all OLTC top gearboxes with a horizontal shaft.
3. Switch the OLTC to the desired operating position. Each time a tap-change is performed, the selector switch emits a distinct "click" sound.
4. After hearing the sound of selector switch operating, continue rotating the driving shaft in the same direction for another two and a half turns to complete the tap-change.



Notice: An incomplete tap-change operation may damage the OLTC.

5. Check the tap position through the inspection window (see the figure below) on the OLTC head to avoid exceeding the OLTC limits.

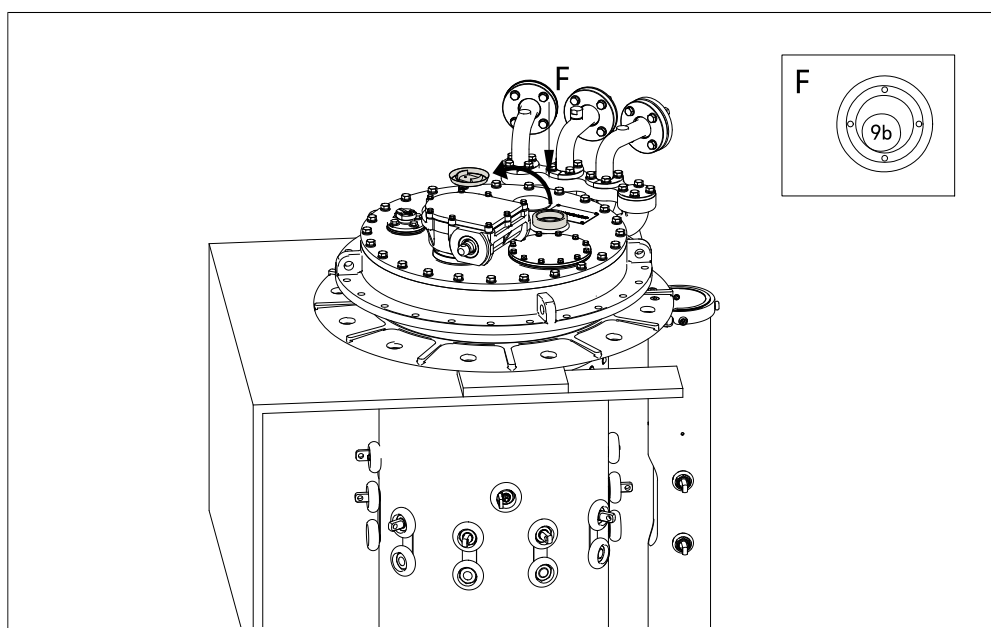


Figure 6-2 Location of the OLTC inspection window

6. Perform transformer turns ratio tests for all operating tap positions.
7. Return the OLTC to the set position.

6.2. DC winding resistance measurement

NOTICE

Damage to the OLTC!

Incorrect operations can lead to damage and malfunctions on the OLTC.

- ✓ Do not operate the OLTC more than 10 cycles without oil before drying. Operating the OLTC without oil can cause damage.
- ✓ Perform each tap-change operation completely.

To perform a DC winding resistance measurement, follow these steps:

1. Perform the DC winding resistance measurement with the OLTC.



Notice: If you perform the measurement in a no-oil state, minimize OLTC operations during this procedure.

2. Return the OLTC to its set position after the DC winding resistance measurement.

7. Drying and oil filling

7.1. Vacuum drying



Important:

To ensure proper insulation performance, the OLTC must be thoroughly dried according to the vacuum or vapor-phase drying procedures.

7.1.1. Vacuum drying in a drying oven

NOTICE

Risk of OLTC damage!

Improper drying operation may cause component damage or fault, affecting the normal operation of the OLTC.

- ✓ When drying in a drying oven, the following components must not be placed inside the oven: top cover, top gearbox, MDU, protective relay, top cover sealing gasket, pressure relief device, bevel gearbox, and other accessories.
- ✓ Prevent components from falling into the selector switch oil compartment during disassembly and assembly. Check whether all components are present.

To perform vacuum drying in a drying oven, follow these steps:

1. If the top cover is installed, remove it. (See [5.3.2 Remove the top cover](#))
2. Perform the drying procedure:
 - a. Heat the OLTC in air at atmospheric pressure at a rate of approximately 10 °C per hour until reaching 110 °C.
 - b. Maintain the OLTC at 110 °C for 20 hours using circulating hot air.
 - c. Dry the OLTC under vacuum for 50 hours, maintaining the temperature between 105 °C and 125 °C. The maximum residual pressure should be 133 Pa.
3. Reinstall the top cover. (See [5.3.8 Install the top cover](#))

7.1.2. Vacuum drying in the transformer tank

NOTICE

Risk of OLTC damage!

Improper drying operation may cause component damage or fault, affecting the normal operation of the OLTC.

- ✓ During the drying process inside the transformer tank, the OLTC top cover must remain closed.
- ✓ Prevent components from falling into the selector switch oil compartment during disassembly and assembly. Check whether all components are present.

Perform vacuum drying inside the transformer tank as follows:

1. Use a 17 mm open-end/socket wrench to remove 8 sets of M10 bolts, each set with spring washer x1 and flat washer x1. Remove pipe connectors E2 and R from the OLTC mounting flange.

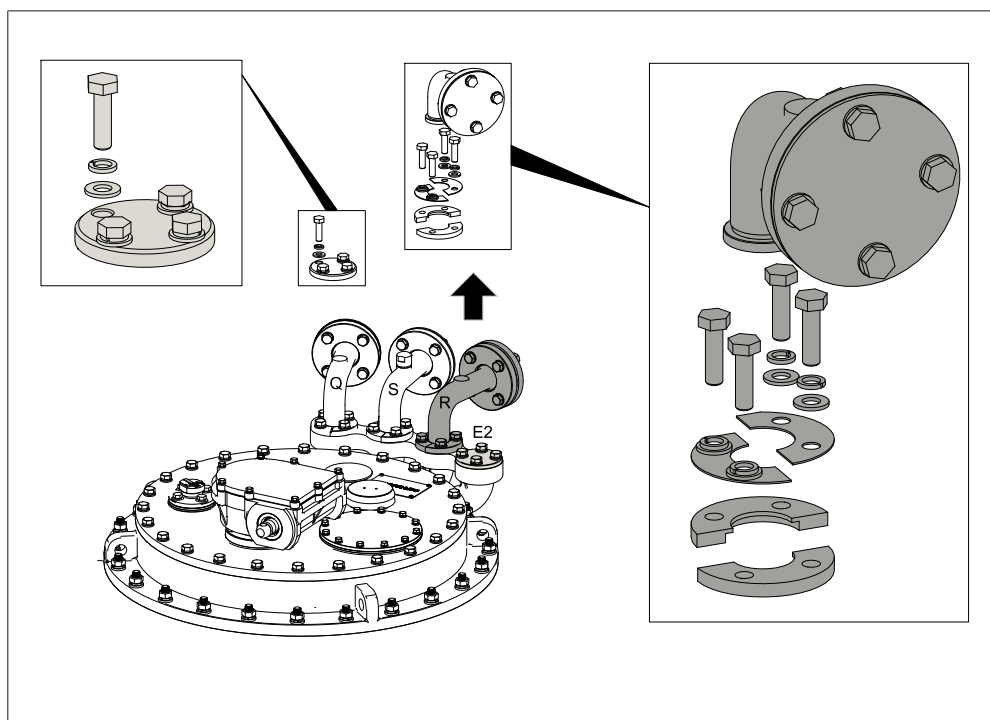


Figure 7-1 Remove the pipe connectors from the OLTC top

2. Use a 17 mm open-end/socket wrench to install 8 sets of M10 bolts, each set with spring washer x1 and flat washer x1. Torque: 28 to 34 Nm. Use a bypass pipe with a minimum inner diameter of 25 mm to connect E2 and R.

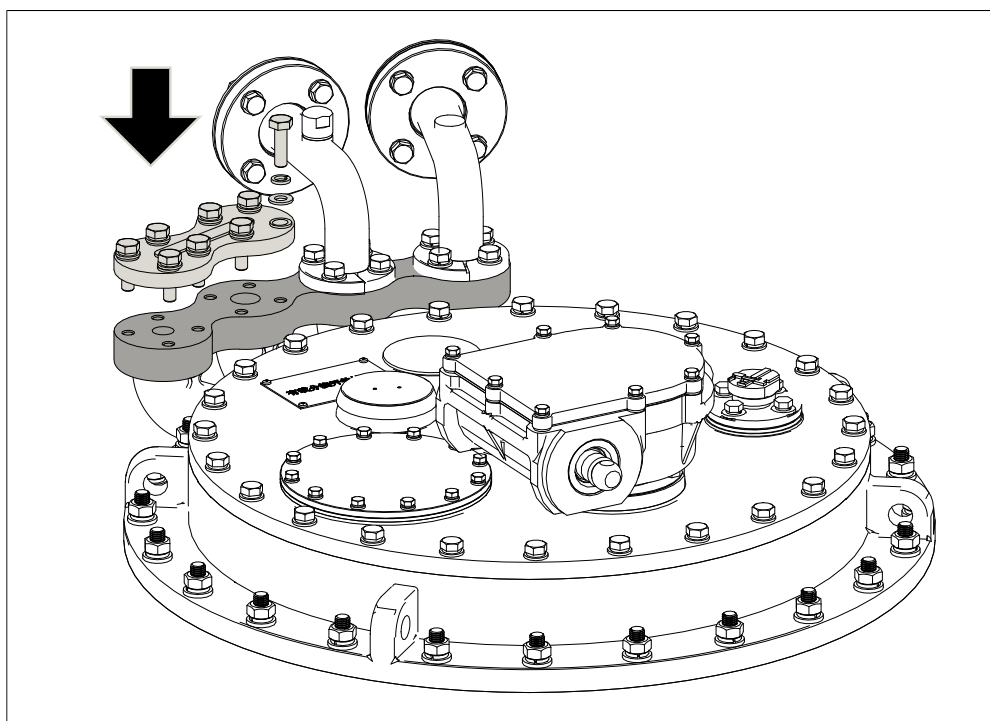


Figure 7-2 Connect E2 and R with a bypass pipe

3. Perform the drying procedure:
 - a. Heat the OLTC in air at atmospheric pressure at a rate of approximately 10 °C per hour until reaching 110 °C.
 - b. Maintain the OLTC at 110 °C for 20 hours using circulating hot air.
 - c. Dry the OLTC under vacuum for 50 hours, maintaining the temperature between 105 °C and 125 °C. The maximum residual pressure should be 133 Pa.
4. Remove the bypass pipe and reinstall pipe connectors E2 and R.

7.2. Vapor-phase drying



Important:

To ensure the insulation performance of the OLTC, thorough drying must be carried out using a vacuum drying or vapor-phase drying process.

1. Use an adjustable open-end wrench to loosen the oil drain bolt. Do not turn it more than 10 revolutions.



Note:

This bolt is located at the bottom of the selector switch oil compartment.

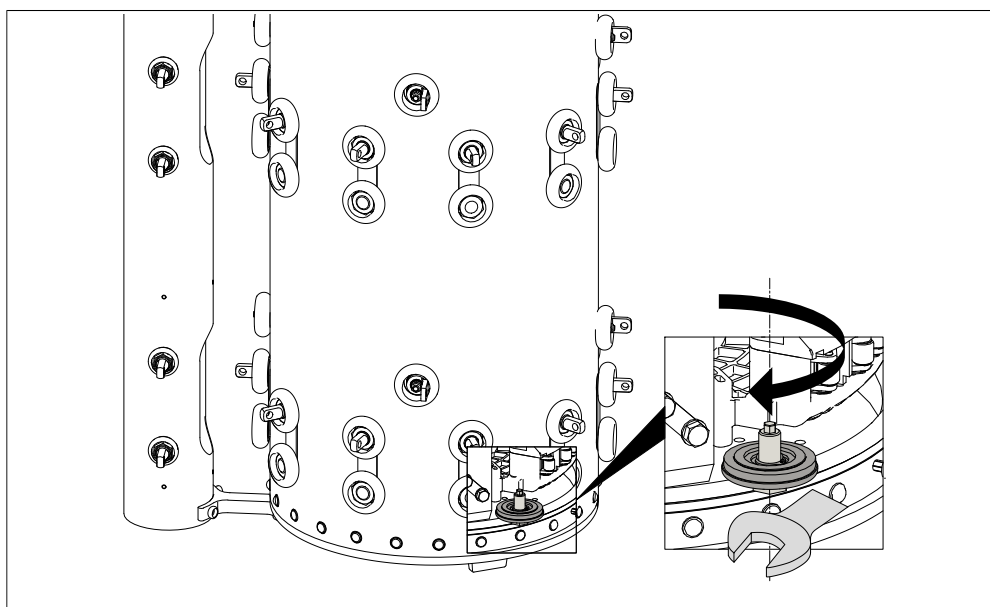


Figure 7-3 Loosen the oil drain bolt

2. After drying treatment, use an adjustable open-end wrench to tighten the oil drain bolt.
Torque: 22 Nm.

7.2.1. Vapor-phase drying in a drying oven

NOTICE

Risk of OLTC damage!

Improper drying operation may cause component damage or fault, affecting the normal operation of the OLTC.

- ✓ When drying in a drying oven, the following components must not be placed inside the oven: top cover, top gearbox, MDU, protective relay, top cover sealing gasket, pressure relief device, bevel gearbox, and other accessories.
- ✓ Prevent components from falling into the selector switch oil compartment during disassembly and assembly. Check whether all components are present.



Important:

- Before drying the OLTC, open the oil drain bolt.
- After drying treatment, tighten the oil drain bolt.

Perform vapor-phase drying in a drying oven as follows:

1. Remove the top cover. (See [5.3.2 Remove the top cover](#))
2. Loosen the oil drain bolt.

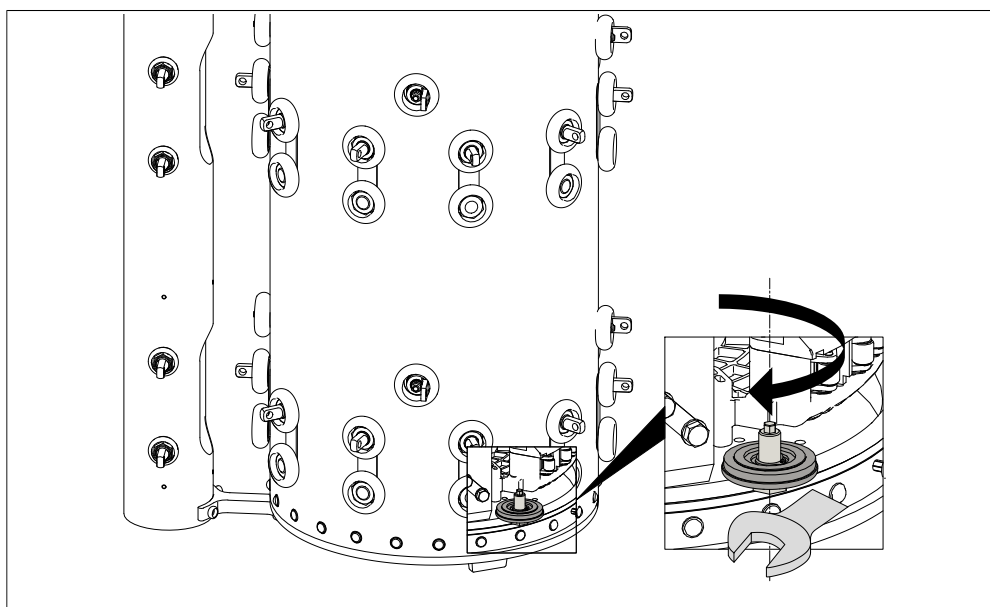


Figure 7-4 Loosen the oil drain bolt

3. Perform the drying procedure:
 - a. Inject kerosene vapor at approximately 90 °C and maintain this temperature for 3–4 hours.
 - b. Increase the kerosene vapor temperature at a rate of 10 °C per hour until the required final temperature is reached. The maximum OLTC temperature should not exceed 125 °C.
 - c. Dry the OLTC under vacuum, maintaining the temperature between 105 °C and 125 °C. The maximum residual pressure should be 133 Pa. The drying time should be the same as that for the transformer, but should not be less than 50 hours.
4. Tighten the oil drain bolt.
5. Reinstall the top cover. (See [5.3.8 Install the top cover](#))

7.2.2. Vapor-phase drying in the transformer tank

NOTICE

Risk of OLTC damage!

Improper drying operation may cause component damage or fault, affecting the normal operation of the OLTC.

- ✓ During the drying process inside the transformer tank, the OLTC top cover must remain closed.
- ✓ Prevent components from falling into the selector switch oil compartment during disassembly and assembly. Check whether all components are present.



Important:

- Before drying the OLTC, open the oil drain bolt.
- After drying treatment, tighten the oil drain bolt.



Note: If the oil drain bolt cannot be loosened from inside the oil compartment, enter the transformer through the transformer manhole to loosen the oil drain bolt. This method does not require removing the switch insert, so you can skip Steps 3 through 7.

Perform vapor-phase drying inside the transformer tank as follows:

1. Use a 17 mm open-end/socket wrench to remove 8 sets of M10 bolts, each set with spring washer x1 and flat washer x1, and remove pipe connectors E2 and R from the OLTC mounting flange.

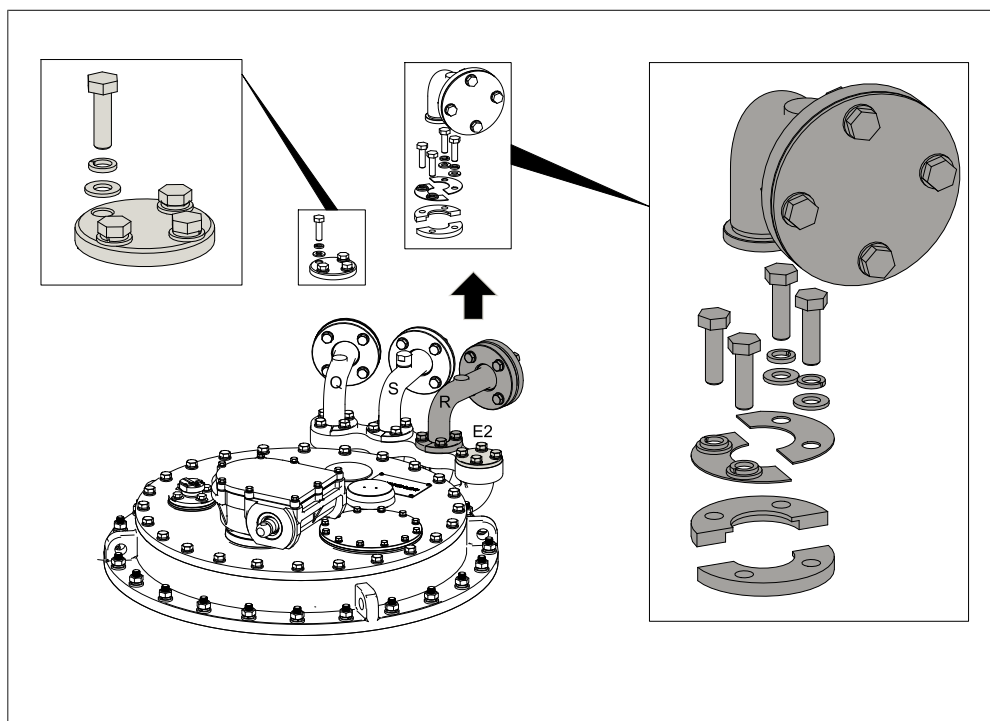


Figure 7-5 Remove the pipe connectors from the OLTC top

2. Connect E2 and R using a bypass pipe (internal diameter at least 25 mm).
3. Remove the top cover. (See [5.3.2 Remove the top cover](#))
4. Remove the selector switch insert. (See [5.5 Remove the selector switch insert](#))
5. Loosen the oil drain bolt.
6. Reinstall the selector switch insert. (See [5.5 Remove the selector switch insert](#))
7. Reinstall the OLTC top cover. (See [5.3.8 Install the top cover](#))
8. Perform the drying procedure:
 - a. Inject kerosene vapor at approximately 90 °C and maintain this temperature for 3–4 hours.
 - b. Increase the kerosene vapor temperature at a rate of 10 °C per hour until the required final temperature is reached. The maximum OLTC temperature should not exceed 125 °C.
 - c. Dry the OLTC under vacuum, maintaining the temperature between 105 °C and 125 °C. The maximum residual pressure should be 133 Pa. The drying time should be the same as that for the transformer, but should not be less than 50 hours.
9. Tighten the oil drain bolt and reassemble the OLTC.

7.3. Oil filling

! **Important:** During disassembly, assembly, or other operations, avoid damaging the sealing surfaces of the OLTC mounting flange. After the transformer has been dried, before operating the OLTC, ensure that the tap selector is completely immersed in transformer oil and that the OLTC oil compartment is filled with insulating oil.



Notice:

- After drying treatment, refill the OLTC oil compartment (with the selector switch insert already installed) with oil as soon as possible to reduce moisture ingress.
- Only use new transformer oil that complies with the relevant standards for filling the oil compartment and its conservator.
- Both the OLTC and the transformer should be filled with new oil simultaneously under vacuum.

The oil filling steps are as follows:

1. Ensure that the top cover is installed.
 - a. Ensure that the sealing ring is in the correct position.
 - b. Use a 17 mm open-end/socket wrench to install 24 sets of M10 bolts, each set with spring washer x1 and flat washer x1. Torque: 28 to 34 Nm.
2. Install the bypass pipe.
 - a. Connect a bypass pipe between pipe connector E2 and R.
 - b. Ensure that both the OLTC and the transformer are evacuated simultaneously.
3. Use one of the spare pipe connectors (S or Q) for oil filling.

8. Installation of protective devices and drive unit

8.1. Install protective relay

Install the protective relay according to the instructions provided in the protective relay manual.

8.2. Install pressure relief valve

Notice: If a pressure relief valve is used, remove the red protection strip from the top of the pressure relief valve before operating the OLTC.

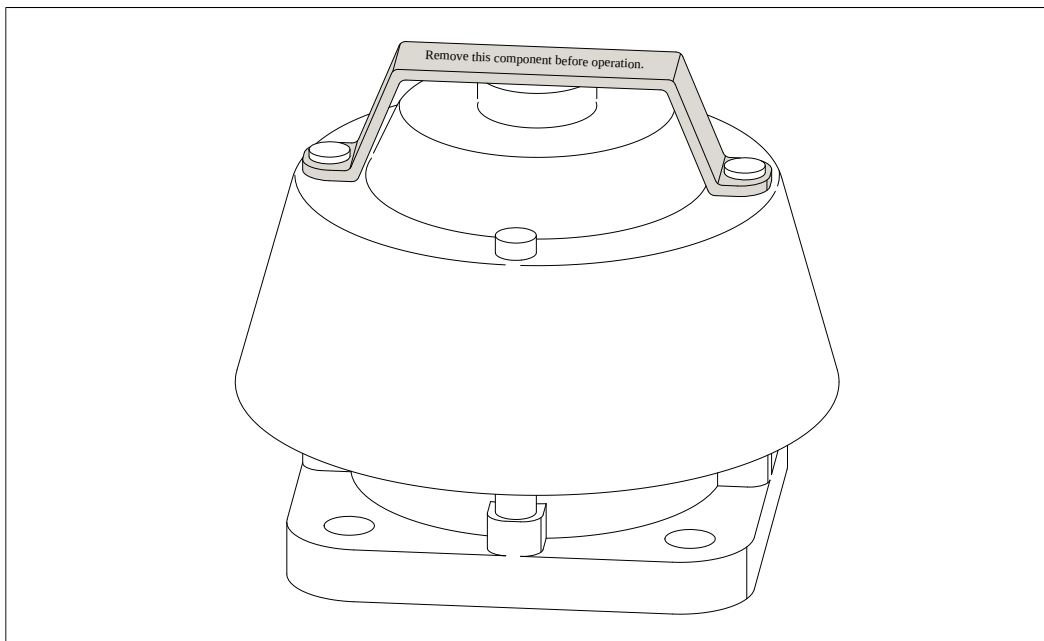


Figure 8-1 Red protection strip on pressure relief valve

8.3. Install Motor Drive Unit (MDU)

NOTICE

Improper MDU installation!

Improper installation of the MDU will affect the operation of the OLTC.

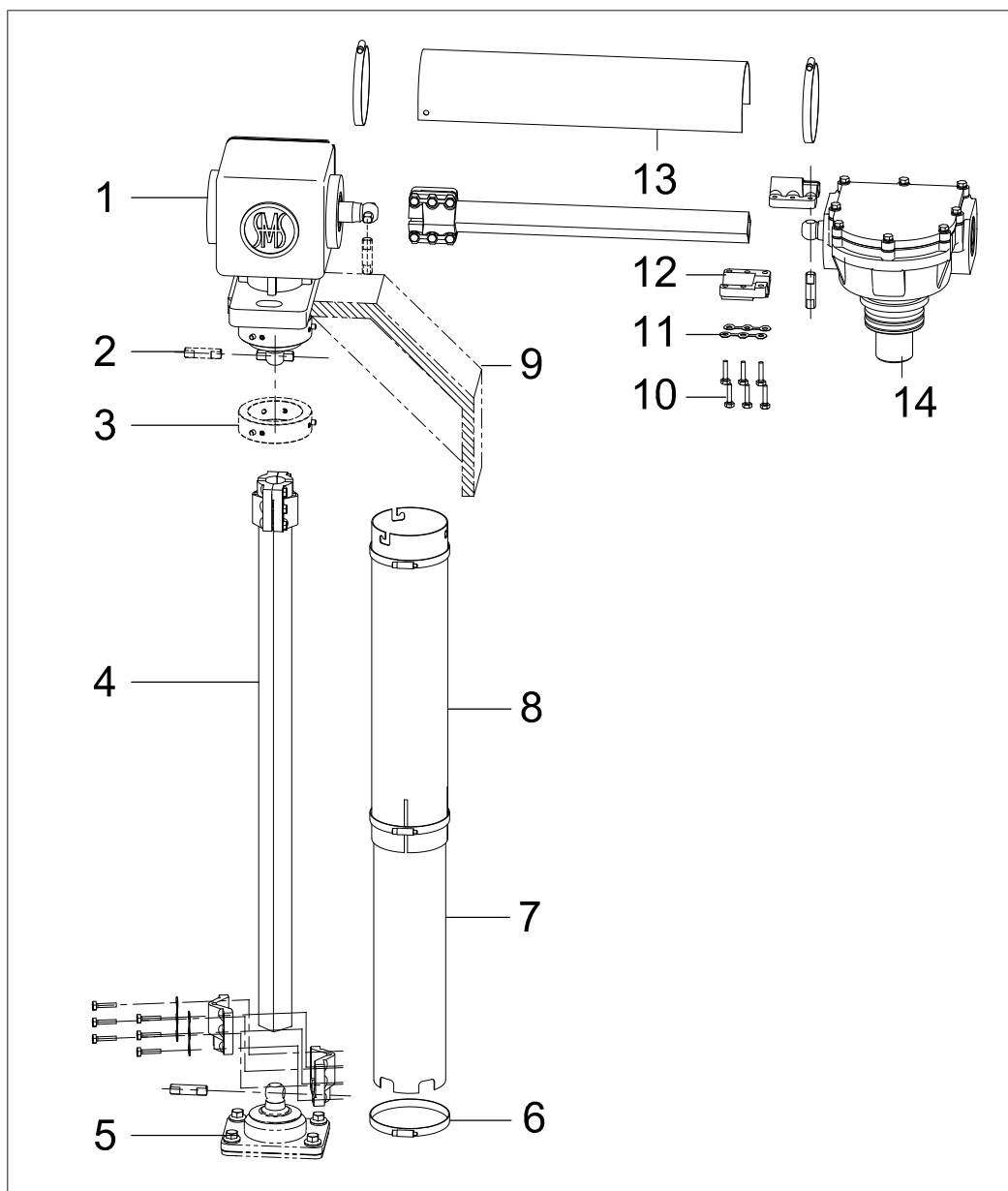
- ✓ Connect the MDU to the OLTC only when both units are in their set positions.
- ✓ Mount the MDU vertically on the transformer tank wall, ensuring accurate alignment.
- ✓ Ensure that the MDU mounting plate is flat to prevent deformation, which can affect operation.

The MDU drives the OLTC during tap-changing operations, either electrically or manually. Refer to the Huaming MDU manual for detailed installation procedures.

8.4. Install drive mechanism components

8.4.1. Drive mechanism components

Drive mechanism components consist of the bevel gearbox, driving shafts, couplings and other accessories.



1. Bevel gearbox	2. Cylindrical pin	3. Bushing
4. Driving shaft	5. Motor Drive Unit (MDU)	6. Clamp
7. Vertical guard 1	8. Vertical guard 2	9. Gearbox bracket (transformer)
10. M6 screw	11. Lock washer	12. Coupling
13. Horizontal guard	14. Top gearbox	

Figure 8-2 Drive mechanism components

8.4.2. Adjust the top gearbox

NOTICE

Mismatched OLTC components!

Mismatched components will affect the normal and stable operation of the OLTC.

✓ Ensure that the nameplate numbers match. Each gearbox is specific to a particular OLTC.



Important:

For driving shafts longer than 2 m, an intermediate support gearbox must be installed.

Adjust the top gearbox according to the following steps:

1. Loosen the six bolts that fix the clamping plates under the top gearbox (do not remove the bolts completely).
2. Pry the three clamping plates outward, then rotate the top gearbox to align its output shaft with the MDU.
3. Use a 13 mm open-end wrench to tighten the M8 bolt to a torque of 23 to 26 Nm. Reposition the clamping plate to ensure that the top gearbox is securely installed without looseness.

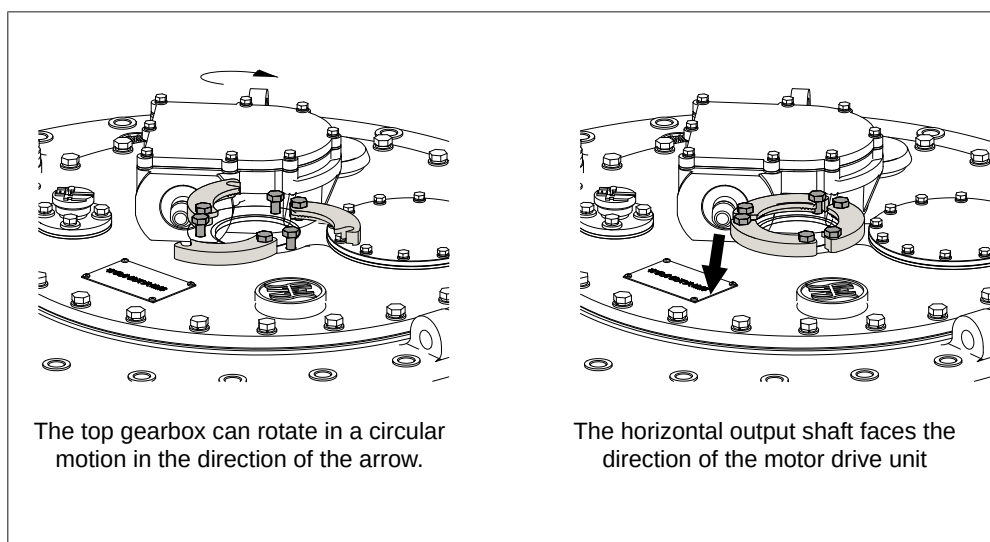


Figure 8-3 Top gearbox adjustment and fixing

8.4.3. Install the bevel gearbox

NOTICE

Mismatched OLTC components!

Mismatched components can affect the normal and stable operation of the OLTC.

- ✓ Ensure that the nameplate serial number match. Each gearbox corresponds to a specific OLTC.



Important:

For driving shafts longer than 2 meters, install an intermediate support gearbox.

1. Align the shaft of the bevel gearbox vertically with the shaft of the MDU. Make sure that they are precisely aligned.

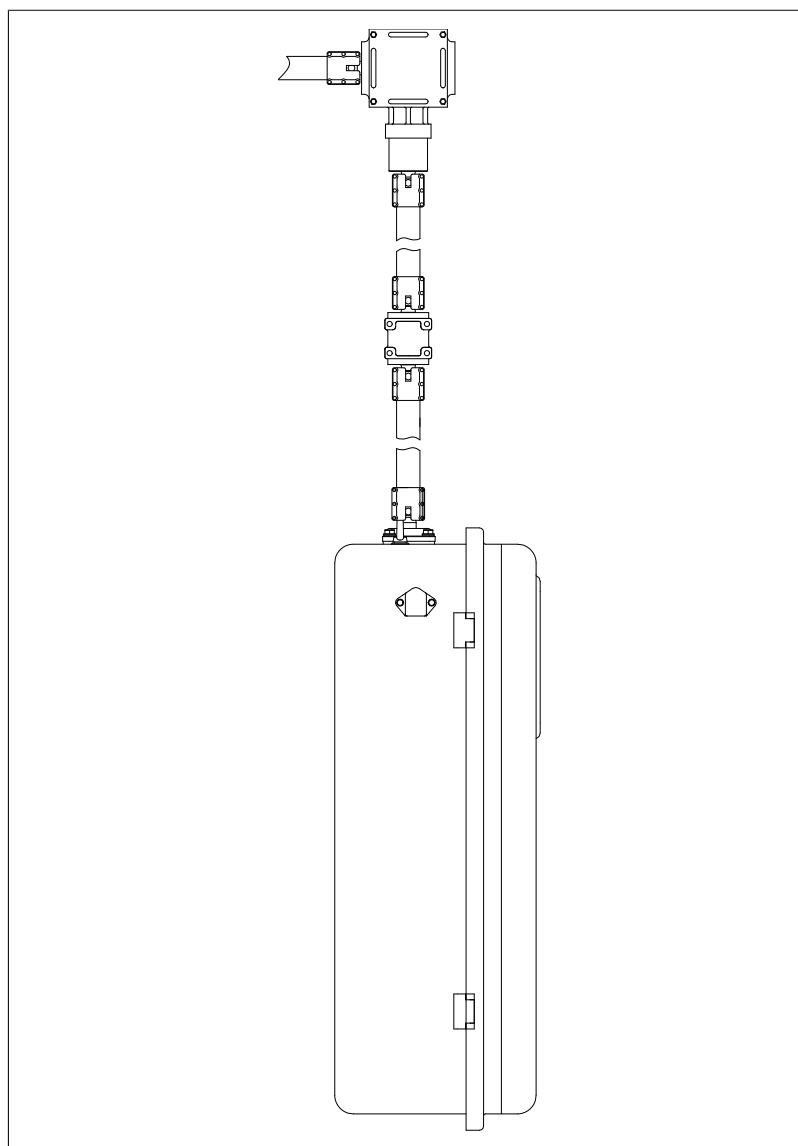


Figure 8-4 Align the shaft of the bevel gearbox and the shaft of the MDU

2. Secure the bevel gearbox on the transformer cover bracket with two bolts (18 mm diameter).

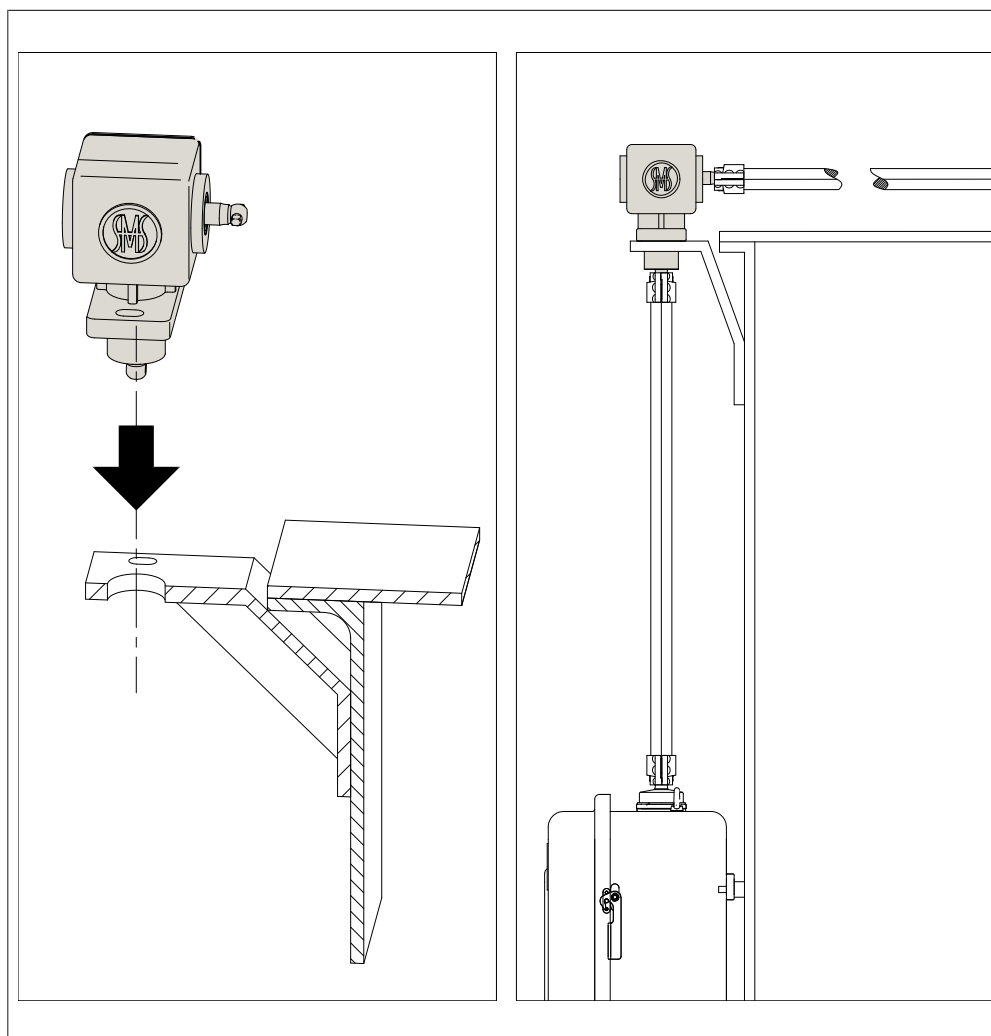


Figure 8-5 Install the bevel gearbox

8.4.4. Install the intermediate support gearbox

When the driving shaft length exceeds 2 m, it is recommended to install intermediate support components to improve drive stability.

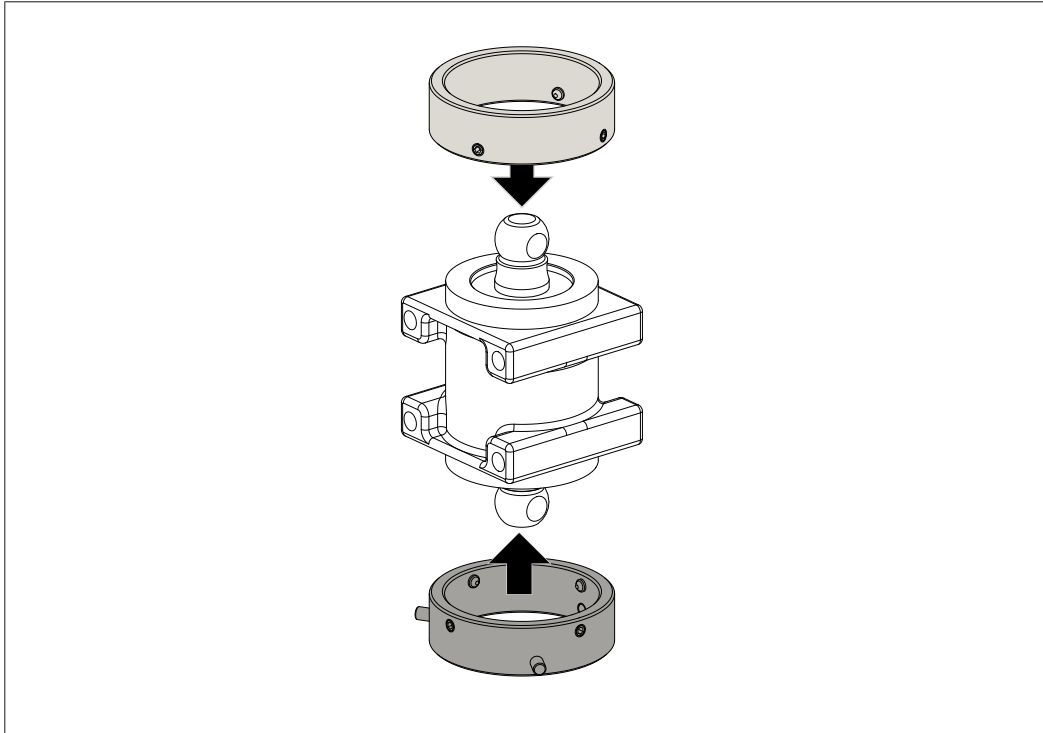


Figure 8-6 Install the vertical shaft intermediate support gearbox

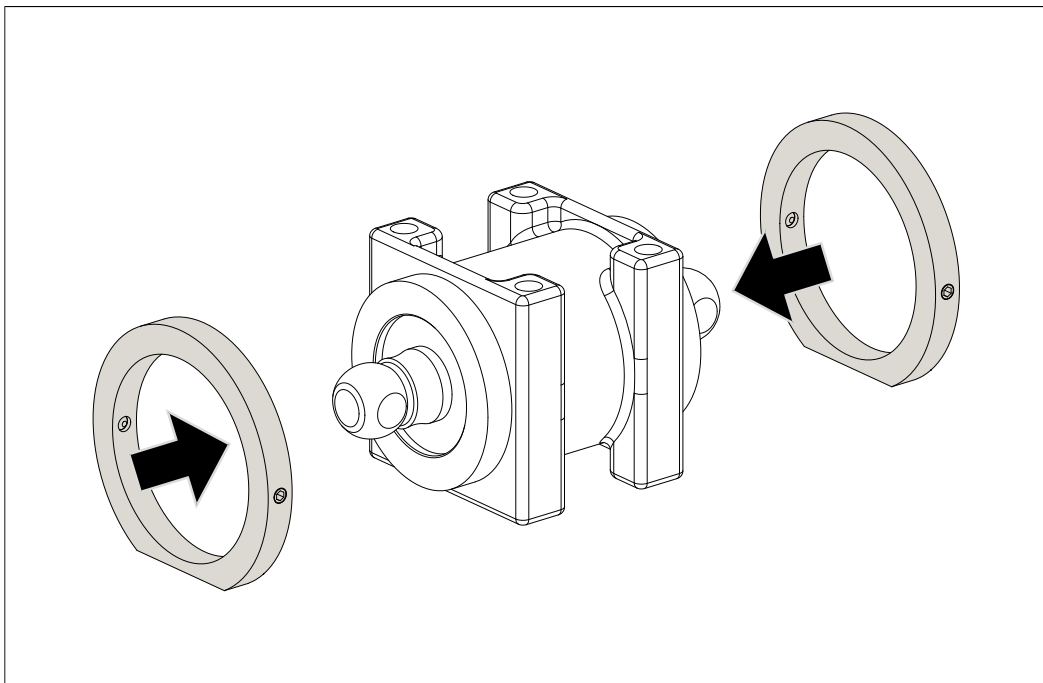


Figure 8-7 Install the horizontal shaft intermediate support gearbox

8.4.5. Install the horizontal driving shaft and vertical driving shaft

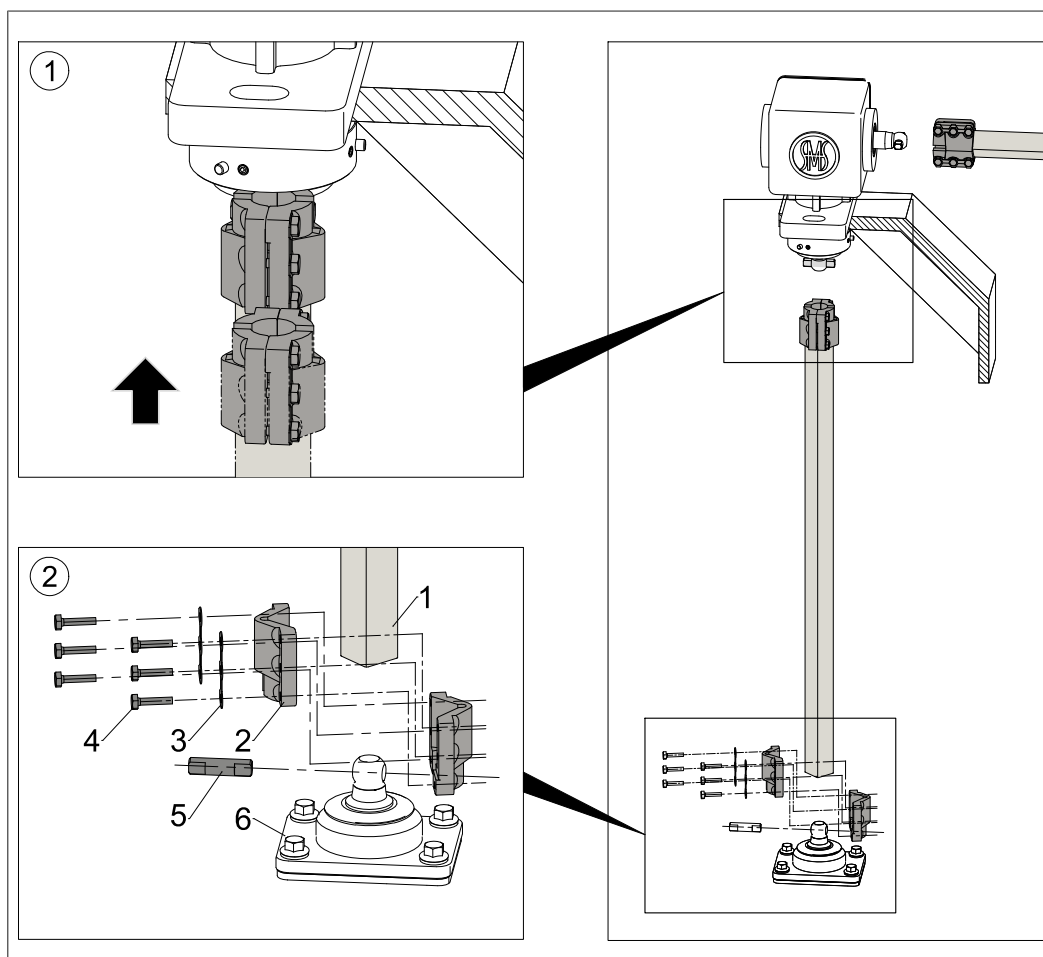
! Important:

- For a driving shaft longer than 2 m, install an intermediate support gearbox.
- To install the vertical guard, install it when installing the vertical driving shaft. See [8.4.6.0 Install the vertical guard](#) for details.

Connect the driving shaft, the MDU and the tap-changer.

The figure below shows the connection of the coupling between the bevel gearbox and the vertical driving shaft. The installation modes of the horizontal driving shaft coupling and vertical driving shaft coupling are the same.

Use a 10 mm open-end/socket wrench to install the M6 screws, torque: 7 to 9 Nm.



1. Driving shaft	2. Coupling	3. Lock washer
4. M6 screw	5. Cylindrical pin	6. Motor Drive Unit (MDU)

Figure 8-8 Install the bevel gearbox and the driving shaft

8.4.6. Install the driving shaft guards



Important:

For a driving shaft longer than 2 meters, install an intermediate support gearbox.

Install the horizontal guard

The length of the horizontal guard is determined by the transformer manufacturer.

1. After installing the driving shaft, place the associated clamps on both ends of the horizontal guard.
2. Position the guard on the mounting surfaces of the top gearbox and the bevel gearbox. Slide the clamps onto the mounting surfaces and tighten them.

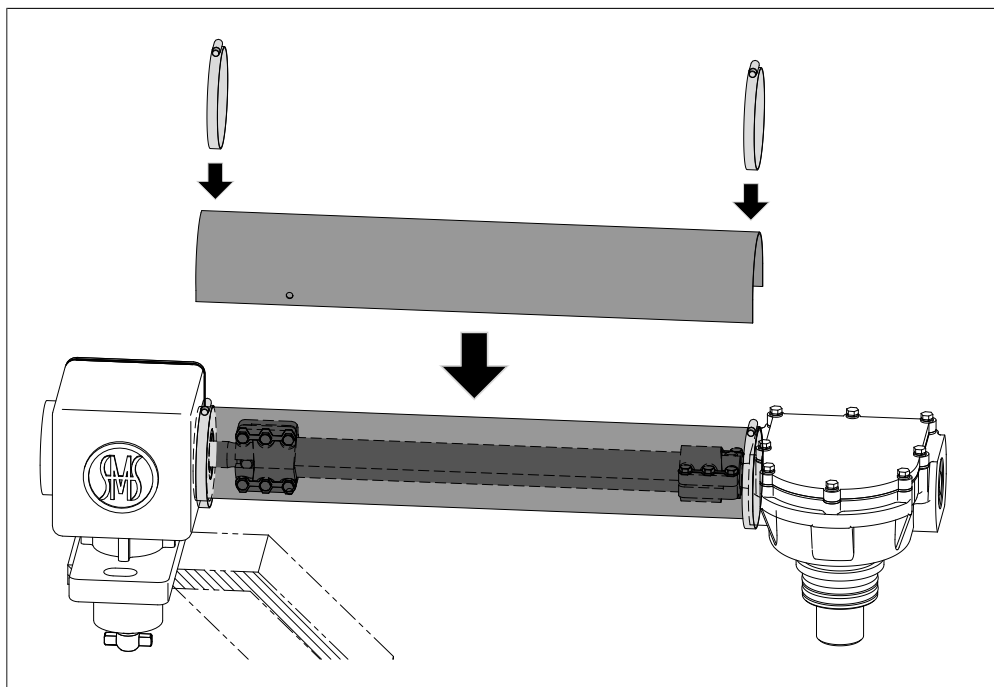


Figure 8-9 Install the horizontal guard

Install the vertical guard

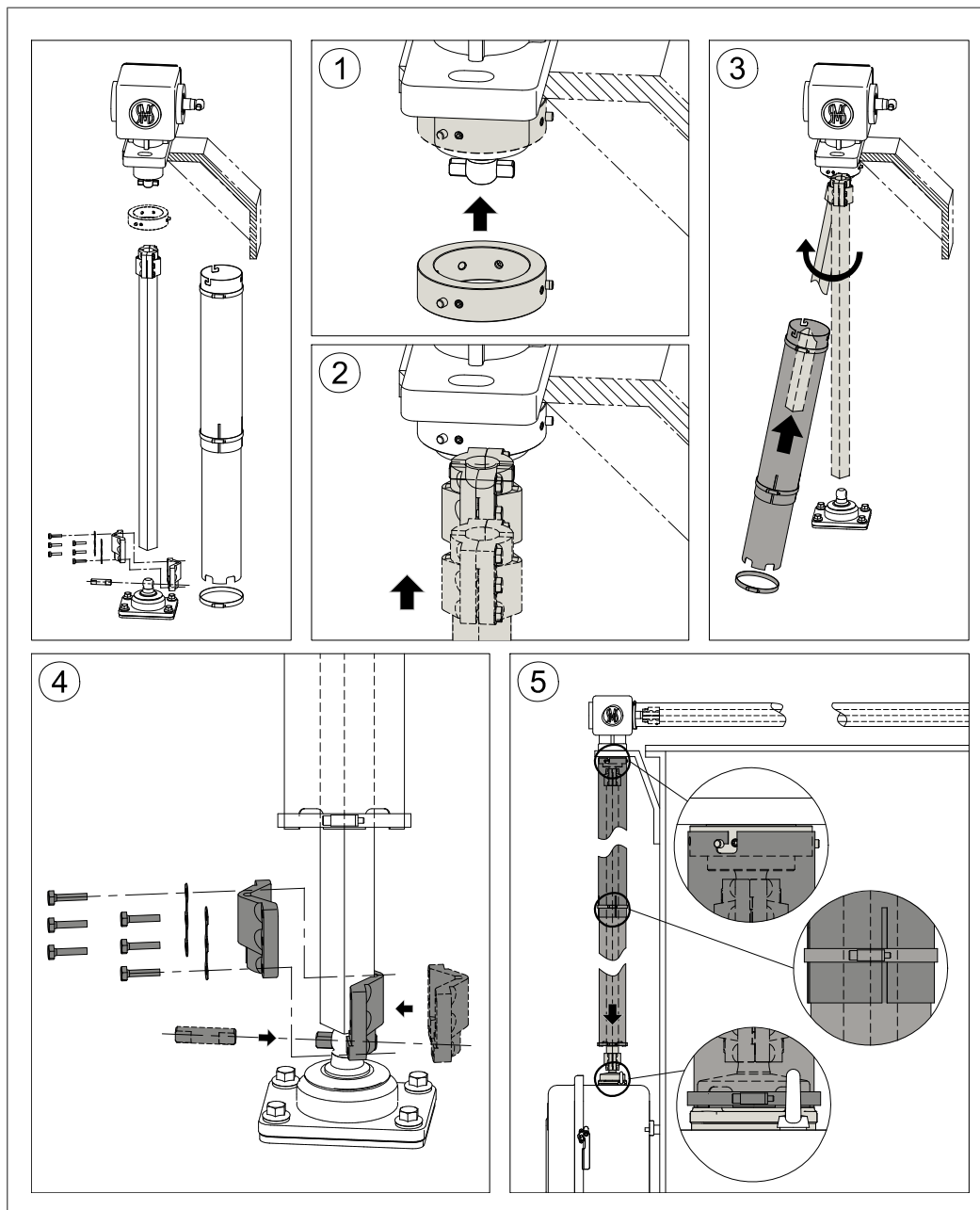


Figure 8-10 Installation steps for the vertical guard

To install the vertical guard, perform these steps:

1. Install the three 6 x 16 stainless steel cylindrical pins into the bushing (evenly spaced at 120° intervals). The pins should protrude approximately 5 mm beyond the outer diameter of the bushing and should not protrude into the inner diameter.
2. Use a 5 mm hex key wrench to tighten 4 sets of M6 screws to a torque of 7 to 9 Nm, and secure the bushing to the lower end of the bevel gearbox.

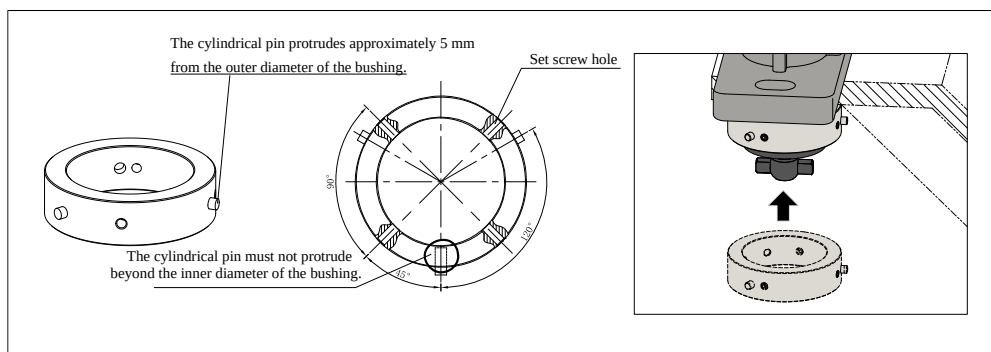


Figure 8-11 Install the bushing

3. Slide the vertical guard upward onto the lower end of the bevel gearbox, ensuring it sleeves over the bushing. Align the three grooves on the guard with the three cylindrical pins. Rotate the guard until the pins snap into the grooves (see [Figure 8-12](#)).
4. Slide the thinner lower guard onto the output shaft of the MDU (see [Figure 8-12](#)).

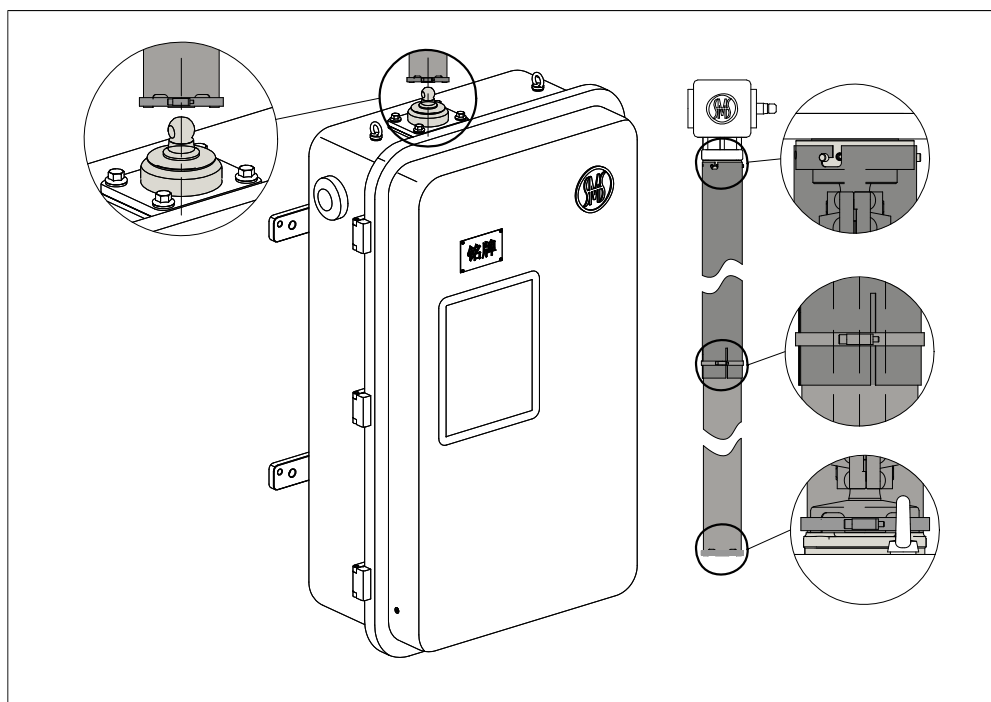


Figure 8-12 Install the vertical guard

5. After positioning both guards in place, secure them with two clamps (see [Figure 8-12](#)).
6. If the vertical driving shaft exceeds 2 m and an intermediate connecting shaft is used, repeat Step 1 to Step 5.

8.5. Installation Precautions for Three Single-phase OLTCs or I+II OLTC Configuration



Important:

Ensure all selector switches are set to the same tap position and operate synchronously.

When installing three single-phase OLTCs or an I+II OLTC configuration on the same transformer, adjusting the gearbox may cause the selector switches to operate. Therefore, after adjusting the gearbox, inspect all selector switches one by one. Ensure they are set to the same tap position and can operate simultaneously during manual operation.

8.6. Calibrate the OLTC Drive System

Before motorized operation of MDU, manually operate the OLTC through one complete cycle to verify whether it is aligned and functioning correctly. The time interval between the operation of the selector switch and the completion of the MDU movement must be the same in both directions of rotation.

To ensure reliable operation of the OLTC, the drive system must be calibrated after disconnecting and reconnecting the horizontal or vertical shaft.

8.6.1. Calibration procedure

Calibrate the drive system according to the following procedure (take the SHM-D calibration process as an example):

1. Use the handle to rotate the driving shaft in the direction of 1→N. When you hear the switching snap of the selector switch, continue rotating and count the number of handle rotations (m), until the center mark in the gray area on the MDU tap position indicator aligns with the arrow.
2. Rotate the handle back to the original set position in the opposite direction (N→1 direction). Count the number of handle turns (k) using the same method.
3. If $m = k$, the connection is correct. If $m \neq k$ and the absolute difference ($|m - k|$) is greater than 1, disconnect the vertical driving shaft and rotate the handle by $1/2 (|m - k|)$ turns in the direction that had more turns. After completing, reconnect the vertical driving shaft to the MDU.
4. Repeat the above steps until the number of turns in both directions is approximately equal.

8.6.2. Calibration example

Calibrate the selector switch for OLTC with model number 10193W:

1. Rotate from tap position 10 (set position) to position 11 ($m = 5$ turns).
2. Rotate back from tap position 11 to tap position 10 ($k = 3$ turns).
3. Calculate:
 - a. Turn difference: $|m - k| = |5 - 3| = 2$ turns
 - b. Adjustment turns: $1/2 (m - k) = 1/2 (5 - 3) = 1$ turn
4. Disconnect the vertical driving shaft from the MDU, rotate the handle one turn from position 10 toward position 11, then reconnect the shaft.
5. Verify whether the turn difference between the two directions is balanced.

Calibration summary:

1. Record the number of turns in both directions (m and k).
2. After disconnecting the coupling, rotate the handle by $1/2 (|m - k|)$ turns in the direction that had more turns.
3. Reconnect and verify until $|m - k| < 1$.

8 - Installation of protective devices and drive unit

9. Factory Acceptance Tests (FAT)

9.1. Test preparations

9.1.1. Degas the OLTC

Before the first operation, it is necessary to exhaust air from the OLTC head and the oil suction pipe connected to the pipe connector S.

The OLTC exhaust steps are as follows:

1. OLTC head exhaust:
 - a. Remove the bolt cap of bleeder valve E1.
 - b. Gently pry the valve stem with a screwdriver to exhaust until the oil overflows (see [Figure 9-1](#)).
 - c. Retighten the bolt cap of bleeder valve E1.

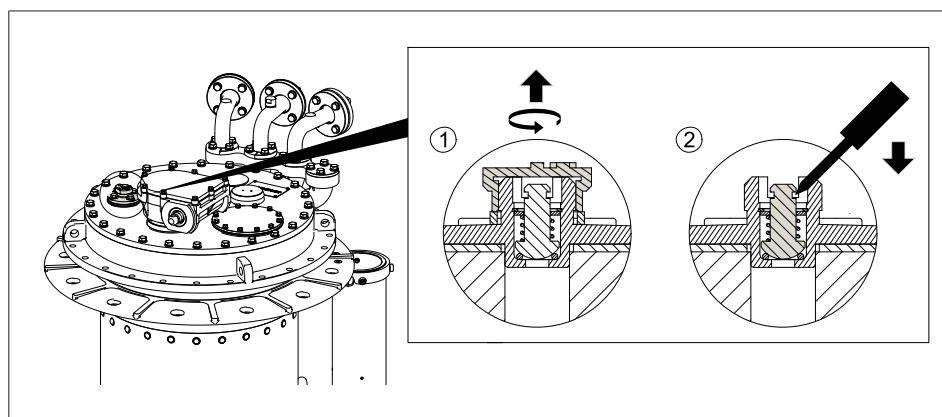


Figure 9-1 Bleeder valve exhaust operation

2. Oil suction pipe exhaust at pipe connector S:
 - a. Use a 24 mm open-end wrench to remove the bolt cap (see [Figure 9-2](#)).
 - b. Use a 10 mm open-end/socket wrench to loosen the M6 air bleed plug and perform the air bleeding operation.
 - c. Use a 10 mm open-end/socket wrench to tighten the M6 air bleed plug. Torque: 7 to 9 Nm.

d. Use a 24 mm open-end wrench to tighten the bolt cap.

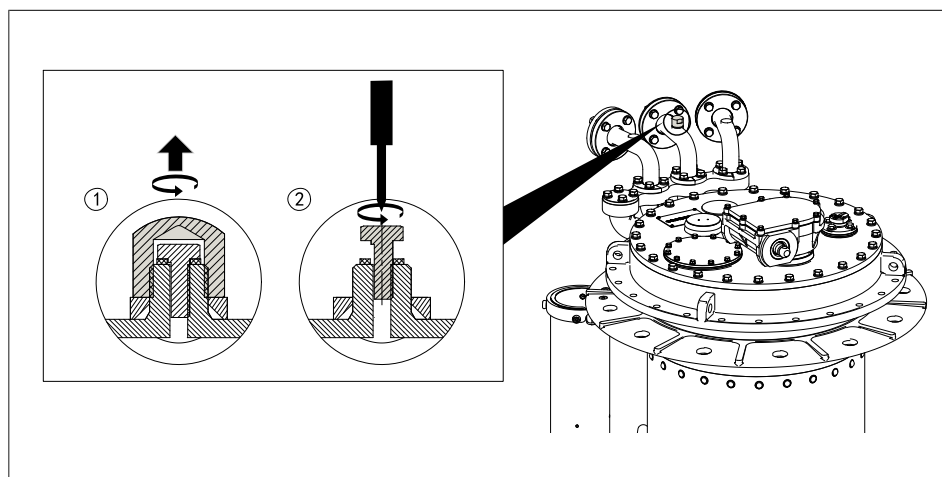


Figure 9-2 Exhaust at pipe connector S

9.1.2. Grounding

The grounding steps are as follows:

1. Use a 19 mm open-end/socket wrench to install the M12 grounding bolt, with a torque of 50 to 60 Nm, and connect the grounding bolt on the OLTC head to the transformer tank cover.

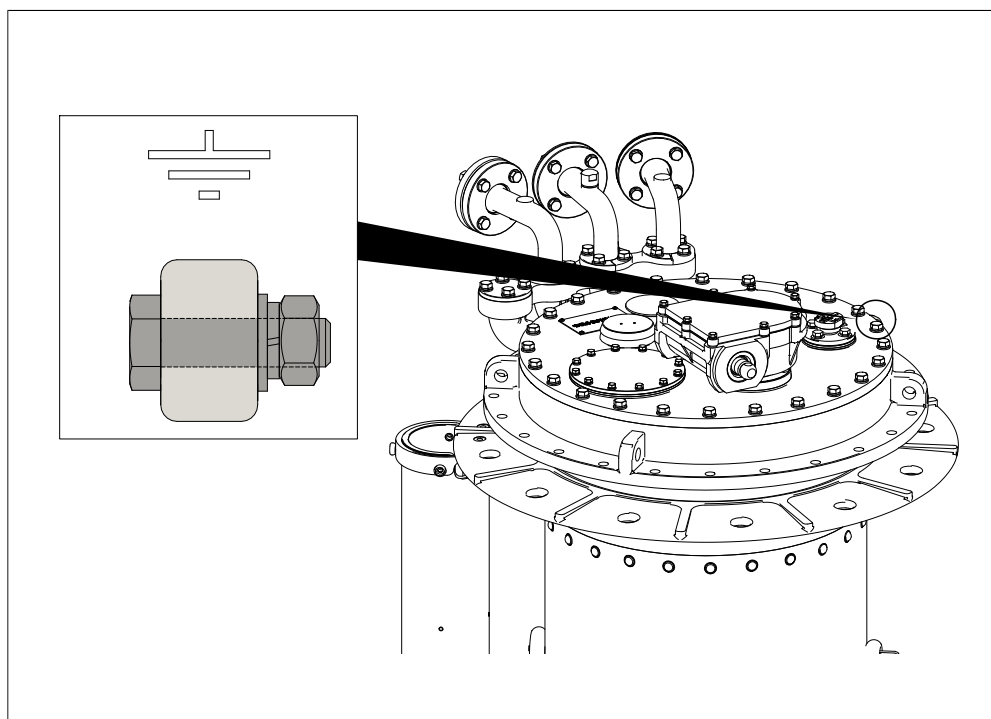


Figure 9-3 Grounding wire connection

2. Use a 19 mm open-end/socket wrench to install the M12 grounding bolt, with a torque of 50 to 60 Nm, and connect the M12 grounding bolt on the MDU housing to the transformer tank.

9.2. Tests at the transformer manufacturer's facility

9.2.1. Functional test

NOTICE

Risk of OLTC damage and fault!

Oil-free operation, incomplete OLTC operating cycles, or incorrect connections may cause damage to OLTC components or operational faults, thereby affecting normal equipment operation.

- ✓ The tap selector must be completely immersed in oil, and the selector switch oil compartment must be completely filled with oil. Never operate the OLTC in an oil-free condition.
- ✓ During the test, the OLTC must perform a full-range operation (one complete cycle).
- ✓ Ensure that the connection between the OLTC and the MDU is correct.



Important: Ensure that the tap position indicators of the MDU and the OLTC (observed through the inspection window on the top cover) display the same tap position.

Before powering on the transformer, perform a functional test to check the mechanical operation of the OLTC and the MDU.

9.2.2. Transformer electrical tests

WARNING



Flammable and explosive gases!

Explosive gases can accumulate inside the OLTC, connection system, conservator, and air dehydrator outlet. Ejected components and hot oil spills can cause serious or fatal injuries.

- ✓ Ensure the environment around the transformer is free of open flames, hot surfaces, sparks (including static electricity), or any potential ignition sources.
- ✓ All OLTC tests must be performed by qualified personnel following established procedures.

10. Transformer transportation and on-site commissioning

10.1. Remove the MDU for transportation

When transporting the transformer from the manufacturer to the site, if the MDU must be removed because the overall dimensions exceed transportation limits, store and secure the tap-changer and the MDU in their as-delivered condition.



Important:

Do not operate the MDU when disconnected from the OLTC.

To remove the MDU for transportation, perform these steps:

1. Set the MDU to its set position.
2. Disconnect the coupling.
3. Remove the MDU.

After transporting the MDU to the site, reinstall the MDU and components of the drive mechanism. For MDU and driving shaft reinstallation, see the Huaming MDU operating manual.

10.2. Transport with oil and no conservator



Important: If the transformer is stored and transported in an oil-filled condition without a conservator for 3-4 weeks, the oil volume inside the OLTC must be reduced by approximately 5 litres.

If the transformer is stored and transported in an oil-filled condition without an installed conservator, connect a bypass pipe between the selector switch oil compartment and the transformer tank to balance pressure generated by oil expansion. This bypass pipe should be connected between the pipe connector E2 on the OLTC top and R.

10.3. Transport without oil

NOTICE

Risk of moisture damage to components!

Moisture may cause component damage, performance degradation, and affect the normal operation of the OLTC.

- ✓ During long-term storage, periodically energize the MDU heater to prevent moisture accumulation.

If the transformer is transported or stored without being filled with oil, completely drain all oil from the selector switch oil compartment. It is recommended to protect the interior of the oil compartment using appropriate methods (such as filling with dry air or inert gas) in accordance with local regulations and actual conditions.

10.4. On-site commissioning

⚠ WARNING



Flammable and explosive gas!

Explosive gases may accumulate under the OLTC top cover, inside the connecting system, within the conservator, and at the air dryer outlet. Ejection of components and splashing of hot oil can cause serious or fatal injury.

- ✓ Ensure that the operating area is free of open flames, high-temperature surfaces, and potential sources of ignition, including static electricity.
- ✓ All OLTC tests must be performed by qualified personnel in accordance with established procedures.

NOTICE

Risk of OLTC damage!

Oil-free operation may cause OLTC damage.

- ✓ Before energizing the transformer, ensure that the tap selector is completely immersed in transformer oil and that the selector switch oil compartment is filled with oil.

To perform on-site commissioning, perform these steps before energizing the transformer:

1. Before initial powering on, ensure that the transformer oil meets the following requirements: breakdown voltage (Ud) > 40 kV/2.5 mm (minimum requirement), water content < 30 µL/L (maximum requirement).
2. Discharge the OLTC in accordance with [9.1.1 Degas the OLTC](#).
3. Perform the functional test in accordance with [9.2.1 Functional test](#).
4. Connect the minimum oil level signal contact of the OLTC conservator to the circuit breaker trip circuit.
5. Check whether the protective relay is functioning properly according to the manual of QJ-25 series or equivalent protective relay products.
6. Before powering on the transformer, remove the red protective strip from the pressure relief valve (if any).

**Important:**

Before performing a tap-change operation, ensure that the magnetising inrush current transient has completely decayed. Magnetising inrush current may reach several times the normal operating current and may overload the OLTC during switching. After powering on the transformer, wait at least 30 seconds for the current to stabilise before operating the OLTC.

All safety precautions and instructions must be observed during commissioning and testing. Refer to the relevant sections of this manual for specific safety regulations.

After energizing the transformer and allowing the inrush current to subside, tap-changing operations can be performed under no-load or on-load conditions.

11. OLTC operation monitoring

Visual inspections of the OLTC should be carried out regularly in conjunction with transformer checks to ensure proper operation.

Voltage regulation mode	Breakdown voltage	Water content
Neutral point voltage regulation	$\geq 30 \text{ kV}/2.5 \text{ mm}$	$< 40 \text{ }\mu\text{L/L}$
Other voltage regulation modes	$\geq 40 \text{ kV}/2.5 \text{ mm}$	$< 30 \text{ }\mu\text{L/L}$

Avoid frequent OLTC operations when the transformer is overloaded. Install an "overcurrent self-resetting contact" to prevent tap-changes when the load current exceeds twice the rated current ($2 \cdot I_n$).

Set the protective relay trip contact to operate at an oil flow velocity of $1 \text{ m/s} \pm 10\%$. Connect this contact to the transformer's circuit breaker trip circuit. When a fault occurs inside the OLTC, gas is generated, increasing the oil flow velocity and triggering the relay to trip the circuit breaker, isolating the transformer. After the protective relay trips, do not manually close the circuit breaker before inspecting the OLTC.

A rupture disk (or pressure relief valve) is mounted on the OLTC top cover. This device does not operate during normal tap-change operations. It only ruptures when an internal failure of the selector switch causes the pressure in the oil compartment to exceed $0.3 \pm 20\% \text{ MPa}$, which prevents further damage to the OLTC.

When installing and maintaining the OLTC, avoid stepping on or striking the pressure relief device.

12. Inspection, maintenance, and disposal

 DANGER	
	Electric shock hazard! Working on an energized transformer can cause death or serious injury. ✓ De-energize and disconnect the transformer's high-voltage and low-voltage terminals. ✓ Lock out the transformer to prevent accidental re-energization. ✓ Ensure that all components are de-energized. ✓ Ground all transformer terminals (including ground wires and grounding disconnect switches) and ensure they are short-circuited. ✓ Cover or isolate adjacent energized components.

 DANGER	
	Electric shock hazard! Working on an energized OLTC can cause death or serious injury. ✓ Do not operate or work on an energized OLTC. ✓ Disconnect all auxiliary circuits, including tap-changer control, pressure relief valve, and pressure monitoring devices. ✓ Verify that all components are de-energized.

 DANGER	
	Explosion hazard! Explosive gases in the OLTC oil compartment, transformer, piping, conservator, and dehydrating breather outlet can ignite or explode, causing death or severe injury. ✓ Ensure that the area around the transformer is free of open flames, hot surfaces, sparks (including static discharge), and any other potential ignition sources. ✓ Do not operate any electrical equipment that could generate sparks (e.g., impact wrenches). ✓ Use only conductive and grounded hoses, piping, and pumping equipment suitable for flammable liquids.

NOTICE	
Damage to the MDU! Condensation in the MDU can cause damage. ✓ Always maintain the MDU seal. ✓ If operation is interrupted for more than two weeks, connect and operate the anti-condensation heater in the MDU. If this is not possible, place sufficient desiccant inside the MDU enclosure.	

**Important:**

For troubleshooting and repairing the OLTC, contact Huaming after-sales service.

12.1. Periodic maintenance inspection

Follow the inspection guidelines below.

Table 12-1 Inspection guidelines

Interval	Inspection item	Procedure
Yearly	Door seals, cable entries, MDU enclosure ventilation, and control panel ventilation	• Visually inspect for aging, damage, or oil leaks. Replace any damaged or deteriorated components. • Perform a ventilation test on the MDU and control panel.
Yearly	OLTC top cover seal, protective relay seal, and all piping connections	• Visually inspect for aging, damage, or oil leaks. Check gasket elasticity using an insulated probe. • Replace any damaged or deteriorated seals.
Yearly	MDU enclosure and control panel heaters	• Visually inspect circuit components for discoloration, damage, or broken pins. If damage is found, contact Huaming After-Sales Service. • Perform a temperature test using a thermometer and verify that the heaters reach their specified operating temperature.
Yearly	Silica gel desiccant in the conservator breather	Place a humidity indicator card inside the conservator for at least 30 minutes. If the humidity exceeds 10%, replace the desiccant; alternatively, as a preventive maintenance measure, replace it once every year.
Synchronized with the transformer maintenance cycle	Protective relay function	Visually inspect for discoloration, damage, or broken pins. Contact Huaming After-Sales Service if damage is found.
Every 5 years	Oil sample testing for Class I OLTC	Drain the sampling pipeline (depending on the length and diameter of the pipeline, drain 1 to 5 liters of oil), then extract a 0.5-liter oil sample from the bottom of the OLTC. If the sample does not meet the parameters in Table 12-2 , replace or recondition the transformer oil.

Table 12-1 Inspection guidelines (continued)

Interval	Inspection item	Procedure
Every 2 years	Oil sample testing for Class II OLTC	Drain 5 liters of oil from the system before taking a sample. Collect a 50-250 ml oil sample from the bottom using a syringe. If the sample does not meet the parameters in Table 12-2 , replace or recondition the OLTC oil.



Note: The above maintenance intervals are based on IEC standards. You may adopt more stringent inspection intervals according to your actual conditions and local regulations.

Table 12-2 Oil sample limits

Parameter		Dielectric strength	Water content
Mineral oil (IEC 60296)			
Initial transformer commissioning		>60 kV/2.5 mm	<15 µL/L
During operation	Class I OLTC	>30 kV/2.5 mm	<40 µL/L
	Class II OLTC	>40 kV/2.5 mm	<30 µL/L
After maintenance		>50 kV/2.5 mm	<15 µL/L
Natural ester oil (IEC 62770), only applicable for vacuum tap-changer			
Initial transformer commissioning		>60 kV/2.5 mm	≤100 µL/L
During operation	Class I OLTC	>30 kV/2.5 mm	≤200 µL/L
	Class II OLTC	>40 kV/2.5 mm	≤200 µL/L
After maintenance		>50 kV/2.5 mm	≤100 µL/L
Synthetic ester oil (IEC 61099), only applicable for vacuum tap-changer			
Initial transformer commissioning		>60 kV/2.5 mm	≤100 µL/L
During operation	Class I OLTC	>30 kV/2.5 mm	≤400 µL/L
	Class II OLTC	>40 kV/2.5 mm	≤400 µL/L
After maintenance		>50 kV/2.5 mm	≤150 µL/L



Note: Class I OLTCs are only suitable for use at the neutral point of windings, while Class II OLTCs suitable for use at any position in the windings (according to IEC 60214-1 classification).

12.2. Maintenance intervals

Table 12-3 Maintenance interval for the tap-changer

Interval	Task
Tap-changer operates 300,000 times (according to the motor drive unit counter)	Check and maintain the tap-changer. Please contact the after-sales service department of Huaming.
Tap-changer operates 600,000 times (according to the motor drive unit counter)	Check the vacuum interrupter chamber. Please contact the after-sales service department of Huaming.
Tap-changer operates 800,000 times (according to the motor drive unit counter)	Replace the selector switch insert. Please contact the after-sales service department of Huaming.

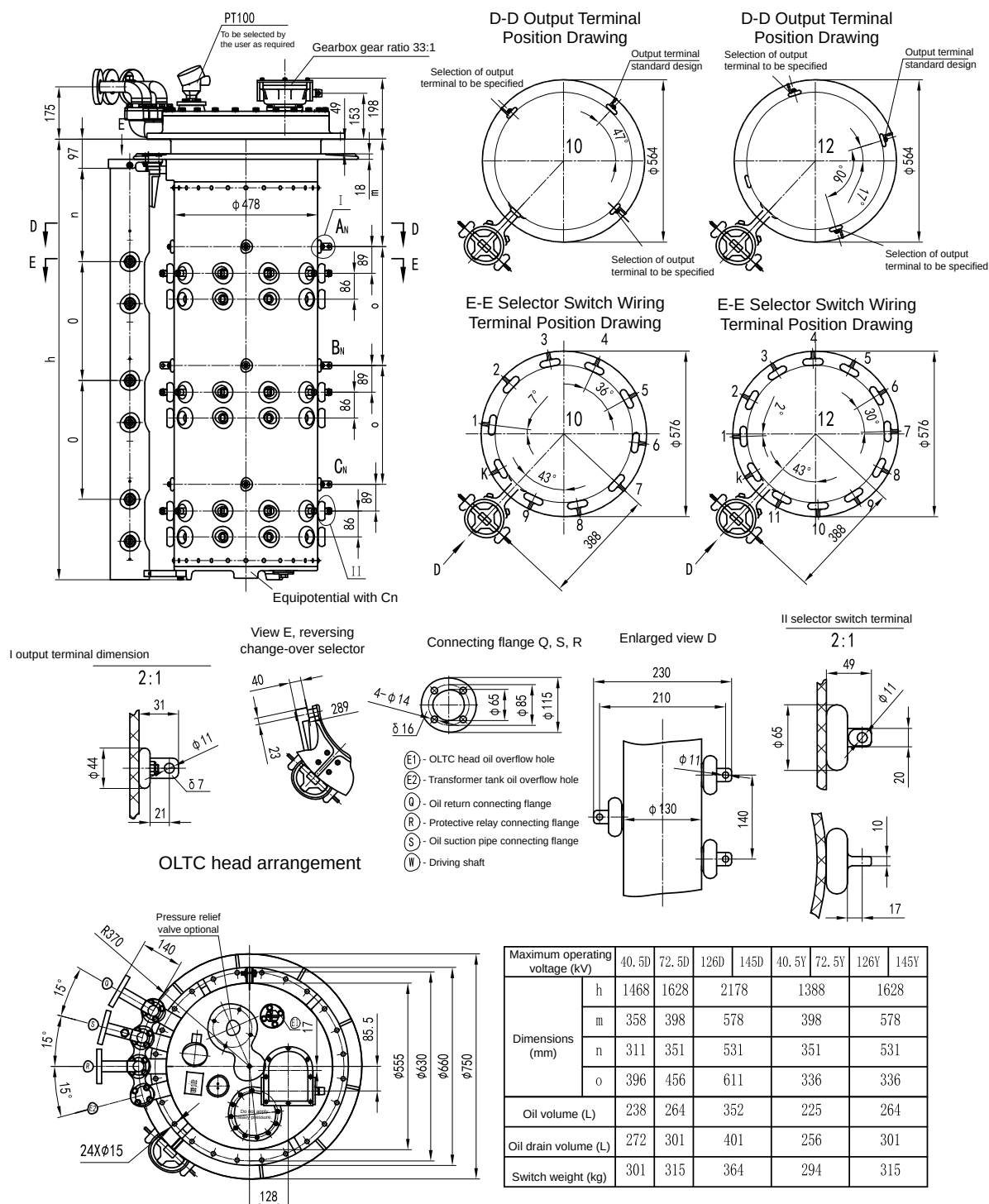
12.3. Disposal

Disposal must obey the applicable national regulations of the country of use. If you have any questions regarding removal or disposal, contact the after-sales service department of Huaming.

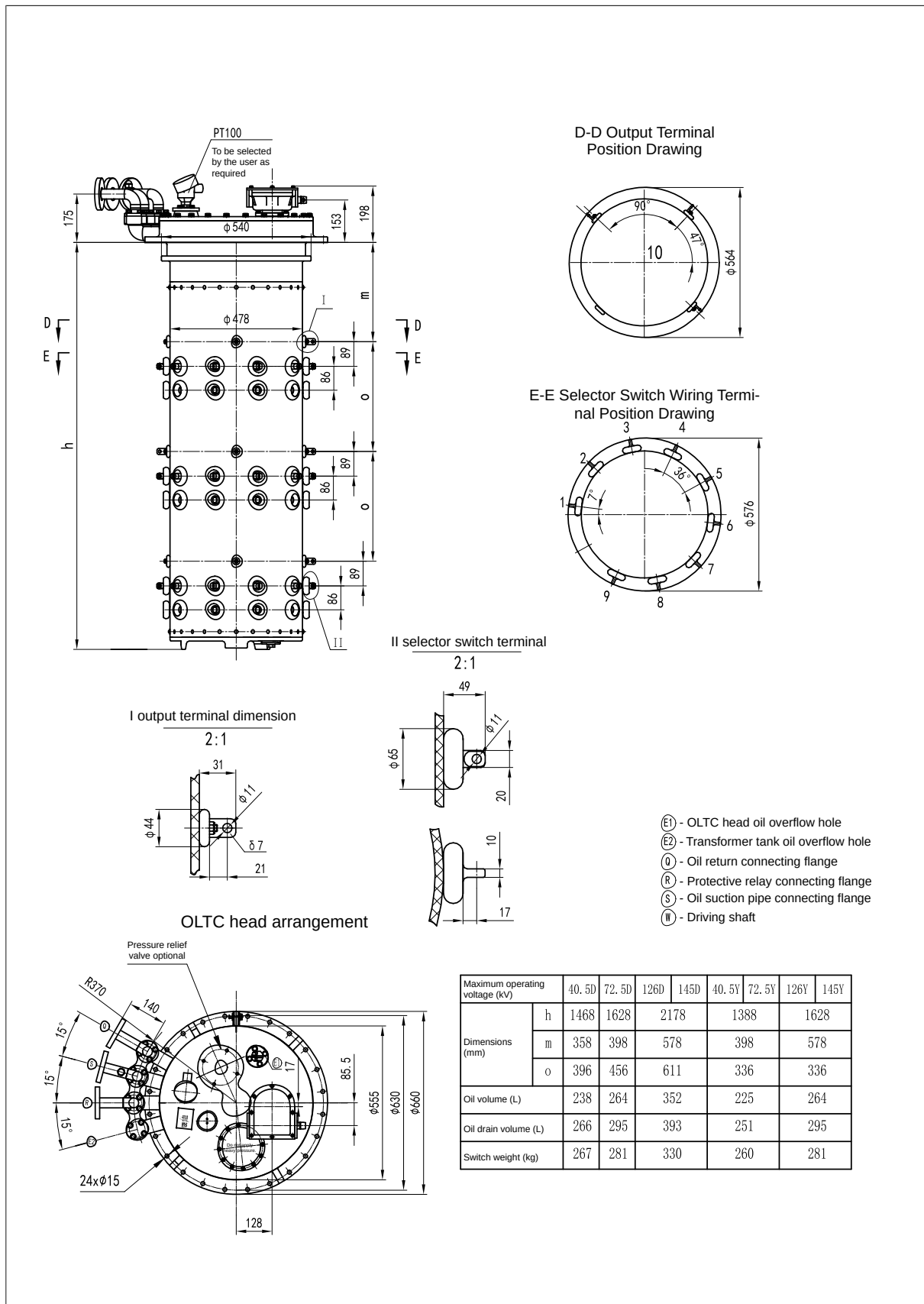
13. Drawings

This chapter contains the main dimensional drawings and partial electrical diagrams for the CV2 type oil-immersed vacuum On-Load Tap-Changer (OLTC).

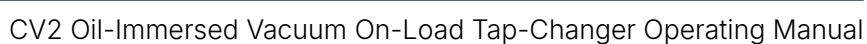
13 - Drawings



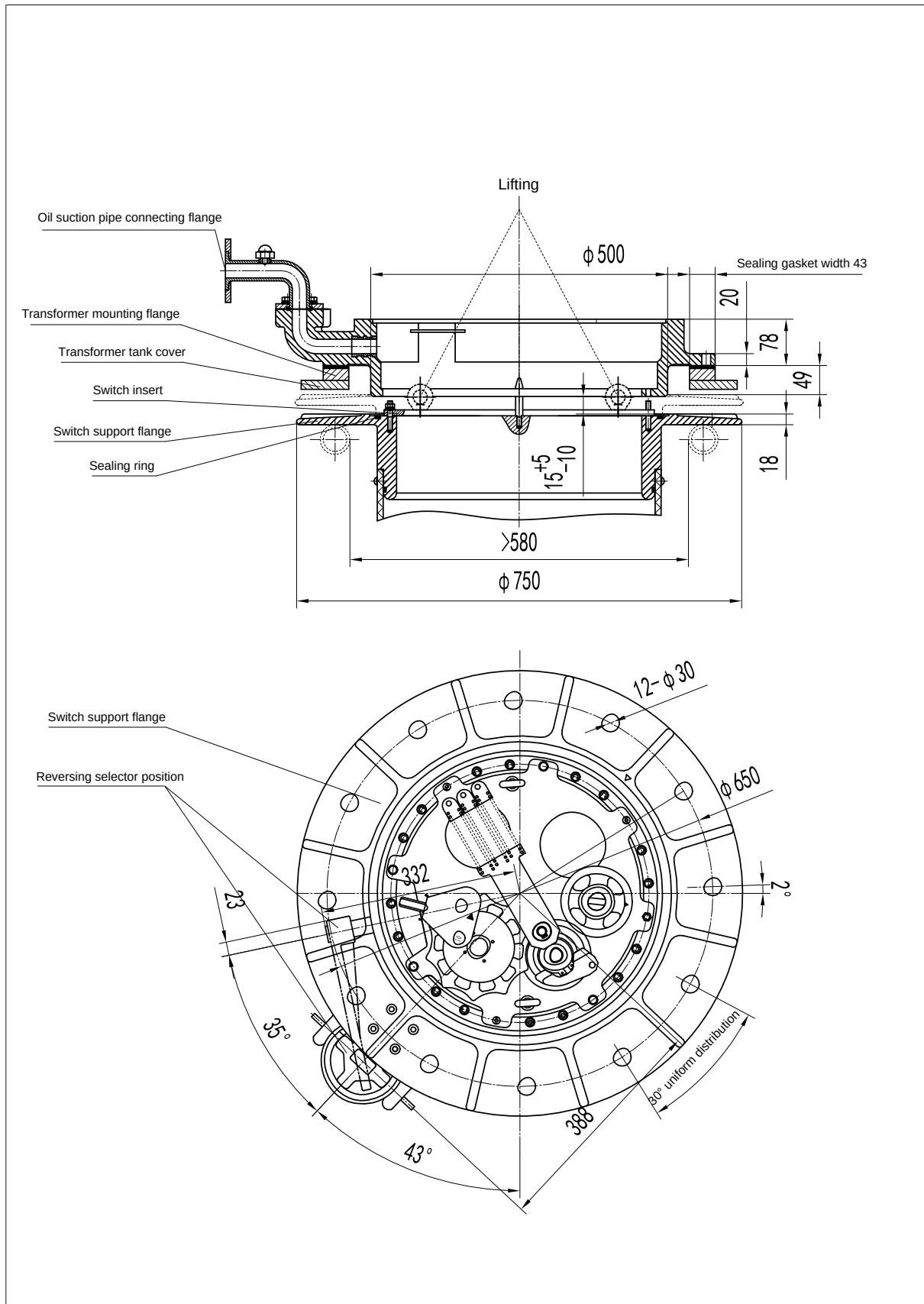
13.2. Overall dimensions of CV2 type OLTC without change-over selector



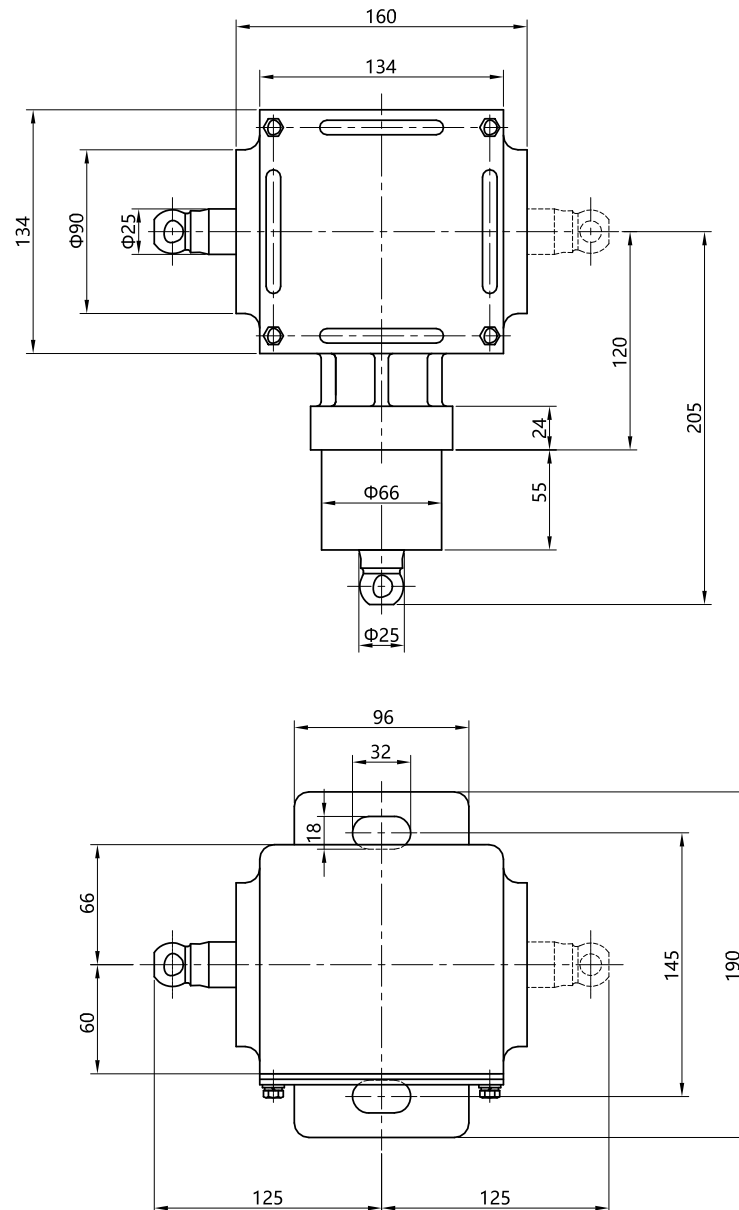
13 - Drawings



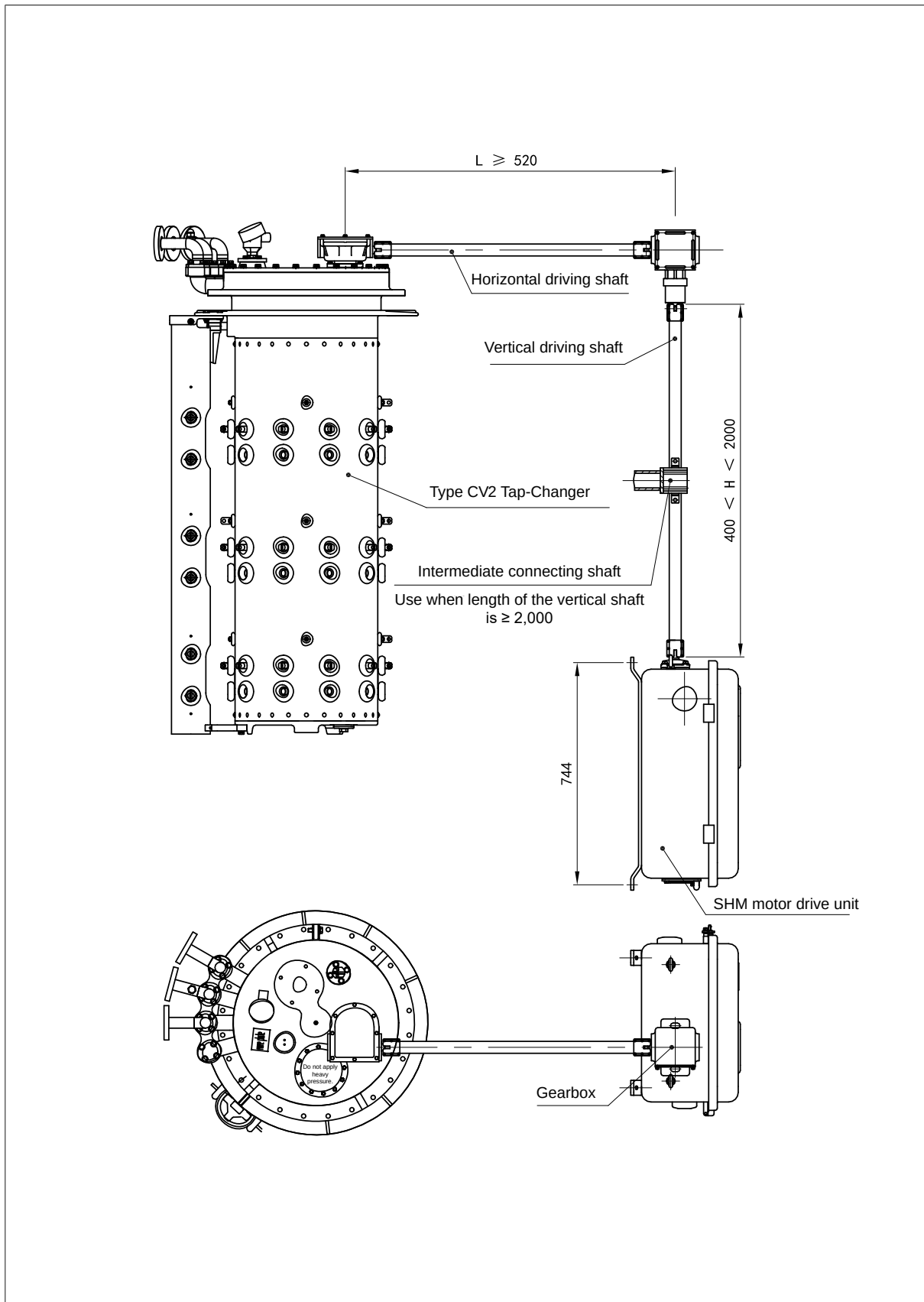
13.4. Bell-type installation diagram for CV2 type OLTC



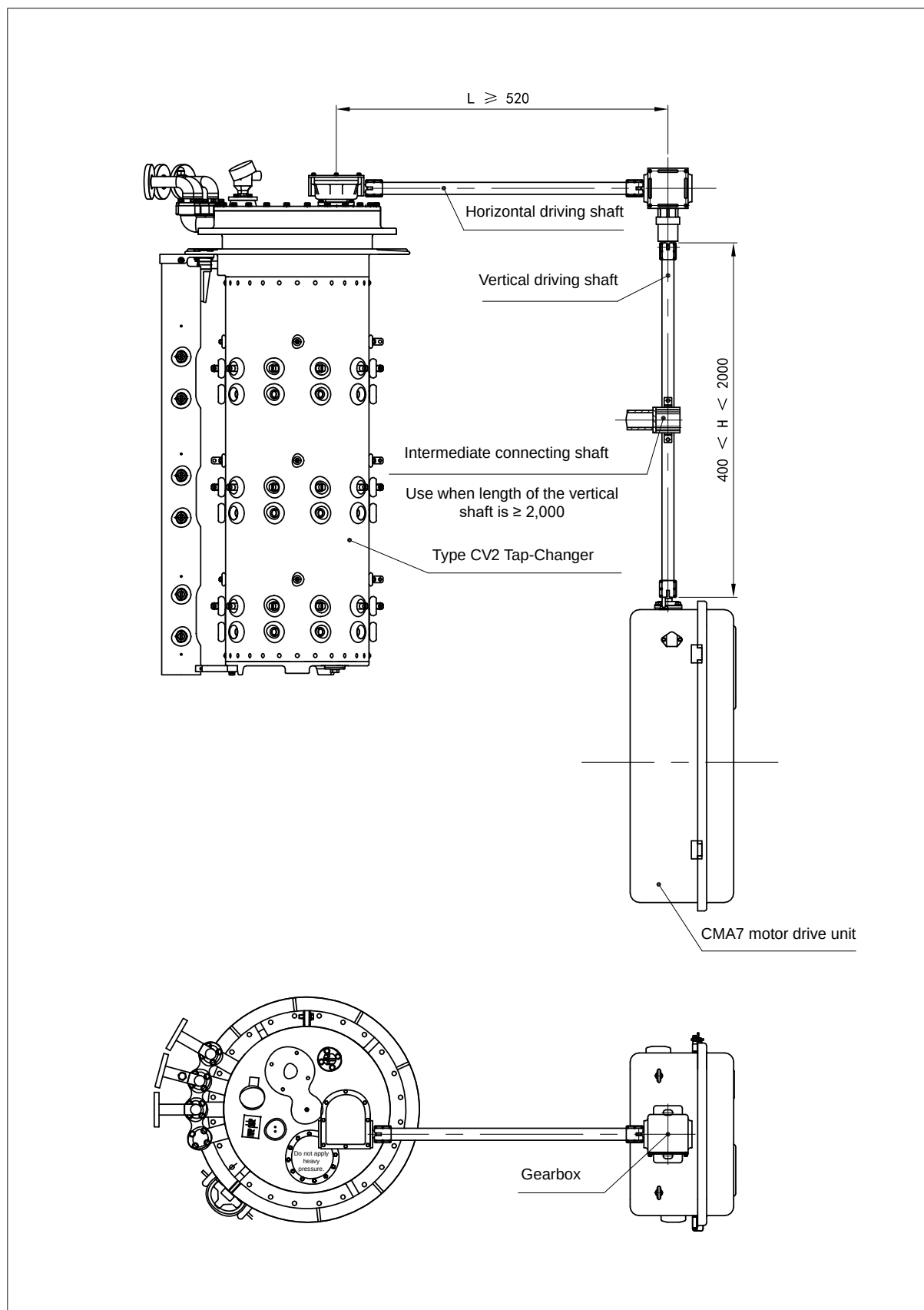
13.5. Dimensional drawing of bevel gearbox



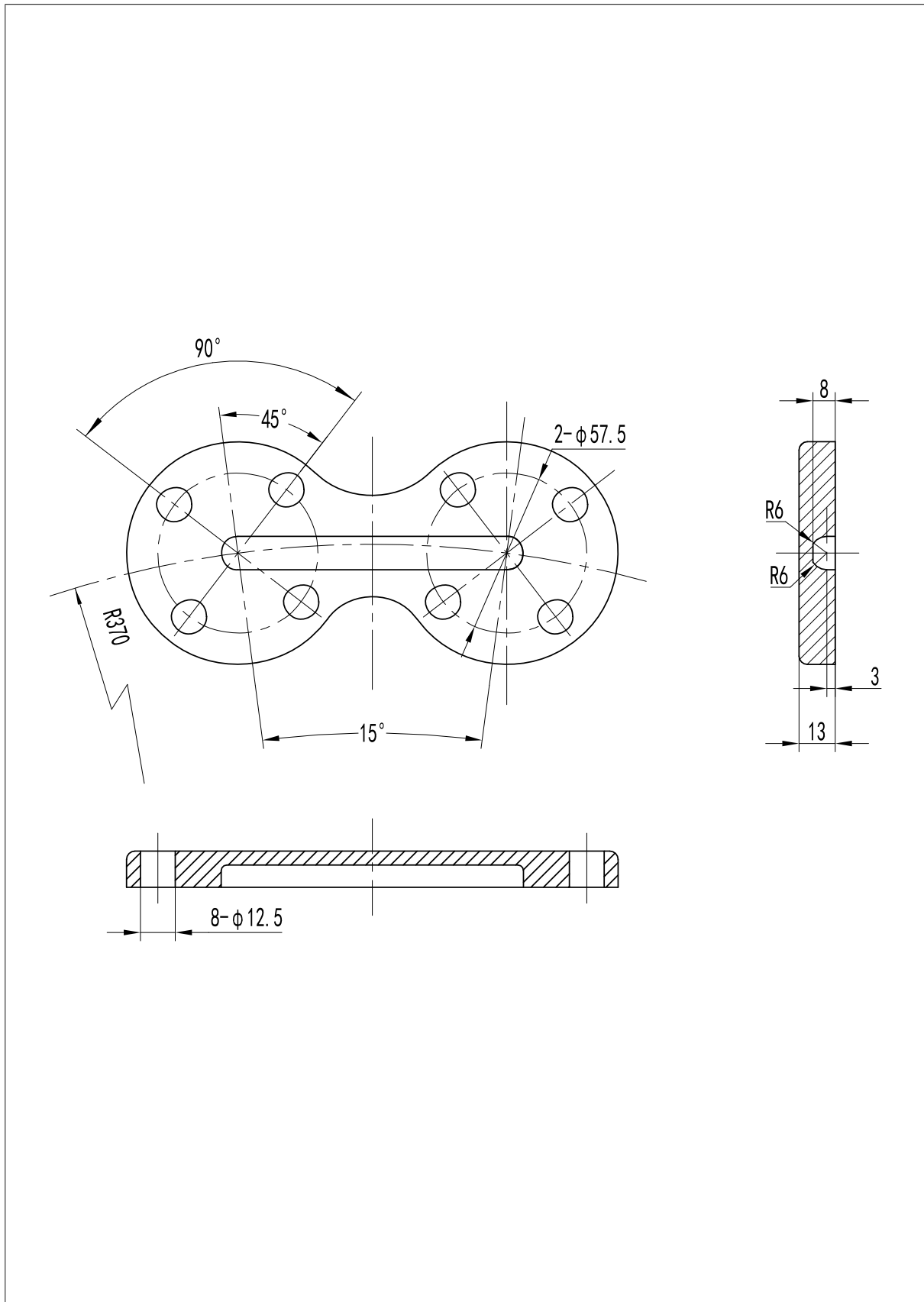
13.6. Connection layout drawing of CV2 type OLTC and SHM series MDU



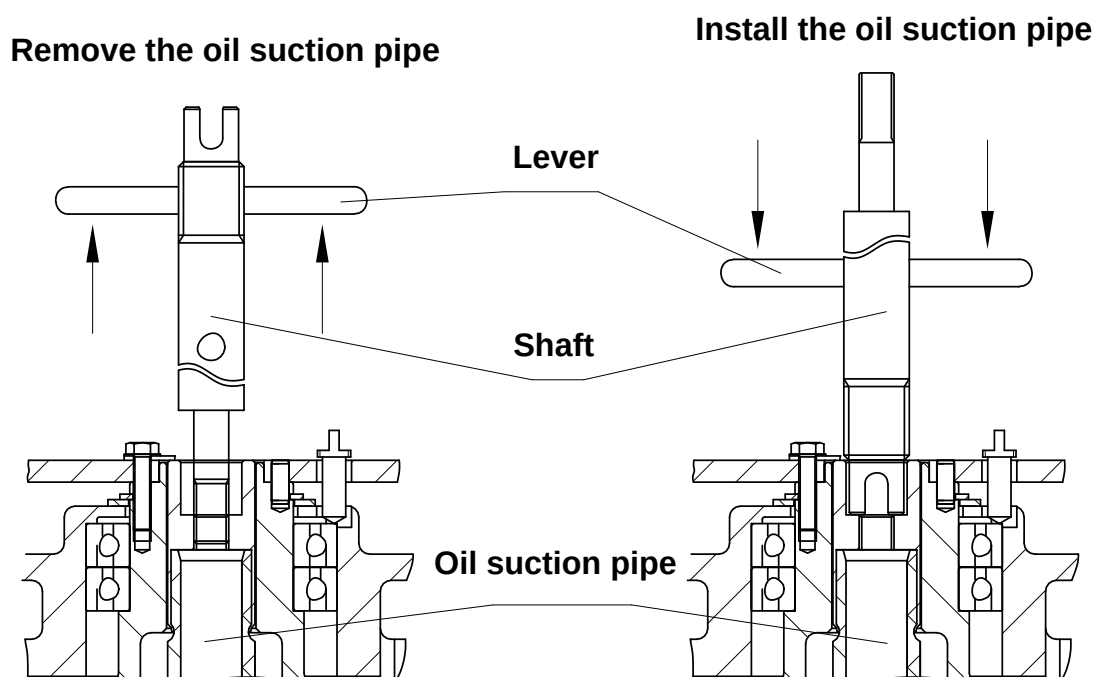
13.7. Connection layout drawing of CV2 type OLTC and CMA7 MDU



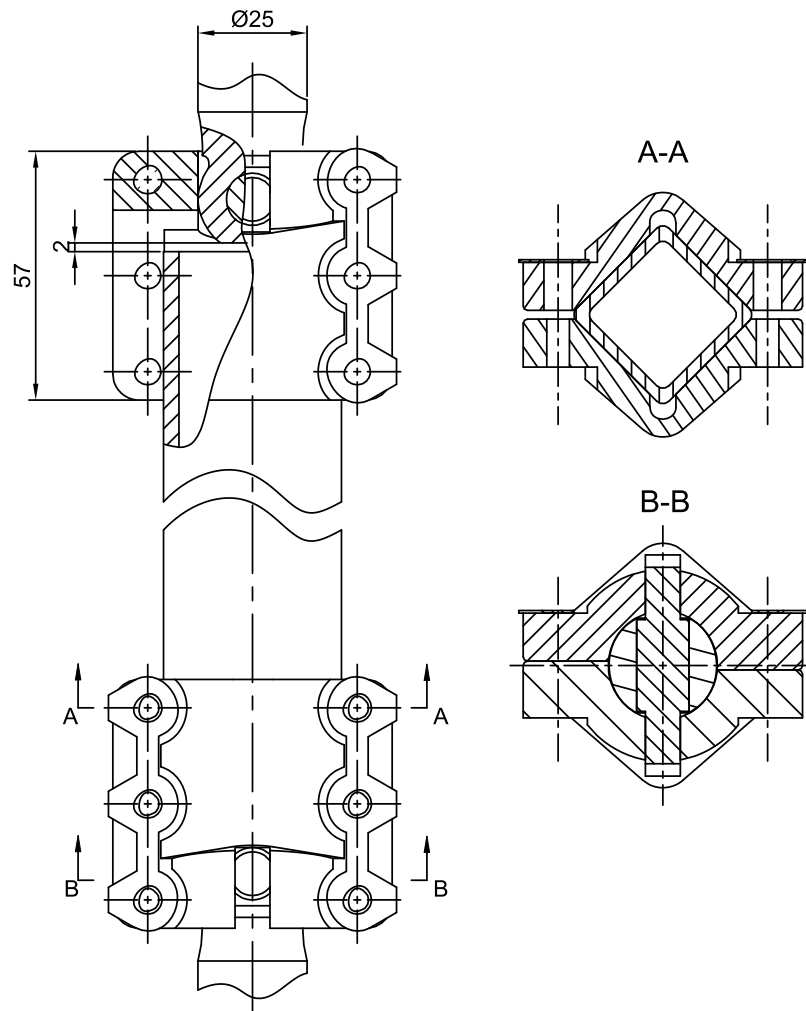
13.8. OLTC bypass pipe structure drawing



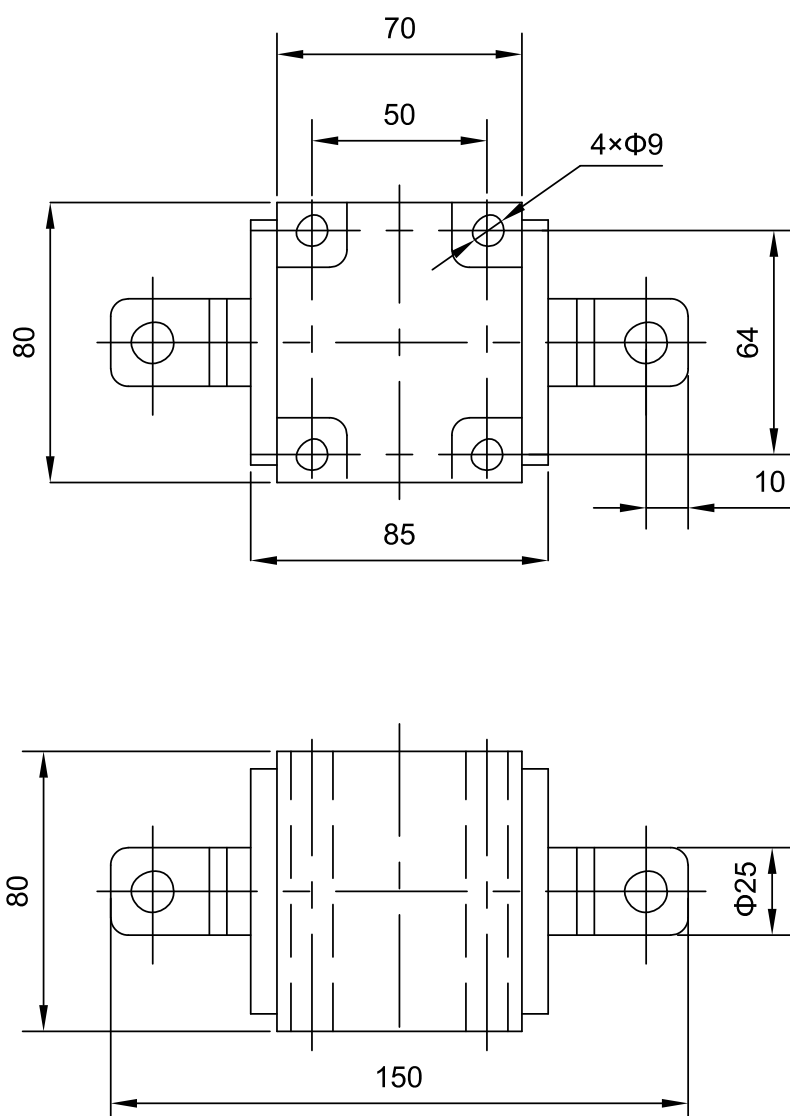
13.9. Special tool for installing and removing oil suction pipe for CV2 type OLTC



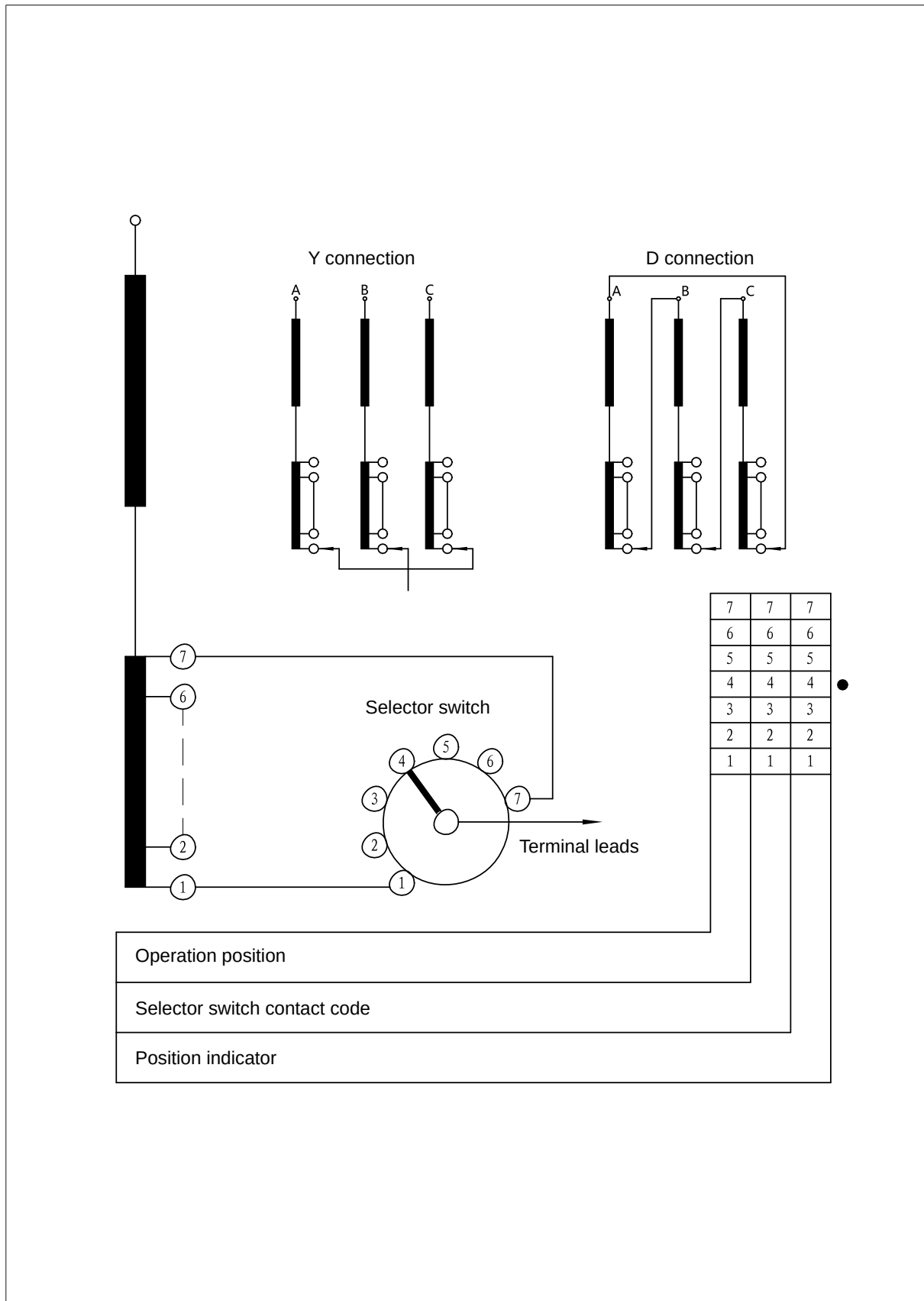
13.10. Horizontal and vertical driving shaft installation



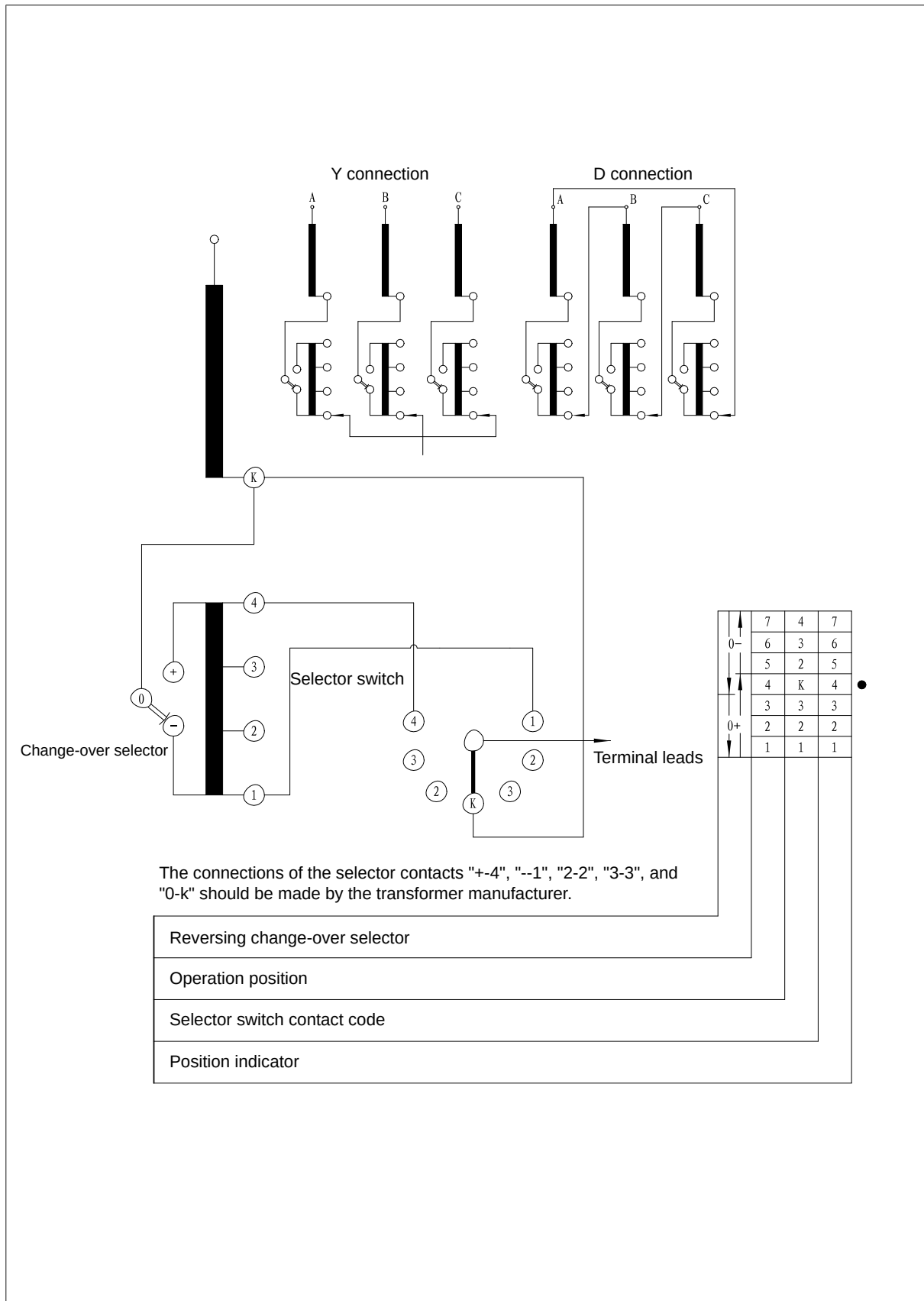
13.11. Dimension drawing of CV2 type intermediate support gearbox



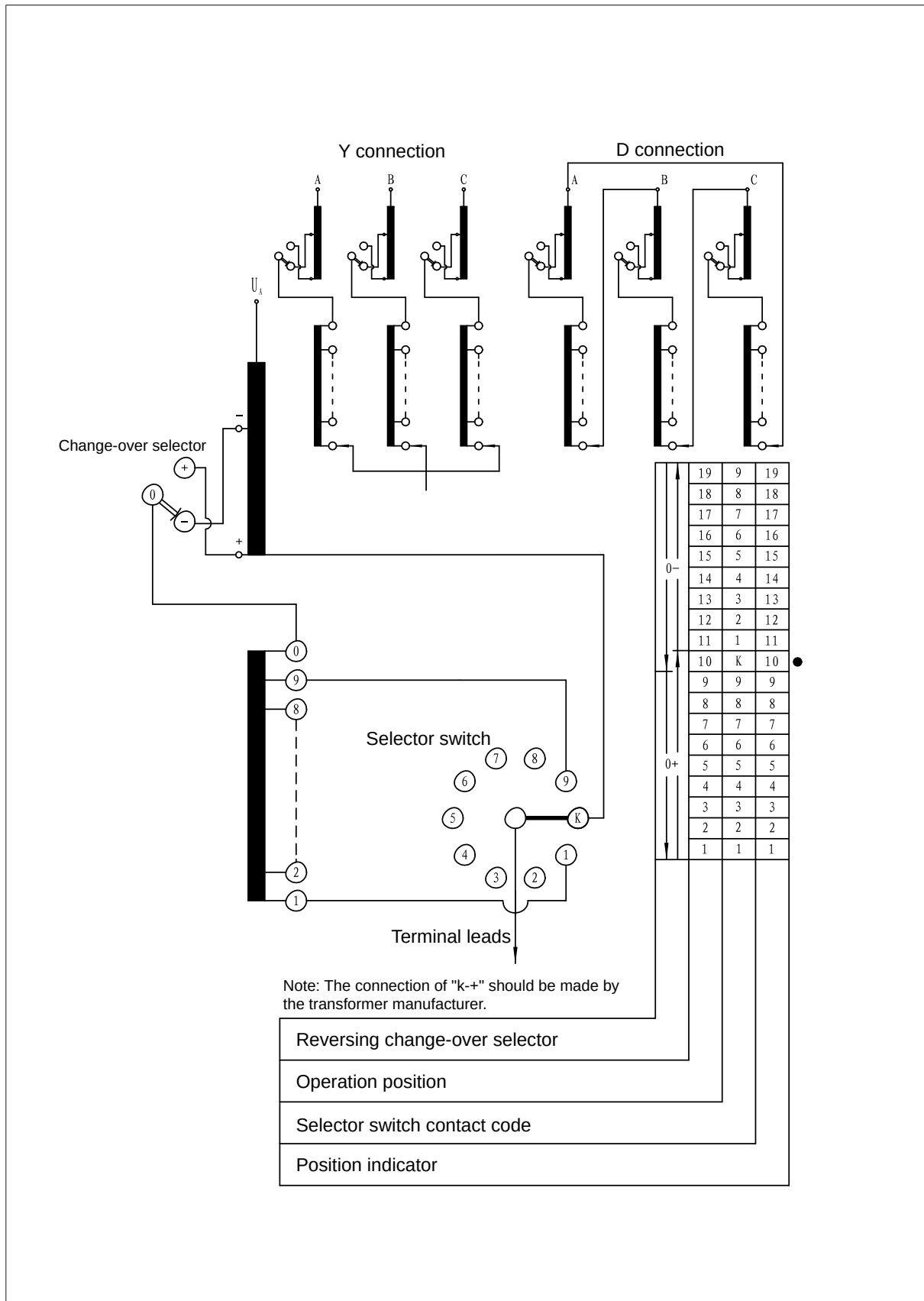
13.12. CV2 type OLTC 10070 operating position table and wiring schematic



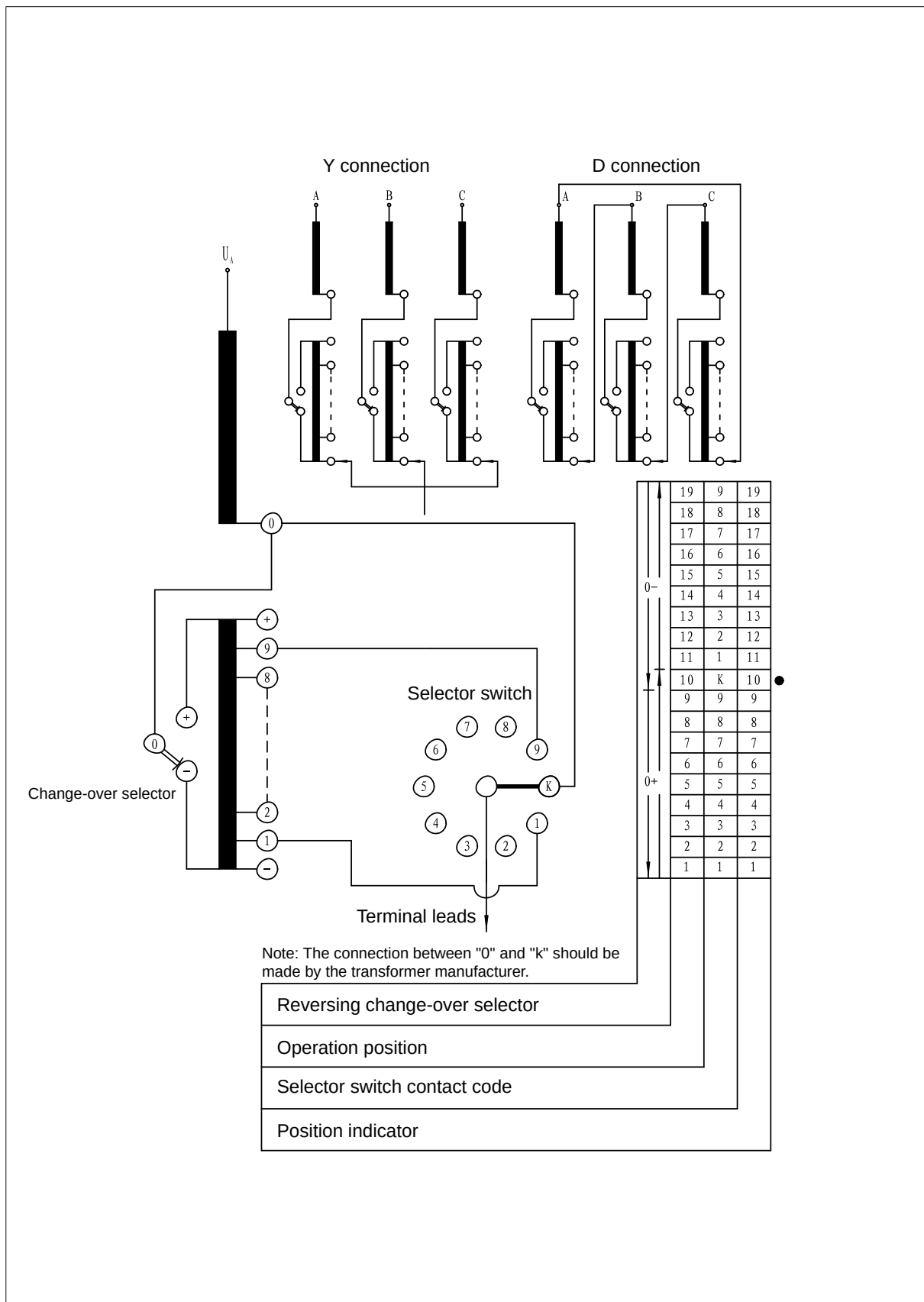
13.13. CV2 type OLTC 10071W operating position table and wiring schematic



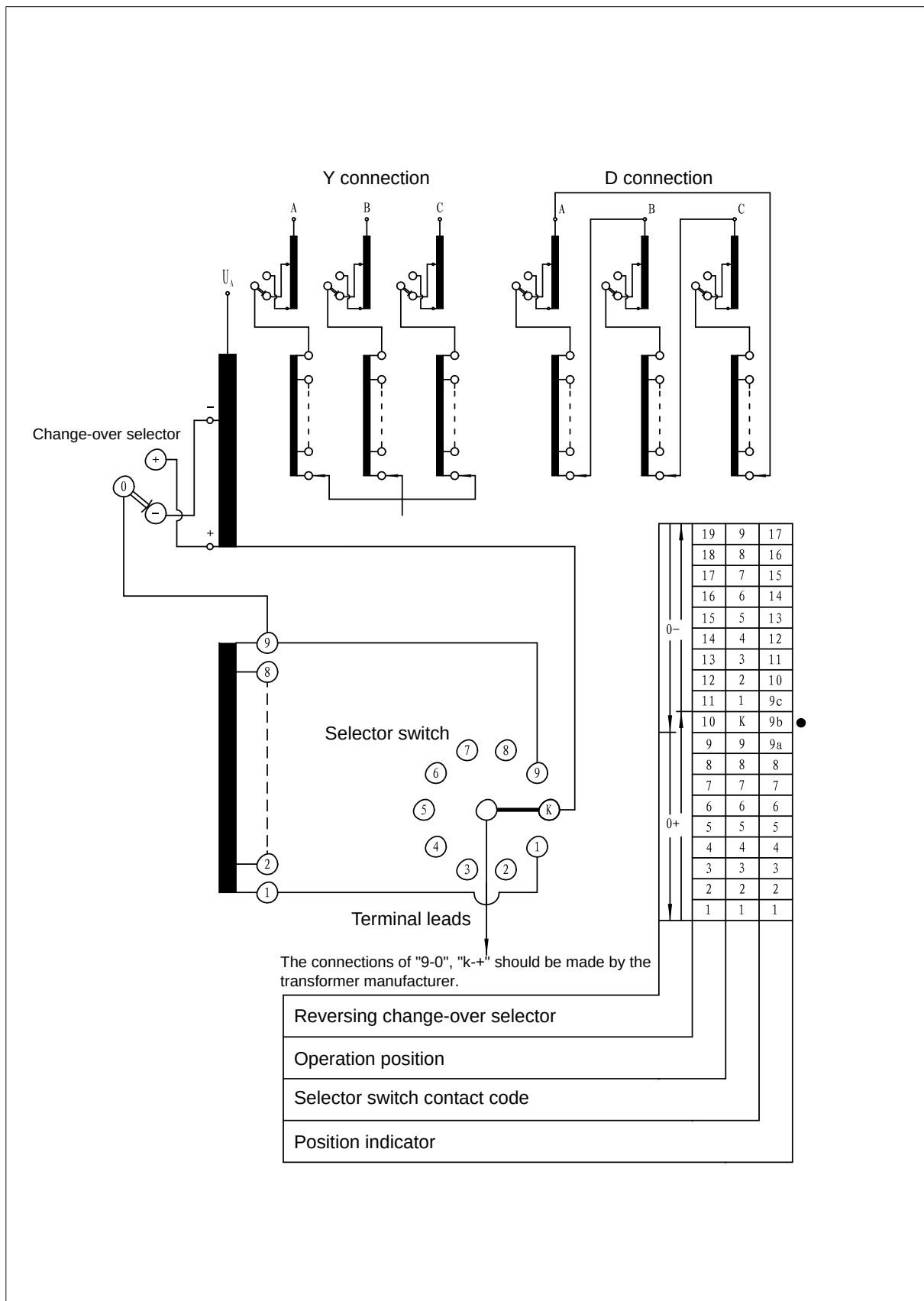
13.14. CV2 type OLTC 10191G operating position table and wiring schematic



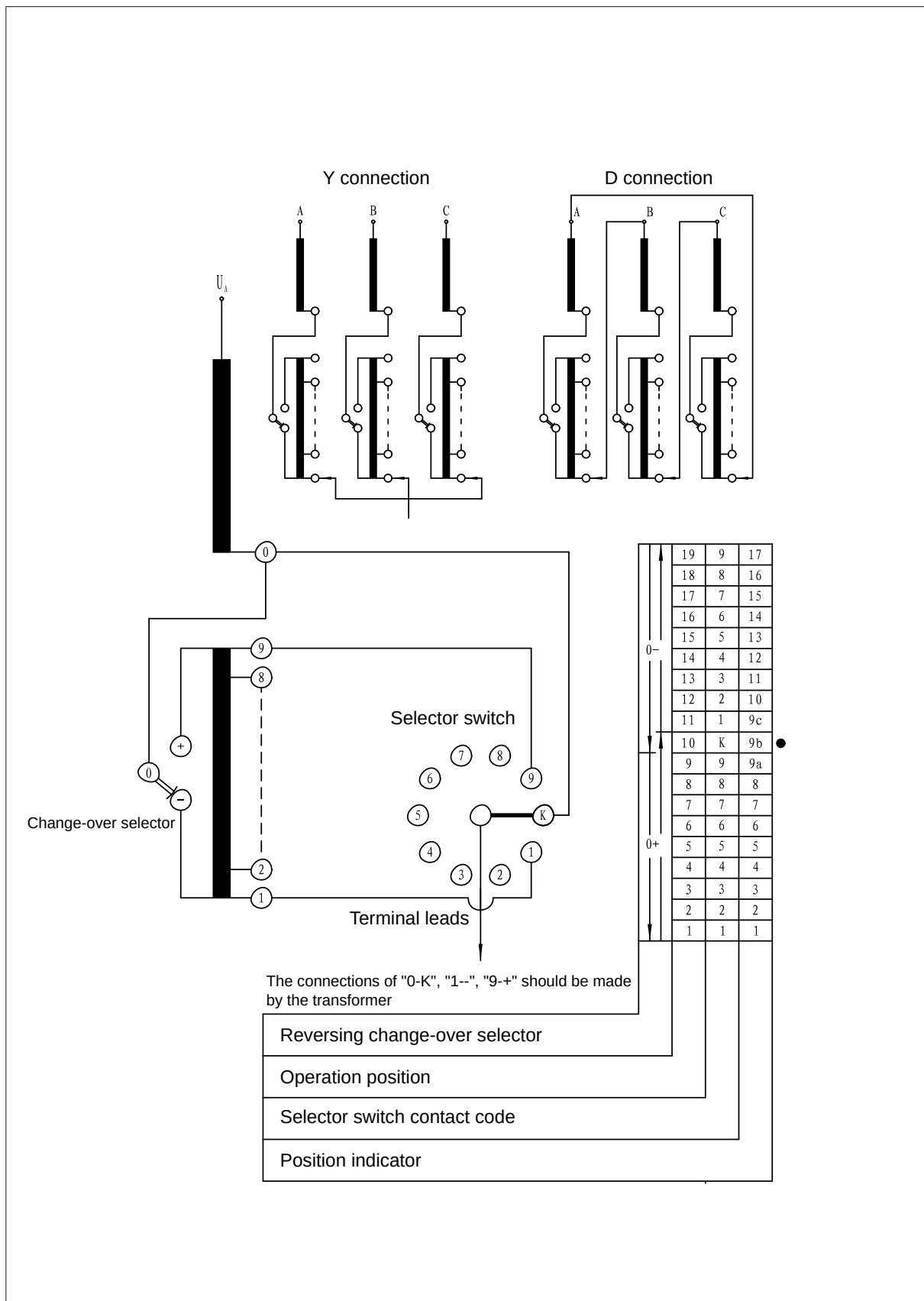
13.15. CV2 type OLTC 10191W operating position table and wiring schematic



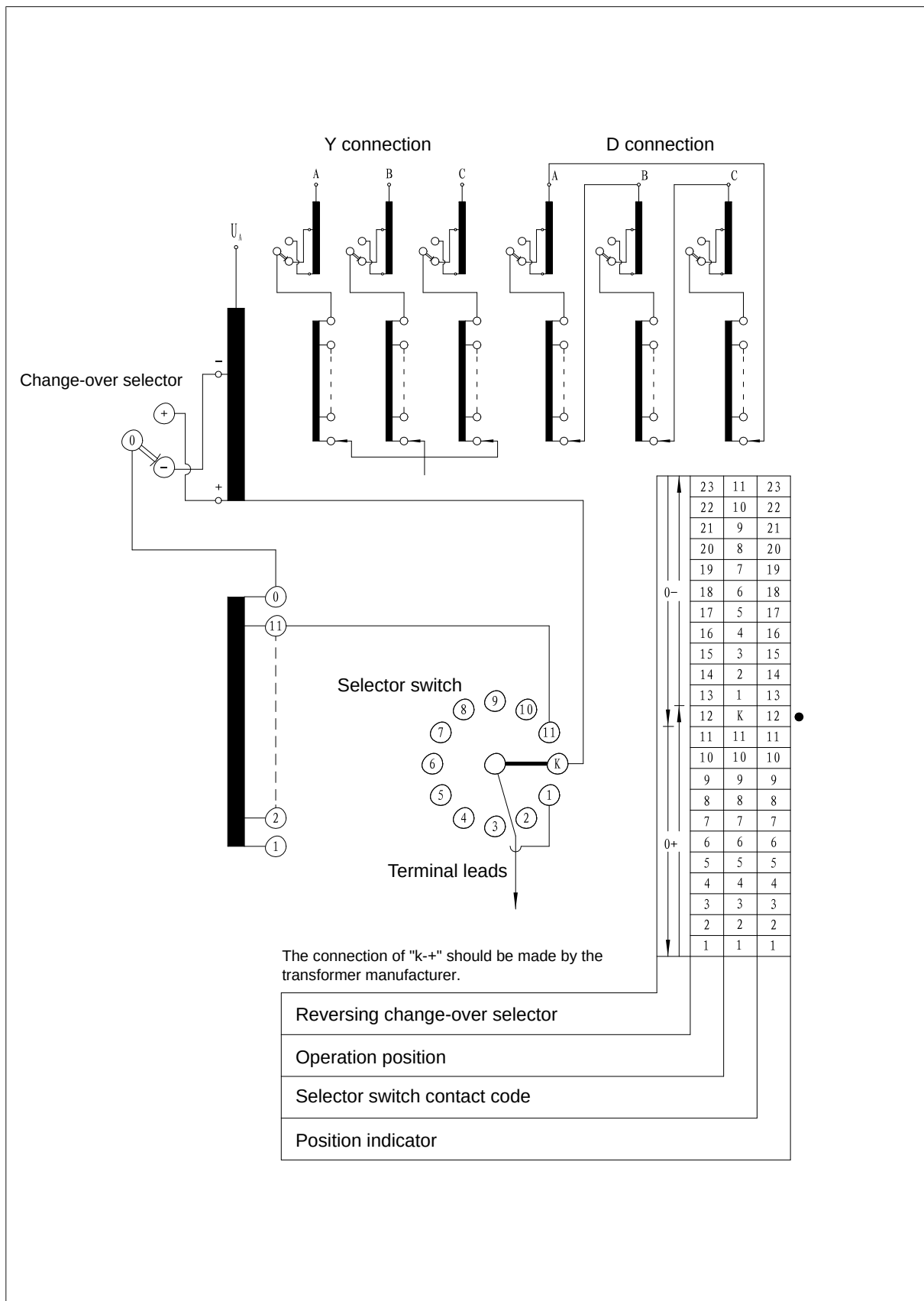
13.16. CV2 type OLTC 10193G operating position table and wiring schematic



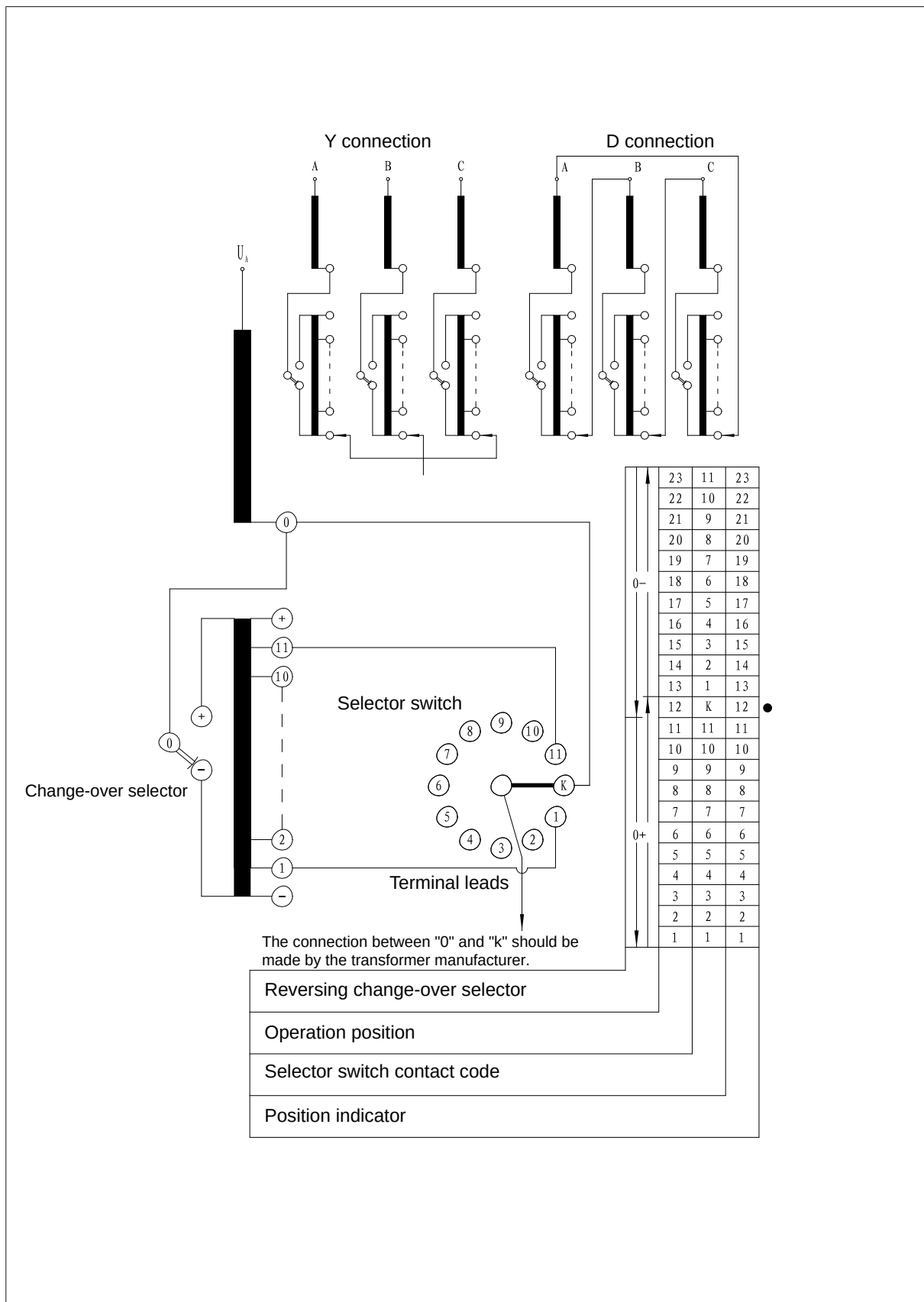
13.17. CV2 type OLTC 10193W operating position table and wiring schematic



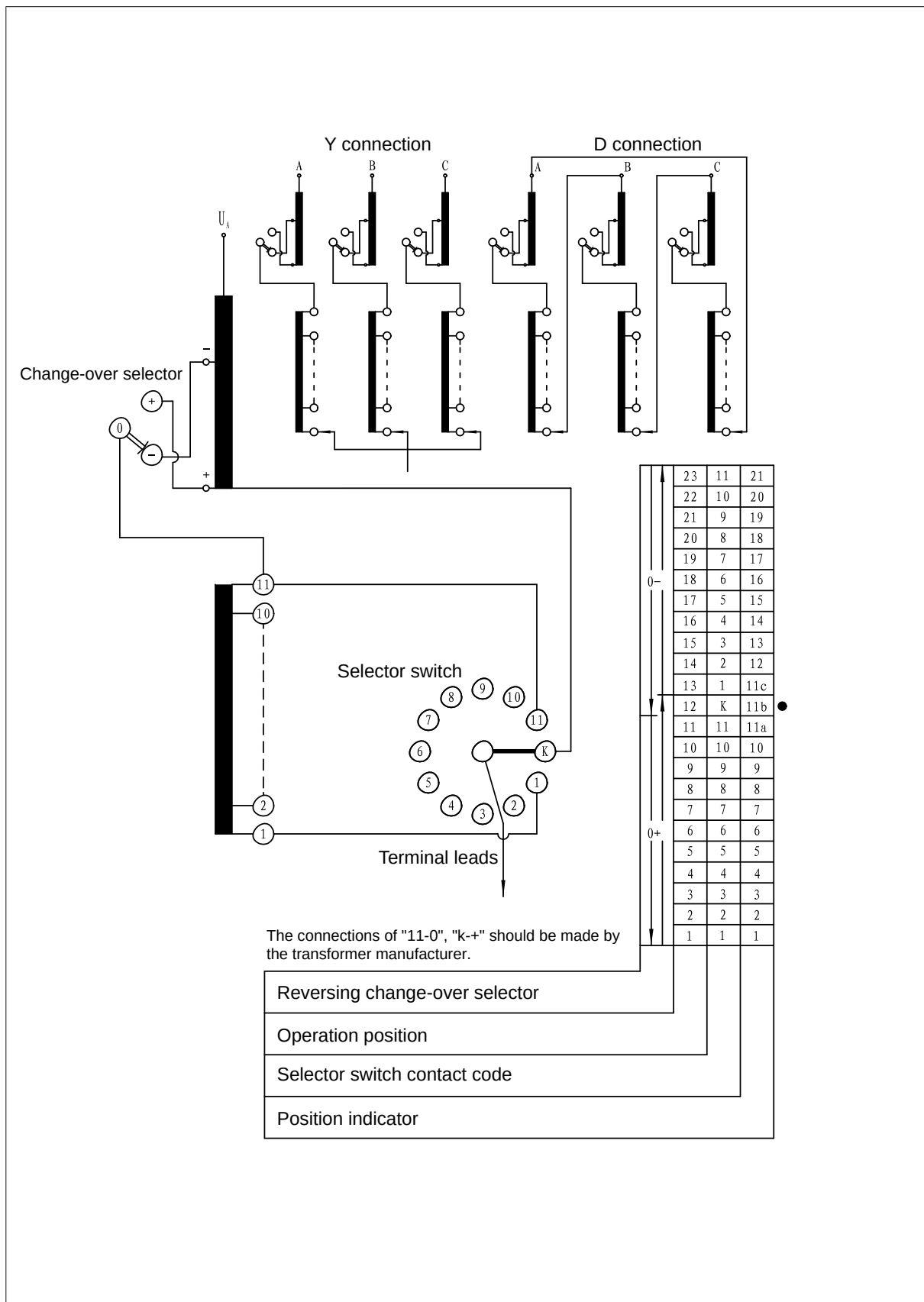
13.18. CV2 type OLTC 12231G operating position table and wiring schematic



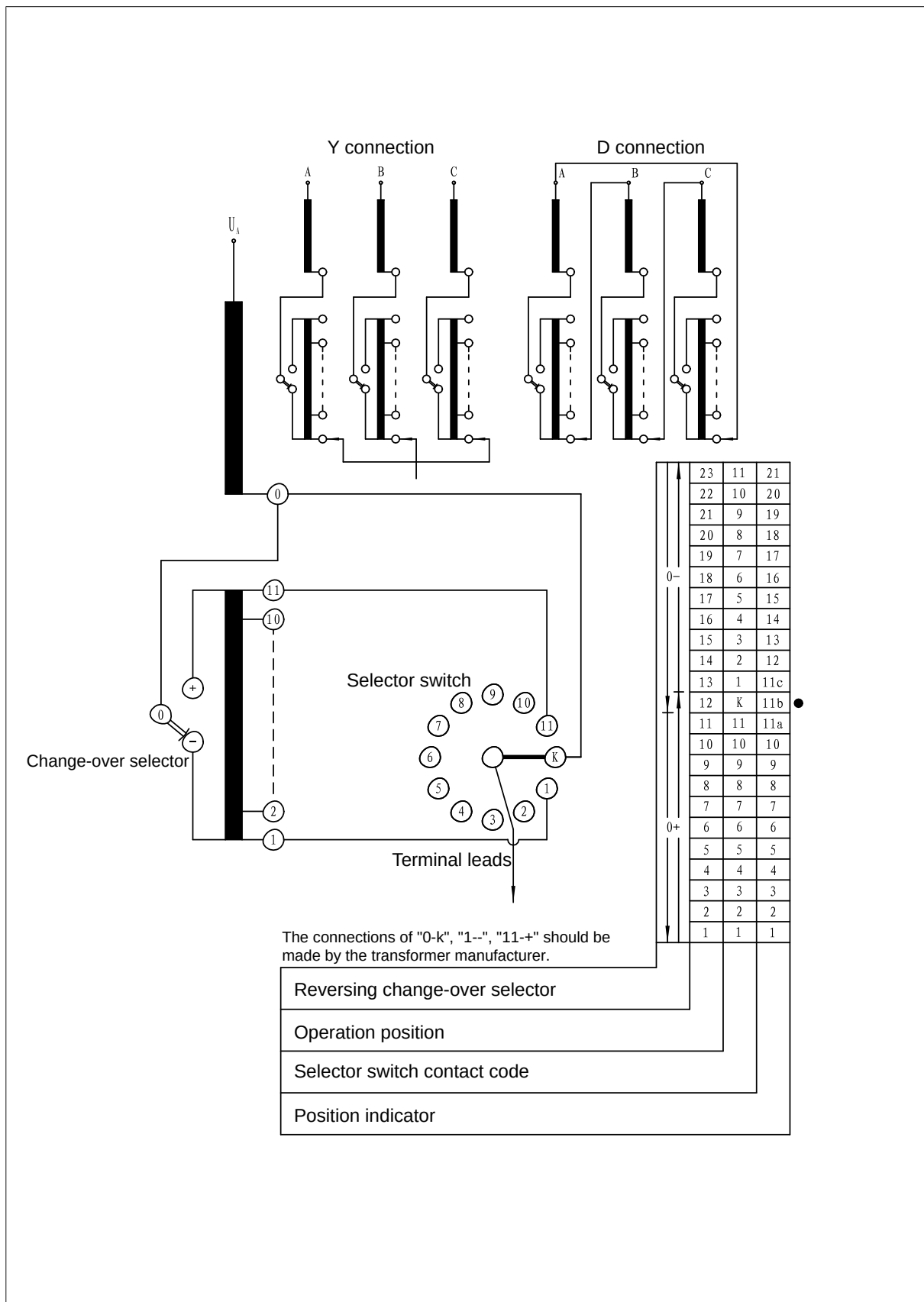
13.19. CV2 type OLTC 12231W operating position table and wiring schematic



13.20. CV2 type OLTC 12233G operating position table and wiring schematic



13.21. CV2 type OLTC 12233W operating position table and wiring schematic





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